PMath 965: Harmonic Maps with Prof. Spiro Karigiannis

A Crash Course in Riemannian Geometry

Riemannian Metrics

Definition
Let M be a smooth m-manifold. A Riemannian metric g on M is a smooth $(0,2)$ tensor which is symmetric and positive definite at every point. A Riemannian manifold is a pair (M,g) where g is a Riemannian metric on M.
The fact it is a $(0,2)$-tensor means $g \in \Gamma(T^*M \otimes T^*M)$, which is to say g is a smooth map $M \to T^*M \otimes T^*M, p \mapsto g_p \in T_p^*M \otimes T_p^*M$, so g_p is a bilinear form on T_pM. The fact g is symmetric and positive definite means for each $p \in M$, g_p is symmetric and positive definite, that is, an inner product on T_pM. In particular, for $X, Y \in \mathfrak{X}(M)$, $g(X,Y) \in \mathcal{C}^\infty(M)$, with $g(Y,X) = g(X,Y)$, and $g(X,X) \geq 0$ with equality if and only if $X = 0$.
Often, we will write $\langle X, Y \rangle = g(X,Y)$.

Fact
Every manifold admits lots of Riemannian metrics (using the dot product on $\mathbb{R}^n$ and blending it together chartwise with a partition of unity).

Let $(e_1, \ldots, e_m)$ be a local frame for TM over some open set $U \subseteq M$. From this, we get a local frame $(e^1, \ldots, e^m)$ for T^*M over U (sometimes called the dual coframe), determined by $e^i(e_j) = \delta^i_j \in \mathcal{C}^\infty(M)$.

Example
If $(x^1, \ldots, x^m)$ are local coordinates for M over U, then taking $e_k = \frac{\partial}{\partial x^k}$ for each k gives a local frame $(\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^m})$ for TM over U. In this case, the dual coframe is given by $e^k = \mathrm{d}x^k$ for each k.

Remark
Not every local frame is a coordinate frame, as we have the necessary condition $[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}] = 0$, which need not hold (for instance take a coordinate frame and multiply it by some nonvanishing smooth function).

Now, given a Riemannian metric g, an open set $U \subseteq M$, and a local frame $(e_1, \ldots, e_m)$ for TM over U, we can describe g in terms of the dual coframe as

$$g = g_{ij} e^i \otimes e^j$$

for some $m \times m$ symmetric positive-definite matrix (g_{ij}) of smooth functions on U.

Connections on TM

Definition
A connection (or covariant derivative) ∇ on TM is an $\mathbb{R}$-linear map $\nabla : \Gamma(TM) \to \Gamma(T^*M \otimes TM)$ such that it satisfies the following Leibniz rule for all $f \in \mathcal{C}^\infty(M), X \in \Gamma(TM)$:

$$\nabla(fX) = f\ \nabla X + df \otimes X$$

We can write this in the following equivalent way. Given $X, Y \in \Gamma(TM)$, define

$$\nabla_X Y := (\underbrace{\nabla Y}_{\in \Gamma(T^*M \otimes TM)})(\overbrace{\ X\ }^{\in \Gamma(TM)})$$

Then, ∇ satisfies:
(1) $\mathcal{C}^\infty(M)$-linear with respect to X (the argument in the subscript).
(2) Additive in Y, and satisfying the following Leibniz rule for all $f \in \mathcal{C}^\infty(M), X, Y \in \Gamma(TM)$:

$$\nabla_X(fY) = (Xf)Y + f\nabla_X Y$$

Facts

These follow from the Leibniz rule and using smooth bump functions.
(1) For all $p \in M$, $(\nabla_X Y)_p$ depends only on X at p and on Y in an open neighbourhood of p. In fact, $(\nabla_X Y)_p$ only depends on the value of Y along a curve α with image in a neighbourhood of p with $\alpha(0) = p, \alpha'(0) = X_p$. Because of this, we often write $\nabla_{X_p} Y$ for $(\nabla_X Y)_p$.
(2) Any manifold admits lots of connections on TM (using a partition of unity).
(3) Given a connection ∇ on TM, it induces connections on all tensor bundles over M in a natural way: On the trivial real line bundle $\mathbb{R}_M := M \times \mathbb{R}$, a section is just a smooth function $f \in \mathcal{C}^\infty(M)$, and we define $\nabla_X f = Xf$ for any ∇.
On T^*M, given a 1-form $\alpha \in \Gamma(T^*M)$, we define $\nabla_X \alpha \in \Gamma(T^*M)$ by

$$(\nabla_X \alpha)(Y) := X(\alpha(Y)) - \alpha(\nabla_X Y)$$

for all $Y \in \Gamma(TM)$. One needs to check $\nabla_X \alpha$ is additive in X, α, $\mathcal{C}^\infty(M)$-linear in X, and satisfies the Leibniz rule $\nabla_X(f\alpha) = (Xf)\alpha + f\nabla_X \alpha$.
Finally, let σ be a (k, ℓ)-tensor on M. We define a (k, ℓ)-tensor $\nabla_X \sigma$ by

$$(\nabla_X \sigma)(\alpha^1, \dots, \alpha^k, Y_1, \dots, Y_\ell) = X(\sigma(\alpha^1, \dots, \alpha^k, Y_1, \dots, Y_\ell))$$

$$-\sum_{i=1}^k \sigma(\alpha^1, \dots, \nabla_X \alpha^i, \dots, \alpha^k, Y_1, \dots, Y_\ell) - \sum_{j=1}^k \sigma(\alpha^1, \dots, \alpha^k, Y_1, \dots, \nabla_X Y_j, \dots, Y_\ell)$$

for all $\alpha^1, \dots, \alpha^k \in \Gamma(T^*M), Y_1, \dots, Y_\ell \in \Gamma(TM)$, and then one must show $\nabla_X \sigma$ is additive in X, σ, $\mathcal{C}^\infty(M)$-linear in X, and satisfies the Leibniz rule $\nabla_X(f\sigma) = (Xf)\sigma + f\nabla_X \sigma$.

Example

Let g a Riemannian metric on M. Let ∇ be a connection on TM. Then in the above setup, $k = 0, \ell = 2$, and we see for $X, Y, Z \in \Gamma(TM)$ that

$$(\nabla_X g)(Y, Z) = X(g(Y, Z)) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z)$$

Definition

∇ is g-compatible if $\nabla g = 0$ if and only if $\nabla_X g = 0$ for all $X \in \Gamma(TM)$ if and only if $X(g(Y, Z)) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$ for all $X, Y, Z \in \Gamma(TM)$. Thinking of ∇ as a directional derivative, this says g is "as constant as it can be".

Another important properties of connections is their torsion:

Definition

Given a connection ∇ on TM, we get a $(1, 2)$-tensor T^∇ on M, called the torsion of ∇, defined by

$$\underbrace{T^\nabla(X, Y)}_{\in \Gamma(TM)} = \nabla_X Y - \nabla_Y X - [X, Y]$$

Note that T^∇ is skew-symmetric in X and Y.
We say ∇ is torsion-free if $T^\nabla = 0$ if and only if $\nabla_X Y - \nabla_Y X = [X, Y]$ for all $X, Y \in \Gamma(TM)$.

Exercise

Show T^∇ is a tensor, which is to say $T^\nabla(fX, Y) = T^\nabla(X, fY) = fT^\nabla(X, Y)$ for all $X, Y \in \Gamma(TM), f \in$

$\mathcal{C}^\infty(M)$.

Theorem (Fundamental Theorem of Riemannian Geometry)
Let g be a Riemannian metric on M. Then there exists a unique connection ∇ on TM such that:
(1) $\nabla g = 0$ (g-compatible).
(2) $T^\nabla = 0$ (torsion-free).
This is called the Levi-Civita connection of g.
Idea of Proof:
Suppose such a ∇ exists. Then you can show that ∇ must satisfy the Koszul formula:

$$2\langle \nabla_X Y, Z\rangle = X\langle Y, Z\rangle + Y\langle X, Z\rangle - Z\langle X, Y\rangle - \langle [X,Z], Y\rangle - \langle [Y,Z], X\rangle + \langle [X,Y], Z\rangle$$

The right hand side depends only on the metric g, and so uniquely determines ∇. Then, because g is positive definite, the left hand side determines $\nabla_X Y$, and then one must check that if we define ∇ this way, ∇ is a connection, g-compatible, and torsion-free, which is a relatively simple computation. ■

Remark
The Levi-Civita connection and the connections it induces on the tensor bundles is the only connection we will consider in this course.

Remark
Why are g-compatibility and torsion-free the right conditions to impose? Well, we can look at the simplest case: $M = \mathbb{R}^m$ with the dot product on each tangent space $T_p\mathbb{R}^m \cong \mathbb{R}^m$ giving a metric on M. For this metric on $\mathbb{R}^m$, the Levi-Civita connection satisfies $\nabla_X(Y^k e_k) = X(Y^k)e_k$, which is to say the directional derivative of a vector field in $\mathbb{R}^m$ is just done by taking the componentwise directional derivatives of its component functions, which is the usual directional derivative. So the Levi-Civita connection coincides with the directional derivative on $\mathbb{R}^m$.

Definition
On the other hand, let $e_1, \dots, e_m$ be a local frame for TM over some open set U. Then we obtain m^3 smooth functions (Γ_{ij}^k) on U by writing

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k$$

called the **Christoffel symbols** for ∇ with respect to $(e_1, \dots, e_m)$ (note that these aren't the components of a $(1,2)$ tensor, since ∇ satisfies a product rule in its second argument and not $\mathcal{C}^\infty(M)$-linearity). Then, if we write $X = X^i e_i$ and $Y = Y^j e_j$, then we get

$$\nabla_X Y = \nabla_{X^i e_i}(Y^j e_j) = X^i \nabla_{e_i}(Y^j e_j)$$

$$= X^i(e_i(Y^j)e_j + Y^j \nabla_{e_i} e_j) = X(Y^j)e_j + X^i Y^j \Gamma_{ij}^k e_k = (X(Y^k) + X^i Y^j \Gamma_{ij}^k)e_k$$

and so we obtain $(\nabla_X Y)^k = (X(Y^k) + X^i Y^j \Gamma_{ij}^k)$.

Example
Let ∇ be the Levi-Civita connection of a metric g, and let $e_1, \dots, e_m$ be a coordinate frame $\{\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^m}\}$. Then the Koszul formula gives

$$g^{\ell m} 2\Gamma_{ij}^k g_{k\ell} = \left(\frac{\partial}{\partial x^i}(g_{j\ell}) + \frac{\partial}{\partial x^j}(g_{i\ell}) - \frac{\partial}{\partial x^\ell}(g_{ij})\right) g^{\ell m} \underbrace{g(\frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^\ell})}_{=g_{k\ell}}$$

and so we obtain,

$$\Gamma_{ij}^m = \frac{1}{2} g^{m\ell}\left(\frac{\partial g_{i\ell}}{\partial x^j} + \frac{\partial g_{j\ell}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^\ell}\right)$$

where we note that $\Gamma_{ij}^m = \Gamma_{ji}^m$ in this case (this is not true for an arbitrary local frame).

Definition
Suppose $u : M \to N$ is a smooth map. A vector field on N along u is a smooth map $X : M \to TN$ such that for all $p \in M$, $X_p \in T_{u(p)}N$. Such a vector field on N along u is the same thing as a smooth section of the pullback bundle u^*TN over M (which has fibres $(u^*TN)_p = T_{u(p)}N$).

Example
Let $M = (a, b) \subseteq \mathbb{R}$ be an open interval, and $\alpha : (a, b) \to N$ a smooth curve. The velocity vector field α' of α is a vector field on N along α. In particular, if $\widehat{\alpha}(t) = (x^1(t), \ldots, x^n(t))$ is its local coordinate representation, then $\alpha'(t) = \frac{dx^k}{dt}\frac{\partial}{\partial x^k}|_{\alpha(t)} \in T_{\alpha(t)}N$.

Definition
Let $u : M \to N$ be a smooth map, and suppose ∇ is a connection on TN. Then there exists a "covariant derivative" $u^*\nabla$ which allows us to covariantly differentiate vector fields on N along u as follows: Let $X_p \in T_pM$, and let $\sigma \in \Gamma(u^*TN)$ be a vector field on N along u. Then we want to define $(u^*\nabla)_{X_p}\sigma \in T_{u(p)}N$, which we do by setting

$$(u^*\nabla)_{X_p}\sigma = \nabla_{(u_*)_p X_p}\sigma$$

(where implicitly on the right, we extend σ to an open neighbourhood of $u(M)$, where the discussion below shows it doesn't depend on the choice of this extension). By properties of ∇ seen above, this only depends on σ only along any curve passing through $u(p)$ with initial velocity $(u_*)_p X_p$. So we can choose any curve α in M with $\alpha(0) = p$, $\alpha'(0) = X_p$, and then $u \circ \alpha$ is the desired curve on N. This $u^*\nabla$ is called the pullback connection.

To see this more concretely, suppose $(x^1, \ldots, x^m)$ are local coordinates on M, and $(y^1, \ldots, y^n)$ local coordinates on N, identifying them with the components of u. Consider the Christoffel symbols for ∇ in these coordinates, $\nabla_{\frac{\partial}{\partial y^a}}\frac{\partial}{\partial y^b} = \Gamma_{ab}^c\frac{\partial}{\partial y^c}$. Then we know

$$(u_*)_p\left(\frac{\partial}{\partial x^i}|_p\right) = \frac{\partial y^a}{\partial x^i}(p)\frac{\partial}{\partial y^a}|_{u(p)}$$

Then,

$$\begin{aligned}(u^*\nabla)_{\frac{\partial}{\partial x^i}}(u^*\frac{\partial}{\partial y^b}) &:= \nabla_{u_*\frac{\partial}{\partial x^i}}\frac{\partial}{\partial y^b} = \nabla_{\frac{\partial y^a}{\partial x^i}\frac{\partial}{\partial y^a}}\frac{\partial}{\partial y^b}\\ &= \frac{\partial y^a}{\partial x^i}\nabla_{\frac{\partial}{\partial y^a}}\frac{\partial}{\partial y^b} = \frac{\partial y^a}{\partial x^i}\Gamma_{ab}^c\frac{\partial}{\partial y^c}\end{aligned}$$

(where strictly speaking, we should write $\Gamma_{ab}^c \circ u$).

Geodesics

Definition
Let (M, g) be a Riemannian manifold, and let ∇ be the Levi-Civita connection. Let $\gamma : (a, b) \to M$ be a smooth curve on M. We say γ is a geodesic if $(\gamma^*\nabla)\gamma' = 0$, where we call $(\gamma^*\nabla)\gamma'$ is called the acceleration of γ. Equivalently, this means for all $t \in (a, b)$, $0 = (\gamma^*\nabla)_{\frac{d}{dt}}\gamma' = \nabla_{\gamma_*\frac{d}{dt}}\gamma' = \nabla_{\gamma'(t)}\gamma'$.

In coordinates, suppose $(x^1, \ldots, x^m)$ are local coordinates for M. Then $\gamma'(t) = \frac{dx^i}{dt}\frac{\partial}{\partial x^i}$, where we see

$$\nabla_{\gamma'(t)}\gamma' = \nabla_{\frac{dx^i}{dt}\frac{\partial}{\partial x^i}}\left(\frac{dx^j}{dt}\frac{\partial}{\partial x^j}\right) = \frac{dx^i}{dt}\nabla_{\frac{\partial}{\partial x^i}}\left(\frac{dx^j}{dt}\frac{\partial}{\partial x^j}\right)$$

$$\overbrace{=}^{\text{Leibniz}} \frac{dx^i}{dt}\left(\left(\frac{\partial}{\partial x^i}\frac{dx^j}{dt}\right)\frac{\partial}{\partial x^j} + \frac{dx^j}{dt}\nabla_{\frac{\partial}{\partial x^i}}\frac{\partial}{\partial x^j}\right) \overbrace{=}^{\text{chain rule}} \frac{d^2x^k}{dt^2}\frac{\partial}{\partial x^k} + \frac{dx^i}{dt}\frac{dx^j}{dt}\Gamma_{ij}^k\frac{\partial}{\partial x^k}$$

$$= \left(\frac{d^2x^k}{dt^2} + \Gamma^k_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt} \right) \frac{\partial}{\partial x^k}$$

and so this is zero if and only if

$$\frac{d^2x^k}{dt^2} + \Gamma^k_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt} = 0$$

for all $k = 1, \dots, n$. This is the "geodesic equation", which we see is a nonlinear second-order ODE.

Remark

We'll see that a geodesic is a special example of a harmonic map, with domain being the Riemannian manifold $((a, b), dt^2)$.

Proposition

Let $p \in M$, and let $v_p \in T_pM$. Then there exists a unique maximal geodesic $\gamma_{v_p} : (a, b) \to M$, where (a, b) is an open interval containing 0, satisfying $\gamma_{v_p}(0) = p$, $\gamma'_{v_p}(0) = v_p$.

Idea of Proof:

This is really the ODE existence and uniqueness theorem applied to the geodesic equation, where maximality follows from uniqueness. ■

Example

Let $M = \mathbb{R}^n$, and let g be the Euclidean metric, which has identically zero Christoffel symbols. Then the geodesic equation becomes $\frac{d^2x^k}{dt^2} = 0$ for all k, which has maximal solution $\gamma_{v_p}(t) = p + tv_p$ for all $t \in \mathbb{R}$.

Fact

On (M, g), geodesics are locally length-minimizing. We won't discuss this much, but we will see geodesics are critical maps for both the length functional and energy functional.

Riemannian Immersions and Submersions

Let (M^m, g) and (N^n, h) be Riemannian manifolds.

Definition

Let $u : M \to N$ be a smooth map.

We say u is a Riemmanian immersion (also called an isometric immersion) if $u^*h = g$. Expanding the definitions, this says for all $p \in M, X_p, Y_p \in T_pM$ that $g_p(X_p, Y_p) = h_{u(p)}((u_*)_p X_p, (u_*)_p Y_p)$, and so for all $p \in M$, the linear map $(u_*)_p : (T_pM, g_p) \to (T_{u(p)}N, h_{u(p)})$ is a linear isometry. In particular, $(u_*)_p$ is injective, and so u is an immersion. Notice that this forces $m \leq n$.

Remark

Suppose $u : M \to (N, h)$ is a smooth immersion, where M need not have a metric on it. Then we can define a Riemannian metric g on M by setting $g = u^*h$. In particular, this gives a metric on any embedded submanifold of a Riemannian manifold, like submanifolds of $\mathbb{R}^n$.

Example

Consider the standard embedding of $\iota : S^n \hookrightarrow \mathbb{R}^{n+1}$. This defines the "round metric" $g_r = \iota^* g_{Euc}$ on S^n.

Definition

Let $u : M \to N$ be a smooth map. We say u is a Riemannian submersion if:

(a) u is a submersion (and so this forces $m \geq n$).

(b) For all $p \in M$, $(u_*)_p : \ker((u_*)_p)^\perp \to T_{u(p)}N$ is a linear isomorphism. Notice that in particular by (a), rank-nullity tells us that $\dim \ker((u_*)_p)^\perp = \dim T_{u(p)}N$, and so $(u_*)_p$ is an isometric isomorphism.

Exercise
Fix $p \in M$. Then the implicit function theorem tells us $u^{-1}(u(p))$ is a submanifold of M with dimension $n - m$. Show that $\ker((u_*)_p) = T_p(u^{-1}(u(p)))$, which is called the vertical space at p (its orthogonal complement $\ker((u_*)_p)^\perp$ is called the horizontal space).

Definition
Suppose $u : M \to N$ is both a Riemannian immersion and submersion. Then $m = n$, and $(u_*)_p : T_pM \to T_{u(p)}N$ is an isometric isomorphism. The fact the pushforward is an isomorphism for all $p \in M$ shows that u is a local diffeomorphism, and so because it is an isometry at each tangent space, we call u a local isometry. If u is both a local isometry and a diffeomorphism (which we get from the inverse function theorem if it is a local isometry and bijective), we call u an isometry, which is the natural notion of isomorphism in the category of Riemannian manifolds.

Example
A local isometry need not be a global diffeomorphism, for example the map $\mathbb{R} \to S^1$, $t \mapsto e^{it}$.

More on Geodesics

Lemma (Rescaling Lemma)
Let γ_{v_p} be the geodesic with initial data $\gamma_{v_p}(0) = p$, $\gamma'_{v_p}(0) = v_p$. Then,

$$\gamma_{cv_p}(t) = \gamma_{v_p}(ct)$$

for all $c \in \mathbb{R}$ and any t such that either side is defined.
Sketch of Proof:
This is an exercise using the special form of the geodesic equation (quadratic in first derivatives, linear in second derivative) to factor out the c^2 appearing from the chain rule, and then using uniqueness of solutions. ■

Definition
Fix $p \in M$. Then, there exists an open neighbourhood $U \subseteq M$ of p and $\varepsilon > 0$ such that if $q \in U$ and $v_q \in T_qM$ with $|v_q|_g < \varepsilon$, then γ_{v_q} is defined on $(-2, 2)$ at least.
This follows from the precise form of the ODE Theorem - given initial data we have a uniform time interval where our solution exists, so then we can use the rescaling lemma to make the interval bigger by shrinking our vectors.
Now let $\widetilde{U} = \{(q, v_q) \in TM : q \in U, |v_q| < \varepsilon\}$. This is an open subset of TM. Define the **exponential map** $\exp : \widetilde{U} \to M$ via $\exp(q, v_q) := \gamma_{v_q}(1)$.
We denote the restriction of exp to $\widetilde{U} \cap T_pM$ by $\exp_p : \widetilde{U} \cap T_pM \to M$ via $\exp_p(v_p) = \gamma_{v_p}(1)$.

Remark
By smooth dependence of the solution to the ODE on initial data, the exponential map (and its restriction $\exp_p$ to T_pM) is smooth.

Proposition
There exists $W \subseteq T_pM$ open containing 0_p with $W \subseteq \widetilde{U} \cap T_pM$ such that $\exp_p|_W : W \to \exp_p(W) \subseteq M$ is a diffeomorphism onto an open subset of M.
Proof:
By the inverse function theorem, it suffices to show $[(\exp_p)_*]_{0_p} : T_{0_p}(B_\varepsilon(0_p)) \to T_{\exp_p(0_p)}M$ is an isomorphism. However, notice by translation and the fact $B_\varepsilon(0_p)$ is open in T_pM that $T_{0_p}(B_\varepsilon(0_p)) \cong T_pM$, and $\exp_p(0_p) = p$ since $\gamma_{0_p}(t) = p$ for all time. Thus, $T_{\exp_p(0_p)}M = T_pM$. But now to compute the pushforward, we know for any $v_p \in T_{0_p}(T_pM) \cong T_pM$ and any smooth curve $\alpha : (-\delta, \delta) \to B_\varepsilon(p) \subseteq T_pM$ with $\alpha(0) = 0_p$, $\alpha'(0) = v_p$, that

$$[(\exp_p)_*]_{0_p}(v_p) = (\exp_p \circ \alpha)'(0)$$

But now take $\alpha(t) = tv_p$, so by definition and the rescaling lemma,

$$(\exp_p \circ \alpha)(t) = \exp_p(tv_p) = \gamma_{tv_p}(1) = \gamma_{v_p}(t)$$

so $(\exp_p \circ \alpha)'(0) = \gamma_{v_p}'(0) = v_p$. Thus, we see $[(\exp_p)_*]_{0_p}(v_p) = v_p$ is the identity under our isomorphism, and thus is an isomorphism. ■

Definition
Let $Y = \exp_p(W)$ be an open neighbourhood of p from the above proposition, and let $F = (\exp_p|_W)^{-1} : Y \to W$ be our diffeomorphism. Then we know $W \subseteq T_pM$ is an open subset, and we can pick a linear isomorphism $L : T_pM \xrightarrow{\cong} \mathbb{R}^m$ by picking a g_p-orthonormal basis $e_1, \ldots, e_n$ of T_pM. Then, $L \circ F$ is a smooth chart for M centered at p. Such a coordinate chart is called geodesic normal coordinates centered at p (or sometimes Riemannian normal coordinates centered at p).

Proposition
Let $\varphi : Y \to \varphi(Y) \subseteq \mathbb{R}^m$ be a geodesic normal coordinate chart containing p (so $\varphi = L \circ F = L \circ (\exp_p|_W)^{-1}$). For any $v_p = v^k \frac{\partial}{\partial x^k}\big|_p \in T_pM$, the geodesic γ_{v_p} has coordinate representation $\widehat{\gamma}_{v_p}(t) = (tv^1, \ldots, tv^m)$. That is, the geodesics passing through p correspond to straight lines through the origin in these coordinates. Moreover, $g_{ij}(p) = \delta_{ij}$, $\Gamma^k_{ij}(p) = 0$, and $\frac{\partial g_{ij}}{\partial x^k}(p) = 0$ (notice that this holds only at p).
Proof:
By abuse of notation, $\varphi = L \circ \exp_p^{-1}$, and $\varphi^{-1} = \exp_p \circ L^{-1}$. Since the pushforward of a linear map is itself, by the chain rule,

$$[(\varphi^{-1})_*]_0 = \underbrace{[(\exp_p)_*]_{0_p}}_{=\mathrm{Id}} \circ L^{-1} = L^{-1}$$

using what we showed in the above proposition. Thus, by definition,

$$\frac{\partial}{\partial x^k}\bigg|_p = [(\varphi^{-1})_*]_0(E_k) = L^{-1}(E_k) = e_k|_p$$

where E_k is the kth standard basis vector of $\mathbb{R}^m$, though of as an element of $T_0\mathbb{R}^m \cong \mathbb{R}^m$, and e_k is the orthonormal basis of T_pM defining L chosen above. Thus, by orthonormality, $g_{ij}(p) = \langle \frac{\partial}{\partial x^i}|_p, \frac{\partial}{\partial x^j}|_p \rangle = \langle e_i, e_j \rangle = \delta_{ij}$. Moreover, the coordinate representation of a geodesic is given by

$$\widehat{\gamma}_{v_p} = \varphi \circ \gamma_{v_p} = L \circ \exp_p^{-1} \circ \gamma_{v_p}$$

and so

$$\widehat{\gamma}_{v_p}(t) = L(\exp_p^{-1}(\underbrace{\gamma_{v_p}(t)}_{=\gamma_{tv_p}(1) = \exp_p(tv_p)})) = L(tv_p) = (tv^1, \ldots, tv^n)$$

Moreover, writing $\widehat{\gamma}_{v_p}(t) = (x^1(t), \ldots, x^m(t))$ with $x^k(t) = tv^k$, inserting this into the geodesic equation

$$\underbrace{\frac{d^2x^k}{dt^2}}_{=0} + \Gamma^k_{ij}(x^1(t), \ldots, x^m(t)) \underbrace{\frac{dx^i}{dt}}_{=v^i} \underbrace{\frac{dx^j}{dt}}_{=v^j} = 0$$

and so at $t = 0$ we obtain $\gamma^k_{ij}(p)v^iv^j = 0$ for all choices of $v^1, \ldots, v^m$, and so $\Gamma^k_{ij}(p) = 0$. Finally, from the formula we had for the Christoffel symbols in terms of the metric, we can manipulate it to

$$\frac{\partial g_{ij}}{\partial x^k} = \Gamma^\ell_{ik} g_{\ell j} + \Gamma^\ell_{jk} g_{ki}$$

which is zero at p. ■

Curvature of a Riemannian Metric

Let (M^m, g) be a Riemannian manifold.

Definition
Given $X, Y \in \Gamma(TM)$, we define a $(1,1)$ tensor $R(X,Y) \in \Gamma(\mathrm{End}(TM)) = \Gamma(TM \otimes T^*M)$ by, for all $Z \in \Gamma(TM)$,

$$R(X,Y)Z = \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X,Y]Z}$$

Exercise
For all $f \in \mathcal{C}^\infty(M)$, we have $R(X,Y)(fZ) = R(fX,Y)Z = R(X,fY)Z = f(R(X,Y)Z)$, so $R \in \Gamma(TM \otimes T^*M \otimes T^*M \otimes T^*M)$ is a $(1,3)$ tensor, called the Riemann curvature tensor of g. Moreover, one can show $R(X,Y) = -R(Y,X)$, and so thinking of $\Lambda^2(T^*M)$ as a subbundle of $T^*M \otimes T^*M$, we can think of R as an $\mathrm{End}(TM)$-valued 2-form.
In local coordinates, we thus write

$$R = R^\ell_{ijk} \frac{\partial}{\partial x^\ell} \otimes \mathrm{d}x^k \otimes \mathrm{d}x^i \otimes \mathrm{d}x^j$$

where $R^\ell_{ijk} = R(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j})\frac{\partial}{\partial x^k}$, satisfying $R^\ell_{ijk} = -R^\ell_{jik}$ (so i,j are the 2-form indices, and k, ℓ are the endomorphism indices).

Exercise
From $\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \Gamma^k_{ij} \frac{\partial}{\partial x^k}$, one can calculate

$$R^\ell_{ijk} = \frac{\partial \Gamma^\ell_{jk}}{\partial x^i} - \frac{\partial \Gamma^\ell_{ik}}{\partial x^j} + \Gamma^\ell_{im}\Gamma^m_{jk} - \Gamma^\ell_{jm}\Gamma^m_{ik}$$

which we see is 2nd-order and very nonlinear in g_{ij}.

Remark
If $n = 1$, then $R = 0$ since there are no 2-forms on a 1-manifold.

Example
In $(\mathbb{R}^n, g_{Euc})$ with standard coordinates, $\Gamma^k_{ij} = 0$ implies $R = 0$. In other coordinates, we may have $g_{ij} \neq \delta_{ij}$, $\Gamma^k_{ij} \neq 0$ but $R = 0$ always for this metric (since it's a tensor).

Definition
We say (M, g) is flat if $R = 0$.

Fact
(M, g) is flat if and only if it is locally isometric to $(\mathbb{R}^m, g_{Euc})$.

Definition
We define a $(0,4)$-tensor, also denoted by R, by

$$R(X,Y,Z,W) = g(R(X,Y)Z, W)$$

for all $X, Y, Z, W \in \Gamma(TM)$, where it's easy to check its components satisfy $R_{ijk\ell} = R^m_{ijk} g_{m\ell}$.

Properties of the Riemann curvature tensor
(1) $R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0$ for all $X, Y, Z \in \Gamma(TM)$, called the 1st Bianchi identity, which is true for any torsion-free connection.
(2) $R(X,Y,Z,W) = -R(X,Y,W,Z)$, and so $R_{ijk\ell} = -R_{ij\ell k}$ (true for any g-compatible connection).
(3) $R(X,Y,Z,W) = R(Z,W,X,Y)$, so $R_{ijk\ell} = R_{k\ell ij}$. (This uses (1) and (2), so is true only for the Levi-Civita connection).

Sectional Curvature

Let (M^m, g) be a Riemannian manifold with $m \geq 2$.

Definition
Given $p \in M$ and $L_p \subseteq T_pM$ a 2-dimensional subspace, we define the sectional curvature $K(L_p) \in \mathbb{R}$ to be
$$K(L_p) = \frac{R(X_p, Y_p, Y_p, X_p)}{|X_p|^2|Y_p|^2 - \langle X_p, Y_p\rangle^2}$$
for any basis X_p, Y_p of L_p. The denominator is positive by Cauchy-Schwarz, and this is well-defined, independent of the choice of basis (exercise, basically the numerator is only a function of $X_p \wedge Y_p$ by skew-symmetry of R, and the denominator is $|X_p \wedge Y_p|^2$). In particular, the denominator is 1 if we picked an orthonormal basis.

Soon, we'll see a geometric meaning for sectional curvature using the Gauss-Codazzi equations and the exponential map.

Definition
Let $k \in \mathbb{R}$. We say (M, g) has constant sectional curvature (commonly abbreviated csc) equal to k if $K(L_p) = k$ for all $p \in M$ and all 2-dimensional subspaces L_p of T_pM. One can show this is equivalent to
$$R(X, Y, Z, W) = k(\langle X, W\rangle\langle Y, Z\rangle - \langle X, Z\rangle\langle Y, W\rangle)$$
which in local coordinates is equivalent to
$$R_{ijk\ell} = k(g_{i\ell}g_{jk} - g_{ik}g_{j\ell})$$

Examples
$(\mathbb{R}^n, g_{Euc})$ has constant sectional curvature $k = 0$, (S^n, g_{round}) has constant sectional curvature $k = +1$, and hyperbolic space $(\mathbb{H}^n, g_{hyp})$ has constant sectional curvature $k = -1$.
A deep theorem of Riemannian geometry says that any space of constant sectional curvature is isometric to one of these spaces quotiented out by a discrete group of isometries of the metric.

Ricci and Scalar Curvature

Let (M^n, g) be Riemannian.

Definition
The Ricci curvature $\mathrm{Ric} \in \Gamma(T^*M \otimes T^*M)$ is a $(0, 2)$-tensor defined by
$$\mathrm{Ric}(X, Y) = g^{i\ell}R(e_i, X, Y, e_\ell) = X^jY^kR_{ijk\ell}g^{i\ell} = X^jY^k\underbrace{R_{jk}}_{=\mathrm{Ric}(e_j,e_k)=R_{ijk\ell}g^{i\ell}}$$

for any local frame $(e_1, \ldots, e_m)$ on M. More invariantly,
$$\mathrm{Ric}(X, Y) = \mathrm{tr}(U \mapsto R(U, X)Y)$$

Lemma
$\mathrm{Ric}(X, Y) = \mathrm{Ric}(Y, X)$.
Proof:
By the first Bianchi identity,
$$R_{jk} = g^{i\ell}R_{ijk\ell}(\cancelto{0}{-R_{jki\ell}} - R_{kij\ell}) = g^{i\ell}(R_{ikj\ell}) = R_{kj}$$

where the cancellation is because $g^{i\ell}$ is symmetric in $i\ell$ but $R_{jki\ell}$ is skew in $i\ell$.

Definition
We say g is Einstein if $\mathrm{Ric} = \lambda g$ for some $\lambda \in \mathbb{R}$.

Example
We'll see that $\mathbb{CP}^n$ with the Fubini-Study metric is Einstein.

Example
If g has constant sectional curvature k, then g is Einstein with $R_{jk} = k(n-1)g$.

Remark
Geometrically, one can interpret Ric is the average of sectional curvatures, which uses that Ric is a bilinear form on each tangent space, and so is determined by its associated quadratic forms. The scalar curvature, introduced below, is the average of Ricci curvatures.

Definition
The scalar curvature of (M, g) is $s \in \mathcal{C}^\infty(M)$ given by

$$s = g^{jk} R_{jk} = g^{jk} \mathrm{Ric}(e_j, e_k) = \sum_{k=1}^{n} \mathrm{Ric}(e_k, e_k)$$

in an orthonormal frame.

Proposition (Riemannian 2nd Bianchi Identity)
For any g,

$$(\nabla_X R)(Y, Z) + (\nabla_Y R)(Z, X) + (\nabla_Z R)(X, Y) = 0$$

In a local frame, we write

$$\nabla_i R_{jk\ell m} + \nabla_j R_{ki\ell m} + \nabla_k R_{ik\ell m} = 0$$

where $\nabla_i R_{jk\ell m} = (\nabla_{e_i} R)(e_j, e_k, e_\ell, e_m)$.
Proof:
The statement is tensorial, so it's enough to prove the left hand side is 0 at any point $p \in M$. Choose geodesic normal coordinates centered at p. By tensoriality, it suffices to show the statement holds for $X = \partial_i, Y = \partial_j, Z = \partial_k, V = \partial_\ell, W = \partial_m$, where then we know $\Gamma_{ij}^k(p) = 0$, and by definition $\nabla_i(\partial_j) = \Gamma_{ij}^k \partial_k$, where $\nabla_i = \nabla_{\partial_i}$. Then, at p,

$$\partial_i(R_{jk\ell m}) \overbrace{=}^{\text{at p}} X(R(Y, Z, V, W)) = (\nabla_X R)(Y, Z, V, W) + R(\nabla_X Y, Z, V, W) + 3 \text{ more terms} \overbrace{=}^{\text{at p}} 0$$

where everything vanishes at p because the Christoffel symbols vanish at p. On the other hand,

$$\begin{aligned} X(R(Y, Z, V, W)) &= X(g(R(Y, Z)V, W)) \\ &= X(g(\nabla_Y(\nabla_Z V) - \nabla_Z(\nabla_Y V) - \nabla_{[Y,Z]} V, W)) \\ &\overbrace{=}^{g\text{-compatible}} g(\nabla_X(\nabla_Y(\nabla_Z V)), W) - g(\nabla_X(\nabla_Z(\nabla_Y V)), W) + g(\ldots, \underbrace{\nabla_X W}_{=0 \text{ at } p}) \end{aligned}$$

Then, if we cyclically permute X, Y, Z and add, we get zero. ■

Musical Isomorphisms

Using the metric g, we get canonical isomorphisms $\Gamma(TM) \cong \Gamma(T^*M), TM \cong T^*M, T_pM \cong T_p^*M$.

Definition
Let $X \in \Gamma(TM)$. Then $X^\flat = g(X, \cdot) \in \Gamma(T^*M)$. The map $\flat$ is injective: if $g(X, \cdot) = 0$, then $g(X, Y) = 0$ for all $Y \in \Gamma(TM)$ if and only if $X = 0$ by positive-definiteness, so $X \mapsto X^\flat$ is an isomorphism because we have an injective linear map pointwise on each tangent space. We write the inverse as $\sharp$, so for $\alpha \in \Gamma(T^*M)$, then $\alpha^\sharp \in \Gamma(TM)$.
In a local frame $(e_1, \dots, e_n)$ for TM with dual coframe $(e^1, \dots, e^n)$ for T^*M, then if we write $e_k^\flat = A_{kl}e^\ell$, then we have $g_{kj} = g(e_k, e_j) = e_k^\flat(e_j) = A_{k\ell}e^\ell(e_j) = A_{kj}$, so $e_k^\flat = g_{k\ell}e^\ell$, and similarly $(e^k)^\sharp = g^{k\ell}e_\ell$. Notice that if $e_1, \dots, e_n$ are orthonormal, this shows $(e_k)^\flat = e^k$ and $(e^k)^\sharp = e_k$.
Given $X = X^k e_k$ and $X^\flat = X_\ell e_\ell$, we have $X^\flat = X^k e_k^\flat = X^k g_{k\ell}e^\ell$, so $X_\ell = X^k g_{k\ell}$, hence why the musical isomorphisms are called lowering an index, and if $\alpha = \alpha_k e^k$, then $\alpha^\sharp = \alpha^\ell e_\ell$ where $\alpha^\ell = g^{\ell k}\alpha_k$.
This musical isomorphism allows us to define a smoothly varying family of positive definite inner products on the fibres of T^*M as follows: $g(X^\flat, Y^\flat) = g(X, Y)$, i.e. we define g_p on T_p^*M by requiring that $\flat : T_pM \to T_p^*M$ is an isometry. In a local frame,

$$g_{ij} = g(e_i, e_j) = g(e_i^\flat, e_j^\flat) = g(g_{i\ell}e^\ell, g_{jk}e^k)$$

and so $g_{ij} = g_{i\ell}g_{jk}g(e^\ell, e^k)$, and so $g(e^\ell, e^k) = g^{\ell k}$.
Using the musical isomorphisms, we can identify $(1,1)$ tensors with $(0,2)$ tensors (i.e. identify endomorphisms with bilinear forms) via

$$\mathrm{End}(V) = V \otimes V^* \overbrace{\cong}^{\text{musical}} V^* \otimes V^* = \mathrm{Bil}(V)$$

Explicitly, $P \in \Gamma(TM \otimes T^*M)$ given by $P = P_j^k e_k \otimes e^j$ is sent to $P = P_{ij}e^i \otimes e^j$, where $P_{ij} = g_{ik}P_j^k$. If we write $Y = Y^a e_a$, then $P(Y) = P_j^k Y^j e_k$, so $P(X, Y) = P_{ij}X^iY^j$, where then

$$\langle X, PY \rangle = \langle X^a e_a, P_j^k Y^j e_k \rangle = X^a Y^j \underbrace{P_j^k g_{ak}}_{P_{aj}} = P(X, Y)$$

and so

$$\langle X, PY \rangle = P(X, Y)$$

under this identification. Under this identification, self-adjoint endomorphisms with respect to g are symmetric, and skew-adjoint endomorphisms with respect to g are skew-symmetric. This allows us to take the trace (with respect to g) of a $(0,2)$-tensor:

$$\mathrm{tr}(P) = P_i^i = g^{ij}P_{ij} = \mathrm{tr}_g(P)$$

Example
The scalar curvature s satisfies $s = g^{ij}R_{ij} = \mathrm{tr}_g(\mathrm{Ric})$, the trace of the Ricci-endomorphism, which satisfies $g(X, \mathrm{Ric}(Y)) = \mathrm{Ric}(X, Y)$ (self-adjoint with respect to g), and so there exists an orthonormal basis $e_1, \dots, e_n$ such that $\mathrm{Ric}(e_k, e_\ell) = \lambda_k \delta_{k\ell}$ where the λ_k's are real by the spectral theorem.

Definition
Let $f \in \mathcal{C}^\infty(M)$. Then $\mathrm{grad}\, f$, called the gradient of f, is a smooth vector field on M, defined by $\mathrm{grad}\, f = (df)^\sharp$, so

$$\langle \mathrm{grad}\, f, Y \rangle = (df)(Y) = Yf$$

which is the directional derivative of f in the Y direction, which agrees with a formula from multivariable calculus like $D_v f = \mathrm{grad}\, f \cdot v$.
In local coordinates, write $df = \frac{\partial f}{\partial x^i}\mathrm{d}x^i$, and so

$$\mathrm{grad}\, f = (df)^\sharp = \frac{\partial f}{\partial x^i}(\mathrm{d}x^i)^\sharp = \left(\frac{\partial f}{\partial x^i}g^{ik}\right)\frac{\partial}{\partial x^k}$$

Definition
More generally, by repeatedly lowering indices, the musical isomorphisms allow us to identify (k,ℓ) tensors with $(0,k+\ell)$ tensors:

$$S^{i_1\dots i_k}_{j_1\dots j_\ell} \leftrightarrow S_{p_1\dots p_k j_1\dots j_\ell} = g_{i_1p_1}\dots g_{i_kp_k}S^{i_1\dots i_k}_{j_1\dots j_\ell}$$

Example
Previously, we saw $R_{ijk\ell} = R^m_{ijk}g_{m\ell}$ where we lowered an index of the Riemann curvature tensor.

Definition
Let S be a $(0,k)$ tensor. Then ∇S is a $(0,k+1)$ tensor, where

$$(\nabla S)(X, Y_1,\dots,Y_k) = (\nabla_X S)(Y_1,\dots,Y_k)$$

where we write

$$(\nabla_S)_{pi_1\dots i_k} = (\nabla_{e_p}S)(e_{i_1},\dots,e_{i_k}) =: \nabla_p S_{i_1\dots i_k}$$

Definition
The divergence of S, denoted $\operatorname{div} S$, is the $(0,k-1)$ tensor given by

$$(\operatorname{div} S)(Y_1,\dots,Y_{k-1}) := g^{ij}(\nabla_{e_i}S)(e_j, Y_1,\dots,Y_{k-1})$$

where in components, we see

$$(\operatorname{div} S)_{i_1\dots i_{k-1}} = g^{ab}\nabla_a S_{bi_1\dots i_{k-1}}$$

In an orthonormal frame,

$$(\operatorname{div} S)(Y_1,\dots,Y_{k-1}) = \sum_{p=1}^n (\nabla_{e_p}S)(e_p, Y_1,\dots,Y_{k-1})$$

Definition
Let (M,g) be a Riemannian manifold and $f \in \mathcal{C}^\infty(M)$, so $(df) \in \Gamma(T^*M)$ is $(0,1)$-tensor. Then $\operatorname{div}(df) \in \mathcal{C}^\infty(M)$ is a smooth function called the Laplacian of f, denoted Δf. In a local frame,

$$\begin{aligned}\Delta f = g^{pq}(\nabla_{e_p}df)(e_q) &= g^{pq}\Big(\nabla_{e_p}((df)(e_q)) - (df)(\nabla_{e_p}e_q)\Big)\\ &= g^{pq}\underbrace{\Big(\nabla_{e_p}(\nabla_{e_q}f) - \nabla_{\nabla_{e_p}e_q}f\Big)}_{:=\nabla_p\nabla_q f}\end{aligned}$$

where in general, if S is $(0,k)$ tensor, ∇S is a $(0,k+1)$-tensor, and $\nabla\nabla S$ is a $(0,k+2)$ tensor, where by definition,

$$\begin{aligned}(\nabla_a\nabla_b S)(\dots) &:= (\nabla(\nabla S))(e_a, e_b, \dots) = (\nabla_{e_a}(\nabla S))(e_b,\dots)\\ &= e_a((\nabla S)(e_b,\dots)) - (\nabla S)(\nabla_{e_a}e_b,\dots) - (\nabla S)(e_b, \nabla_{e_a}(\dots))\\ &= e_a((\nabla_{e_b}S)(\dots)) - (\nabla_{\nabla_{e_a}e_b}S)(\dots) - (\nabla_{e_b}S)(\nabla_{e_a}\dots)\\ &= (\nabla_{e_a}(\nabla_{e_b}S))(\dots) - (\nabla_{\nabla_{e_a}e_b}S)(\dots)\end{aligned}$$

so $\nabla_a\nabla_b S_{i_1\dots i_k}$ means

$$(\nabla\nabla S)(e_a, e_b, e_{i_1},\dots,e_{i_k)} = (\nabla_{e_a}(\nabla_{e_b}S))(e_{i_1},\dots,e_{i_k}) - (\nabla_{\nabla_{e_a}e_b}S)(e_{i_1},\dots,e_{i_k})$$

and

$$\nabla_a\nabla_b S = \nabla_a(\nabla_b S) - (\nabla_{\nabla_{e_a}e_b}S)$$

On assignment 1, we show

$$\Delta f = g^{ij}\left(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma^k_{ij}\frac{\partial f}{\partial x^k}\right)$$

which is linear in derivatives of f.
Finally, note $\Delta : \mathcal{C}^\infty(M) \to \mathcal{C}^\infty(M)$ is linear, and div : $\Gamma(\bigotimes^k(T^*M)) \to \Gamma(\bigotimes^{k-1}(T^*M))$ is also linear.

Definition
$f \in \mathcal{C}^\infty(M)$ is called harmonic if $\Delta f = 0$.

Remark
We will see that map $f : (M, g) \to (\mathbb{R}, g_{Euc})$ is harmonic map between Riemannian manifolds if and only if it is a harmonic function. Similarly, $\gamma : ((a, b), g_{Euc}) \to (M, g)$ is a harmonic map between Riemannian manifolds if and only if it is a geodesic.

Proposition
Covariant differentiation commutes with contraction using the metric g. That is, if S is a $(0, k)$-tensor for $k \geq 2$, then if U is the $(0, k-2)$-tensor given in some local frame by

$$U_{i_1 \dots i_{k-2}} = g^{ab} S_{ab i_1 \dots i_{k-2}}$$

Then $g^{ab} \nabla_m S_{ab i_1 \dots i_{k-2}} = \nabla_m U_{i_1 \dots i_{k-2}}$.
Proof:
Both sides are tensorial, so it suffices to show they are equal pointwise. Let $p \in M$, and choose geodesic normal coordinates centered at p. We compute at p:

$$g^{ab}(\nabla_{e_m} S)(e_a, e_b, \dots) = \sum_{\ell=1}^{n} (\nabla_{e_m} S)(e_\ell, e_\ell, \dots)$$

$$\overbrace{=}^{\text{normal coords at } p} \quad e_m(U(\dots)) \quad \overbrace{=}^{\text{normal coords at } p} \quad (\nabla_{e_m} U)(\dots) \quad \blacksquare$$

Example
The following is called the twice contracted Riemannian second Bianchi identity. Starting with the second Bianchi identity

$$\nabla_i R_{jkab} + \nabla_j R_{kiab} + \nabla_k R_{ijab} = 0$$

we take the trace in indices k, a to obtain

$$\nabla_i R_{jb} - \nabla_j R_{ib} + g^{ka} \nabla_k R_{ijab} = 0$$

and then trace in j, b to obtain

$$\nabla_i s - \operatorname{div}(\mathrm{Ric})_i - \operatorname{div}(\mathrm{Ric})_i = 0$$

implying

$$\operatorname{div}(\mathrm{Ric})_i = \frac{1}{2}(ds)_i$$

As an exercise, this implies that if Ric $= fg$ for $n \geq 3$ and $f \in \mathcal{C}^\infty(M)$, then f is a constant.

Oriented Riemannian Manifolds

Now, assume that M is oriented, with nowhere vanishing n-form μ determining an orientation $[\mu]$. Recall a coordinate chart $(x^1, \dots, x^n)$ is called oriented if $\mu(\frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n}) > 0$.

Definition
Because $(M, g, [\mu])$ has a metric, there exists a preferred representative (volume form) μ_g of $[\mu]$ defined by $\mu_g(e_1, \dots, e_n) = 1$ whenever $e_1, \dots, e_n$ is an oriented orthonormal frame. In oriented local coordinates $(x^1, \dots, x^n)$, $\mu_g = \sqrt{\det g_{ij}} \mathrm{d}x^1 \wedge \dots \wedge \mathrm{d}x^n$. We call μ_g the Riemannian volume form, and henceforth will denote it by μ.

Lemma
We have $\nabla\mu = 0$.
Proof:
Let $\{e_1, \dots, e_n\}$ be any oriented orthonormal frame

$$0 = e_m(\underbrace{g(e_i, e_j)}_{=\delta_{ij}}) \overbrace{=}^{\nabla g = 0} g(\nabla_{e_m} e_i, e_j) + g(e_i, \nabla_{e_m} e_j) = g(\Gamma^k_{mi} e_k, e_j) + g(e_i, \Gamma^k_{mj} e_k)$$

implying $0 = \Gamma^j_{mi} + \Gamma^i_{mj}$. Thus, by the Leibniz rule,

$$(\nabla_{e_m}\mu)(e_1, \dots, e_n) = \overbrace{e_m(\underbrace{\mu(e_1, \dots, e_n)}_{=1})}^{=0} - \mu(\nabla_{e_m} e_1, e_2, \dots, e_n) - \dots - \mu(e_1, \dots, e_{n-1}, \nabla_{e_m} e_n)$$

$$= -\Gamma^1_{m1} \overbrace{\mu(e_1, \dots, e_n)}^{=1} - \dots - \Gamma^n_{mn} \underbrace{\mu(e_1, \dots, e_n)}_{=1} = -\sum_{\ell=1}^{n} \Gamma^\ell_{m\ell} = 0$$

where the last equality is identity derived above. ■

Lemma
Let S be a $(0,k)$-tensor on M. Then

$$(\mathcal{L}_X S)(Y_1, \dots, Y_k) = (\nabla_X S)(Y_1, \dots, Y_k) + \sum_{j=1}^{k} S(Y_1, \dots, \nabla_{Y_j} X, \dots, Y_k)$$

Proof:
Let's calculate:

$$(\mathcal{L}_X S)(Y_1, \dots, Y_k) = X(S(Y_1, \dots, Y_k)) - \sum_{j=1}^{k} S(Y_1, \dots, \underbrace{\mathcal{L}_X Y_j}_{=[X, Y_j]}, \dots, Y_k)$$

$$\overbrace{=}^{\text{torsion-free \& Leibniz rule}} (\nabla_X S)(Y_1, \dots, Y_k) + \sum_{j=1}^{k} S(Y-1, \dots, \nabla_X Y_j, \dots, Y_k) - \sum_{j=1}^{k} S(Y_1, \dots, \cancel{\nabla_X Y_j} - \nabla_Y X_j, \dots, Y_k) \quad ■$$

Corollary
Let (M^n, g, μ) be an oriented Riemannian manifold. Then $\mathcal{L}_X\mu = (\operatorname{div} X)\mu$ for all $X \in \Gamma(TM)$.
Proof:
Take $S = \mu$ in the lemma and use $\nabla\mu = 0$, so that when we write $\mathcal{L}_X\mu = f\mu$ for some $f \in \mathcal{C}^\infty(M)$, we have

$$f = (\mathcal{L}_X\mu)(e_1, \dots, e_n) = \sum_{j=1}^{k} \mu(e_1, \dots, \overbrace{\nabla_{e_j} X}^{=(\nabla_{e_j} X)^\ell e_\ell = (\nabla_j X^\ell) e_\ell}, \dots, e_n)$$

$$= (\nabla_j X^j)\mu(e_1, \dots, e_n) = \operatorname{div} X \quad ■$$

Definition
Let (M, g, μ) be oriented Riemannian. Let $f \in \mathcal{C}^\infty(M)$ be compactly supported. We can define the integral of f over M by $\int_M f := \int_M f\mu$.

Theorem (Divergence Theorem)
Let (M, g, μ) be oriented Riemannian. Let $X \in \Gamma(TM)$ be compactly supported. Then $\int_M \operatorname{div} X = 0$.
Proof:
We calculate

$$\int_M (\operatorname{div} X) = \int_M (\operatorname{div} X)\mu = \int_M \mathcal{L}_X\mu \overbrace{=}^{\text{Cartan}} \int_M (X \lrcorner \overset{0}{\cancel{d\mu}}) + d(X \lrcorner \mu) = \int_M d(X \lrcorner \mu) \overbrace{=}^{\text{Stokes}} 0 \quad ■$$

Formal Adjoints

Let (M, g, μ) be oriented Riemannian, and E, F be tensor bundles over M. Let $P : \Gamma(E) \to \Gamma(F)$ be $\mathbb{R}$-linear. For example, we could take $P = d$ to be the exterior derivative, or $P = (X \mapsto \mathcal{L}_X g)$ for $E = TM, F = T^*M \otimes T^*M$.

Definition
The formal adjoint $P^* : \Gamma(F) \to \Gamma(E)$ is an $\mathbb{R}$-linear map that satisfies

$$\int_M \langle P\sigma, \eta \rangle = \int_M \langle \sigma, P^*\eta \rangle$$

for all $\sigma \in \Gamma(E), \eta \in \Gamma(F)$ with at least one compactly supported. We can think of these integrals as L^2 inner products.

Remark
P^* is unique if it exists. Indeed, we can see it's unique on compactly supported sections by observing that if Q is another such operator,

$$\int \langle \sigma, P^*\eta \rangle = \int_M \langle \sigma, Q\eta \rangle$$

implying

$$\int_M \langle \sigma, (P^* - Q)\eta$$

so taking $\sigma = (P^* - Q)\eta$, we obtain $(P^* - Q)\eta = 0$ for all η compactly supported, so $P^* = Q$ on compactly supported sections. Then by doing some analysis (some compactly supported things are dense in L^2 argument), we get uniqueness everywhere.

Lemma
$\operatorname{div}(fX) = \langle \operatorname{grad} f, X \rangle + f \operatorname{div} X$.
Proof:

$$\operatorname{div}(fX) = \nabla_i((fX)^i) = \nabla_i(fX^i) = (\nabla_i f)X^i + f\nabla_i X^i = \underbrace{(df)(X)}_{=\langle \operatorname{grad} f, X\rangle} + f \operatorname{div} X \quad \blacksquare$$

Example
Let's compute the formal adjoint of $\operatorname{grad} : \Omega^0 \to \Gamma(TM)$ that sends f to $(df)^\sharp$. By the lemma,

$$\int_M \langle \operatorname{grad} f, X \rangle = \int_M (\operatorname{div}(fX) - f \operatorname{div} X) = -\int_M \langle f, \operatorname{div} X \rangle$$

implying $(\operatorname{grad})^*(X) = -\operatorname{div} X$.

Example
Let $P : \Gamma(TM) \to \Gamma(\mathcal{S}^2(T^*M))$ be the map that sends $X \in \Gamma(TM)$ to $\mathcal{L}_X g$ with components $(\mathcal{L}_X g)_{ij} = \nabla_i X_j + \nabla_j X_i$ (using g-compatibility of ∇). Then

$$\int_M \langle PX, h \rangle = \int_M \langle \mathcal{L}_X g, h \rangle \overbrace{=}^{\text{in an o.n. frame}} \int_M (\nabla_i X_j + \nabla_j X_i) h^{ij}$$

$$= 2\int_M \nabla_i X_j h^{ij} = 2\int_M (\underbrace{\nabla_i (X_j h^{ij})}_{=\text{div of something}} - X_j \nabla_i h^{ij}) \overbrace{=}^{\text{divergence thm}} -2\int_M \langle X, (\operatorname{div} h)^\sharp \rangle$$

and so $P^* = -2(\operatorname{div}(-))^\sharp$.

Example

On Assignment 1, we show $d^*\alpha = -g^{ab}e_a \lrcorner (\nabla_{e_b}\alpha)$.

Definition

Let (M^n, g, μ) be oriented Riemannian. We can define the Hodge star operator $\star : \Omega^k \to \Omega^{n-k}$ determined uniquely by, for all $\alpha, \beta \in \Omega^k$,

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \mu$$

where this inner product on k-forms is the one such that if $e_1, \ldots, e_n$ is an orthonormal frame of TM, then $e^{i_1} \wedge \ldots \wedge e^{i_k}$ for $i_1 < \cdots < i_k$ is an orthonormal frame of $\Lambda^k(T^*M)$.

Facts

(1) We have $\star^2 = (-1)^{k(n-k)}$ on Ω^k, and thus squares to the identity for all k when n is odd, and squares to $(-1)^k$ if n is even.
(2) $\langle \star\alpha, \star\beta \rangle = \langle \alpha, \beta \rangle$ for all $\alpha, \beta \in \Omega^k$, so $\star$ is an isometry.
(3) We show on Assignment 1, that on Ω^k, $d^* = (-1)^{nk+n+1} \star d \star$.

Riemannian Immersions and the Volume Functional

Definition

Let $u : (M^m, g) \to (N^n, h)$ be smooth. We say u is a Riemannian immersion (also called an isometric immersion) if and only if $u^*h = g$ if and only if for all $p \in M$, $X_p, Y_p \in T_pM$, $g_p(X_p, Y_p) = (u^*h)_p(X_p, Y_p) = h_{u(p)}((u_*)_p X_p, (u_*)_p Y_p)$. (this forces $m \leq n$ and u is an immersion.)

Remark

Any immersion is locally an embedding. So when we consider local properties of immersion, we can assume u is an embedding and identify M with its image $u(M)$, i.e. we can identify $T_pM \cong (u_*)T_pM \subseteq T_{u(p)}N$.

Since we think of u as inclusion, we think of T_pM as a linear subspace of T_pN, where we then have an orthogonal decomposition $T_pN = T_pM \oplus \nu_pM$, where ν_pM is the normal space to M at p with respect to h_p, and we think of $g_p = h_p|_{T_pM}$. In this case, we say M is a (locally) isometrically embedded submanifold of codimension $n - m$.
Given this decomposition, given $V \in \Gamma(u^*TN)$, we have $V_p \in T_{u(p)}N = T_pM \oplus \nu_pM$, and thus can decompose V into components V^T (tangential) and $V^\perp$ (normal) by orthogonally projecting onto the two factors. These give smooth sections $V^T \in \Gamma(TM)$ and $V^\perp \in \Gamma(\nu M)$.

Definition

A smooth section $\eta \in \Gamma(\nu M)$ of νM is called a normal vector field on M.

Let ∇^N be the Levi-Civita connection on (N, h).

Lemma

For $X \in \Gamma(TM)$, let $D_X : \Gamma(TM) \to \Gamma(TM)$ be the map $D_X Y = (\nabla^N_X Y)^T$. Then D is the Levi-Civita connection on M with respect to $g = u^*h$.
Proof is on the homework.

By the lemma, we have $\nabla^N_X Y = \nabla^M_X Y + (\nabla^N_X Y)^\perp$.

Definition

The second fundamental form A of the Riemannian immersion $u : M \to N$ is a section $A \in \Gamma(T^*M \otimes$

$T^*M \otimes \nu M)$ defined by $A(X,Y) = (\nabla^N_X Y)^\perp$.

Remark
To verify that A lies in $\Gamma(\nu M)$, we need to show $A(fX,Y) = fA(X,Y) = A(X,fY)$ for all $f \in \mathcal{C}^\infty(M)$. To see this,

$$A(fX,Y) = (\nabla^N_{fX} Y)^\perp = (f\nabla^N_X Y)^\perp = f(\nabla^N_X Y)^\perp = fA(X,Y)$$

and

$$A(X,fY) = (\nabla^N_X (fY))^\perp = ((Xf)Y + f\nabla^N_X Y)^\perp = fA(X,Y)$$

Lemma
A is symmetric, i.e. $A(X,Y) = A(Y,X)$ for all $X,Y \in \Gamma(TM)$.
Proof:

$$A(X,Y) - A(Y,X) = \nabla^N_X Y - \nabla^N_Y X)^\perp = ([X,Y])^\perp = 0$$

because if X,Y are tangent to M, so too is $[X,Y]$. ■

Definition
Let $u : (M^m, g) \to (N^n, h)$ be a Riemannian immersion. We say u is totally geodesic if $A = 0$.

Remark
Suppose $\gamma : (-\varepsilon, \varepsilon) \to M$ is smooth on M, and let $\gamma' = u \circ \gamma : (-\varepsilon, \varepsilon) \to N$ on N, then:

$$\nabla^N_{\gamma'} \gamma' = \nabla^M_{\gamma'} \gamma' + A(\gamma', \gamma')$$

Proposition
The following are equivalent for a Riemannian immersion $u : (M,g) \to (N,h)$:
(1) u is totally geodesic ($A = 0$).
(2) Every geodesic in (M,g) is a geodesic in (N,h)
(3) For all $p \in M$, if $X_p \in T_pM$, then the geodesic γ in (N,h) with initial data $\gamma(0) = p, \gamma'(0) = X_p$ stays in M and is a geodesic in M.
Proof:
(1) implies (2) by the remark above, and (2) implies (1) since any $X_p \in T_pM$ is $\gamma'(0)$ for some geodesic γ in M (and thus N) passing through p.
(2) if and only if (3) is just an application of the existence and uniqueness theorem for geodesics. ■

Example
$\mathbb{R}^m \to \mathbb{R}^n$ by $x \mapsto (x,0)$ is a Riemannian immersion, and $g|_{Euc\ in\ \mathbb{R}^n}|_{\mathbb{R}^m} = g|_{Euc\ in\ \mathbb{R}^m}$.
One can check that $\nabla^{\mathbb{R}^n}_X Y = \nabla^{\mathbb{R}^m}_X Y$ if $X,Y \in \Gamma(T\mathbb{R}^m)$, as for the standard global frame $e_1, \ldots, e_n$ with $Y = \sum_{i=1}^m Y^i e_i$, then

$$\nabla^{\mathbb{R}^m}_X Y = \sum_{i=1}^m X(Y^i)e_i = \sum_{i=1}^n X(Y^i)e_i = \nabla^{\mathbb{R}^n}_X Y$$

Notice that if γ is a geodesic in $\mathbb{R}^n$, then it is a straight line. If γ is initially tangent to $\mathbb{R}^m$, it stays in $\mathbb{R}^m$, and is a geodesic in $\mathbb{R}^m$, so that $\mathbb{R}^m \to \mathbb{R}^n$ is totally geodesic.

Example
Consider $S^n \to (\mathbb{R}^{n+1}, g_{Euc})$ by $\iota^* g_{Euc} = g_{round}$. Let's compute the second fundamental form. Let $X,Y \in \Gamma(TS^n)$, we have $\nabla^{\mathbb{R}^{n+1}}_X Y = \nabla^{S^n}_X Y + A(X,Y)$.
Since $|p| = 1, T_p(S^n) = \{v \in \mathbb{R}^{n+1} : \langle v, p\rangle = 0\}$, then:

$$A_p(X_p, Y_p) = \langle \nabla^{\mathbb{R}^{n+1}}_{X_p} Y, p\rangle p$$

Let's define $R \in \Gamma(T\mathbb{R}^{n+1})$ by $R_p = p$ which is the position vector field and $R = x^i \frac{\partial}{\partial x^i}$.
estricted to S^n, R is the outward unit normal vector field.

$$\langle Y, R\rangle = 0$$

since $Y \in \Gamma(TS^n)$ and

$$0 = X\langle Y, R\rangle = \langle \nabla_X Y, R\rangle + \langle Y, \nabla_X R\rangle$$

So at p,

$$\langle \nabla_{X_p} Y, R\rangle = -\langle Y_p, \nabla_{X_p} R\rangle = -\langle Y_p, X_p\rangle$$

Let $X = a^i \frac{\partial}{\partial x^i}$, then,

$$\nabla_X R = X(x^i \frac{\partial}{\partial x^i}) = a^k (\frac{\partial x^i}{\partial x^k}\frac{\partial}{\partial x^i}) = a^i \frac{\partial}{\partial x^i} = X$$

Hence,

$$A_p(X_p, X_p) = -\langle X_p, Y_p\rangle p \neq 0$$

so ι is not totally geodesic, so a a geodesic in S^n is not a geodesic in $\mathbb{R}^{n+1}$.

Theorem (The Gauss Equation)
Let $(M, g) \to (N, h)$ be a Riemannian immersion. Let $X, Y, Z, W \in \Gamma(TM)$, then at $p \in M$,

$$R^N(X, Y, Z, W) = R^M(X, Y, Z, W) - \langle A(X, W), A(Y, Z)\rangle - \langle A(X, Z), A(Y, W)\rangle$$

Proof:
First, note that

$$\begin{aligned}\langle \nabla^N_X(A(Y, Z)), W\rangle &= X\langle A(Y, Z), W\rangle - \langle A(X, Z), \nabla^N_W W\rangle \\ &= -\langle A(Y, Z), \nabla^N_X W\rangle \\ &= -\langle A(Y, Z), A(X, W)\rangle\end{aligned}$$

We compute

$$\begin{aligned}R^N(X, Y, Z, W) &= \langle \nabla^N_X(\nabla^N_Y Z) - \nabla^N_Y(\nabla^N_X Z) - \nabla^N_{[X,Y]} Z, W\rangle \\ &= \langle \nabla^N_X(\nabla^N_Y Z + A(Y, Z)) - \nabla^N_Y(\nabla^N_X Z + A(X, Z)) - \nabla^N_{[X,Y]} Z, W\rangle \\ &= \langle \nabla^M_X(\nabla^N_Y Z) - \nabla^M_Y(\nabla^M_X Z) - \nabla^M_{[X,Y]} Z, W\rangle - \langle A(Y, Z), A(X, W)\rangle + \langle A(X, Z), A(Y, W)\rangle\end{aligned}$$

and the result follows. ■

Remark
This allows us to get the following geometric interpretation of sectional curvature.
Let (M^n, g) be a Riemannian manifold with $n \geq 2$. Let $p \in M$, and let $L_p \subset T_pM$ be a 2-dimensional subspace.
Let U be an open neighbourhood of 0_p in T_pM on which $\exp_p : U \to \exp_p(U)$ is a diffeomorphism onto its image.
Think about $L_p \cap U$: $\exp_p|_{U \cap L_p} : U \cap L_p \to M$ is an embedding: let (Σ^2, g_p) be its image, which is a 2-dimensional manifold. By construction, geodesics on M starting at p initially tangent to Σ stay in Σ, so Σ is totally geodesic at p ($A_p = 0$). Thus, $A(\gamma', \gamma') = 0$ for all γ starting at p, so that $A(X_p, Y_p) = 0$ for all $X_p, Y_p \in L_p$. By the Gauss equation at p, if X_p, Y_p is an orthonormal basis of L_p, then

$$K^M(L_p) = R^M(X_p, Y_p, Y_p, X_p) = R^\Sigma(X_p, Y_p, Y_p, X_p)$$

is the Gaussian curvature of Σ at p.

Definition
Let (M^m, g, μ_g) be a compact oriented Riemannian manifold. Then we can define the volume of M to be $\mathrm{vol}(M, g) = \int_M \mu_g$.
Now, let $(M^m, [\mu])$ be a compact oriented manifold, and let (N^n, h) be a Riemannian manifold with $n \geq m$ (need not be orientable nor compact). Let $\mathrm{Imm}(M, N)$ be the set of all immersions from M to N. We define the volume functional $\mathcal{F}$ on $\mathrm{Imm}(M, N)$ to be the map $\mathcal{F} : \mathrm{Imm}(M, N) \to \mathbb{R}$,

$$\mathcal{F}(u) = \int_M \mu_{u^*h} = \mathrm{vol}(M, u^*h)$$

Theorem
$u_0 \in \mathrm{Imm}(M, N)$ is a critical point of $\mathcal{F}$ (i.e., $\frac{d}{dt}|_{t=0}\mathcal{F}(U_t) = 0$ for any smooth family u_t of immersions with $u_{t=0} = u_0$) if and only if $\sum_{k=1}^m A(e_k, e_k) = 0$, where A is the second fundamental form of u_0 and $e_1, \ldots, e_m$ is a local orthonormal frame for $(M, u_0^*h =: g_0)$. In an arbitrary frame, this is $(g_0)^{ab}A(e_a, e_b) = 0$.
This is the trace of the second fundamental form with respect to g_0, where we recall $A \in \Gamma(\mathcal{S}^2(T^*M) \otimes \nu M)$, so this is a normal vector field. We denote $H = (g_0)^{ab}A(e_a, e_b)$ to be the mean curvature vector field of u.
Proof:
Let u_t be a smooth 1-parameter family of immersions $M \to N$ such that $u_{t=0} = u_0$. That is, we have a smooth map $u : (-\varepsilon, \varepsilon) \times M \to N$ such that $u_t = u(t, \cdot) : M \to N$ is an immersion for all t, and $u(0, \cdot) = u_0$. We need to show that $\frac{d}{dt}|_{t=0}\mathcal{F}(u_t) = 0$ for all such u_t if and only if $H = 0$. First, writing $\mu_{u_t^*h} = f\mu_{g_0}$ for $f \in \mathcal{C}^\infty((-\varepsilon, \varepsilon) \times M)$, we note by differentiating under the integral sign that

$$\frac{d}{dt}\mathcal{F}(u_t) = \frac{d}{dt}\int_M \mu_{u_t^*h} = \int_M \frac{\partial f}{\partial t}\mu_{g_0}$$

We'll compute $\frac{\partial f}{\partial t}(0, p)$ for $p \in M$ by evaluating it in oriented geodesic normal coordinates $x^1, \ldots, x^m$ centered at p, with respect to the metric g_0. Let $\mu_t = \mu_{u_t^*h}$, where then we have in coordinates

$$\mu_t = \overbrace{\sqrt{\det(g_t)_{ij}}\mathrm{d}x^1 \wedge \ldots \wedge \mathrm{d}x^m}^{=f_t\mu_0}$$

where $g_t = u_t^*h$. But we have, using the formula for the derivative of a determinant,

$$\frac{\partial}{\partial t}\sqrt{\det g_t} = \frac{1}{2\sqrt{\det g_t}}\frac{\partial}{\partial t}(\det g_t) = \frac{1}{2\sqrt{\det g_t}}(g_t)^{ab}\frac{\partial}{\partial t}(g_t)_{ab}$$

At p at $t = 0$, we have that $(g_0)_{ij}$ is the identity matrix because we're in normal coordinates, so evaluating the above formula at at p at $t = 0$, we get

$$\frac{\partial}{\partial t}|_{(0,p)}\sqrt{\det g_t} = \frac{1}{2}\sum_{a=1}^m \frac{\partial}{\partial t}|_{(0,p)}(g_t)_{aa}$$

and thus

$$\frac{d}{dt}|_{t=0}\mathcal{F}(u_t) = \frac{1}{2}\int_M k\mu_0$$

where $k \in \mathcal{C}^\infty(M)$ is the function such that $k(p) = \sum_{a=1}^m \frac{\partial}{\partial t}|_{t=0}(g_t)_{aa}$ in geodesic normal coordinates centered at p. But now

$$\frac{\partial}{\partial t}(g_t)_{aa} = \frac{\partial}{\partial t}g_t(\frac{\partial}{\partial x^a}, \frac{\partial}{\partial x^a}) = \frac{\partial}{\partial t}h(u_*\frac{\partial}{\partial x^a}, u_*\frac{\partial}{\partial x^a}) \overbrace{=}^{\nabla^N h=0} 2h((u^*\nabla^N)_{\frac{\partial}{\partial t}}(u_*\frac{\partial}{\partial x^a}), u_*\frac{\partial}{\partial x^a})$$

where we think of $\frac{\partial}{\partial t}$ and $\frac{\partial}{\partial x^a}$ as vector fields on $(-\varepsilon, \varepsilon) \times M$. Now, using torsion-freeness of the pullback connection from A1 with $X = \frac{\partial}{\partial t}$ and $Y = \frac{\partial}{\partial x^a}$, this is equal to

$$= 2h((u^*\nabla^N)_{\frac{\partial}{\partial x^a}}(u_* \frac{\partial}{\partial t}), u_* \frac{\partial}{\partial x^a})$$

Define $V = u_* \frac{\partial}{\partial t} \in \Gamma(u^*TN)$, called the variation vector field of u_t. Locally identifying X with $u_* X$ by thinking of u as inclusion, we have shown $k(p) = \sum_{a=1}^m 2h(\nabla^N_{e_a} V, e_a)$ for any orthonormal frame $e_1, \ldots, e_m$ (since we have shown at every point with geodesic normal coordinates), so that

$$\frac{d}{dt}|_{t=0} \mathcal{F}(u_t) = \int_M \left(\sum_{a=1}^m h(\nabla^N_{e_a} V, e_a) \right) \mu_0$$

Write $V = V^T + V^\perp$, where then

$$\sum_{a=1}^m h(\nabla^N_{e_a}(V^T + V^\perp), e_a) = \sum_{a=1}^m h((\nabla^N_{e_a} V^T)^T, e_a) + \sum_{a=1}^m h((\nabla^N_{e_a} V^\perp), e_a)$$

$$\overbrace{=}^{h = g \text{ on } M} (\operatorname{div}_{(M,g)} V^T) + \sum_{a=1}^m (-h(V^\perp, \nabla^N_{e_a} e_a))$$

where in the last equality we use that e_a is tangent to M, $h(V^\perp, e_a) = 0$ so by metric compatibility, $h(\nabla_{e_a} V^\perp, e_a) = -h(V^\perp, \nabla^N_{e_a} e_a)$. But the normal part of $\nabla^N_{e_a} e_a$ is $A(e_a, e_a)$, and so we have

$$= \operatorname{div}(V^T) - \sum_{a=1}^n h(V^\perp, A(e_a, e_a)) = \operatorname{div}(V^T) - h(V^\perp, H)$$

and thus by the divergence theorem,

$$\frac{d}{dt}|_{t=0} \mathcal{F}(u_t) = -\int_M h(V^\perp, H) \mu_0$$

This equals zero for all u_t if and only if $\int_M h(V^\perp, H)\mu_0 = 0$ for all $V^\perp \in \Gamma(\nu M)$ (exercise, every normal vector field is $V^\perp$ of something) if and only if $H = 0$. ■

Remarks
(1) The equation $H = 0$ is a second order nonlinear PDE for the immersion $u_0 : M \to (N, h)$. The solutions to this PDE are called minimal immersions or minimal submanifolds. However, this is bad terminology: a critical point need not be a local minimum of the volume functional, it could be a saddle point or a local max. Better names are (volume) critical/stationary immersions.
(2) We don't really need compactness of M, then $\operatorname{vol}(M)$ could be infinite, but what makes sense is the difference $\mathcal{F}(u_t) - \mathcal{F}(u_0)$ of $u_t = u_0$ outside a compact set.
(3) You don't really need orientability either by looking at small enough open sets that are orientable.

Calibrated Geometry

Definition
Let (N^n, h) be Riemannian, and let $\alpha \in \Omega^k(N)$. We say α has comass one if

$$\alpha_p(e_1, \ldots, e_k) \leq 1$$

for all $p \in N$ and $e_1, \ldots, e_k$ are orthonormal in T_pN. This holds if and only if $-1 \leq \alpha_p(e_1, \ldots, e_k) \leq 1$.

Definition
We say $\alpha \in \Omega^k(N)$ is a calibration if it has comass one and α is closed.

Definition
Let $\alpha \in \Omega^k(N)$ be a calibration. Let L^k be an oriented k-dimensional immersed submanifold of N with $\iota : L \to N$ the immersion, $g = \iota^* h$ the induced metric, and μ_L the induced Riemannian volume form. Then $\iota^*\alpha = \alpha|_L$ is a top form on L, so we can write $\alpha|_L = \iota^* f \mu_L$ for $f \in \mathcal{C}^\infty(L)$. We claim $-1 \leq f \leq 1$ because α has comass one. Indeed, for $e_1, \ldots, e_k$ an orthonormal basis of T_pM, $\alpha_p|_L(e_1, \ldots, e_k) = f(p)\mu_L(e_1, \ldots, e_k) = \pm f(p)$, and so since the left hand side is in $[-1, 1]$, we have $-1 \leq f \leq 1$.

Definition
We say L is α-calibrated (or calibrated with respect to α) if $f = 1$ if and only if $\alpha|_L = \mu_L$, and so $\iota^*\alpha = \mu_{\iota^* h}$. This is a first order nonlinear PDE for the immersion $\iota : L^k \to (N, h)$.

Why should we care?

Theorem (Harvey-Lawson 1982, the Fundamental Theorem of Calibrated Geometry)
Let L be α-calibrated (and compact for simplicity). Then L is minimal ($H = 0$), and moreover, L is "globally volume-minimizing in its homology class". In particular, it's always a local minimum of the volume functional.
Proof:
Let $L, \widetilde{L}$ be compact oriented k-submanifolds of N. We say L and $\widetilde{L}$ are in the same homology class if and only if there exists a compact oriented submanifold Y^{k+1} with boundary such that $\partial Y = L \cup (-\widetilde{L})$. Observe that by Stokes' theorem,

$$0 = \int_Y d\alpha = \int_{\partial Y} \alpha = \int_{L \cup (-\widetilde{L})} \alpha = \int_L \alpha - \int_{\widetilde{L}} \alpha$$

since α is closed, and so $\int_L \alpha = \int_{\widetilde{L}} \alpha$. On the other hand, since α is L-calibrated, $\mu_L = \alpha|_L$, so

$$\mathrm{vol}(L) = \int_L \mu_L = \int_L \alpha = \int_{\widetilde{L}} \alpha \overbrace{\leq}^{\text{comass one}} \int_{\widetilde{L}} \mu_{\widetilde{L}} = \mathrm{vol}(\widetilde{L}) \quad \blacksquare$$

Manifolds with $U(m)$-structure

These are manifolds where we'll be able to say interesting things about harmonic maps.

Definition
Let M be a smooth manifold. An almost complex structure J on M is a smooth section of $TM \otimes T^*M \cong \mathrm{End}(TM)$ such that $J^2 = -\mathrm{Id}$. This allows us to give each T_pM the structure of complex scalar multiplication (J behaves like i), which forces $\dim M = 2m$ to be even, and then under a choice of coordinates, gives us isomorphisms $(T_pM, J_p) \cong (\mathbb{R}^{2m}, J_0) \cong (\mathbb{C}^m, i)$, where J_0 is some almost complex structure on $\mathbb{R}^{2m}$.

Remark
Such a J does not always exist, there are topological obstructions. For instance, one can show that if J exists, then M is orientable.

Definition
With such a J, we can take the complex vector bundle of rank $2m$ $TM \otimes \mathbb{C}$ and decompose it into a direct sum

$$TM \otimes \mathbb{C} = (TM)^{1,0} \oplus (TM)^{0,1}$$

where $(TM)^{1,0}$ is the $+i$ eigenbundle of J, and $(TM)^{0,1}$ is the $-i$ eigenbundle of J, where for $X_p \in T_pM \otimes \mathbb{C}$, we can write

$$X_p = \frac{1}{2}(X_p - iJ_pX_p) + \frac{1}{2}(X_p + iJ_pX_p)$$

where the former is $1,0$ and the latter is $0,1$. We call $\Gamma((TM)^{1,0})$ the space of complex-valued vector fields of type $(1,0)$ (and similarly for $(0,1)$). We can induce the dual operator $J \in \mathrm{End}(T^*M)$, $(J\alpha)(X) = \alpha(JX)$, which still squares to -1, and we can do the analogous decomposition

$$T^*M \otimes \mathbb{C} = (T^*M)^{1,0} \oplus (T^*M)^{0,1}$$

where sections of $(T^*M)^{1,0}$ annihilate $(0,1)$ vector fields and vice versa.
These allow us to form the space of forms of type (p,q), where

$$\Omega^{p,q}(M) = \Gamma(\Lambda^p((T^*M)^{1,0}) \otimes \Lambda^q((T^*M)^{0,1}))$$

for $0 \leq p, q \leq m$. Explicitly in a local frame, if $f_1, \ldots, f_m$ is a local frame for $(TM)^{1,0}$ and $\overline{f}_1, \ldots, \overline{f}_m$ is a local frame for $(TM)^{0,1}$, then $f^1, \ldots, f^m$ is a local frame for $(T^*M)^{1,0}$ and $\overline{f}^1, \ldots, \overline{f}^m$ is a local frame for $(T^*M)^{0,1}$, then we can write $\alpha \in \Omega^{p,q}(M)$ locally as

$$\alpha = \alpha_{i_1 \ldots i_p j_1 \ldots j_q} f^{i_1} \wedge \ldots \wedge f^{i_p} \wedge \overline{f}^{j_1} \wedge \ldots \wedge \overline{f}^{j_q}$$

Example
Let M^{2m} be a holomorphic manifold (a.k.a. a complex manifold), that is, there exists a cover of M by charts $\varphi_\alpha : U_\alpha \to \mathbb{C}^m$ such that the transition maps $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ are holomorphic between open subsets of $\mathbb{C}^m$. Then there exists a naturally induced almost complex structure J as follows.
For each $p \in M$, we have a real linear isomorphism $(\varphi_\alpha)_*|_p : T_pM \xrightarrow{\cong} \mathbb{C}^m$, and so given $X_p \in T_pM$, we can "multiply it by i" by

$$J_pX_p = [(\varphi_\alpha)_*]_p^{-1}(i[(\varphi_\alpha)_*]_pX_p)$$

where this is well-defined independent of the chart because $\varphi_\beta \circ \varphi_\alpha^{-1}$ is holomorphic, so its pushforward is complex linear, and thus commutes with i, which implies the above definition is well-defined.
In this case, we have local holomorphic coordinates $z^k = x^k + iy^k$, giving a coframe of 1-forms via $dz^k = dx^k + idy^k$ of type $(1,0)$ and $d\overline{z}^k = dx^k - idy^k$ of type $(0,1)$, and then $\frac{\partial}{\partial z^k} = \frac{1}{2}\left(\frac{\partial}{\partial x^k} - i\frac{\partial}{\partial y^k}\right)$ is $(1,0)$ and $\frac{\partial}{\partial \overline{z}^k} = \frac{1}{2}\left(\frac{\partial}{\partial x^k} + i\frac{\partial}{\partial y^k}\right)$.

Definition
When an almost complex structure J comes from such a holomorphic atlas, we say J is integrable.

Examples
$\mathbb{C}^m$ and $\mathbb{CP}^m$ with the standard coordinates give examples of holomorphic manifolds with their normal holomorphic atlas.

Definition
Let (M^{2m}, J) be an almost complex manifold. The Nijenhuis tensor $N_J \in \Gamma(TM \otimes \Lambda^2(T^*M))$ is defined to be

$$N_J(X,Y) = [X,Y] + J[JX,Y] + J[X,JY] - [JX,JY]$$

One can check this is indeed a smooth tensor. If J is integrable, then $N_J = 0$. N_J is measuring the obstruction to the Lie bracket of two $(1,0)$ vector fields to be again of type $(1,0)$.

Theorem (Newlander-Nirenberg)
J is integrable if and only if $N_J = 0$.

Remark
There do exist non-integrable J's.

Now let's add a metric g.

Definition
(M^{2m}, J, g) is called a manifold with $U(m)$ structure if $g(JX, JY) = g(X, Y)$ for all $X, Y \in \Gamma(TM)$. If M admits a J, then it's easy to find a compatible g: choose any metric $\widetilde{g}$ and let $g(X, Y) = \widetilde{g}(X, Y) + \widetilde{g}(JX, JY)$.

Lemma
Let (M, g, J) be a $U(m)$-structure. Define $\omega \in \Gamma(T^*M \otimes T^*M)$ by $\omega(X, Y) = g(JX, Y)$. Then $\omega \in \Omega^2(M)$.
Proof:

$$\omega(Y, X) = g(JY, X) = g(X, JY) = g(JX, J^2Y) = -g(JX, Y) = -\omega(X, Y) \quad \blacksquare$$

Remark
This is equivalent to saying that J is skew-adjoint, since ω is really using a musical isomorphism to lower an index from J.

Definition
ω is called the Kähler form of the $U(m)$-structure.

Lemma
ω is of type $(1, 1)$.
Proof:
First, observe that
$$\omega(JX, JY) = g(J^2X, JY) = g(JX, Y) = \omega(X, Y)$$
Now, suppose X, Y are both type $(1, 0)$, and so
$$\omega(X, Y) = \omega(JX, JY) = \omega(iX, iY) = -\omega(X, Y)$$
implying $\omega(X, Y) = 0$ if X, Y are both type $(1, 0)$. Similarly this holds if both are type $(0, 1)$ since $(-i)^2 = -1$. Thus, ω must be type $(1, 1)$. $\blacksquare$

Let ∇ be the Levi-Civita connection of g.

Definition
Let (M^{2m}, J, g, ω) is called Kähler if and only if $\nabla J = 0$ if and only if $\nabla\omega = 0$, where we use metric compatibility to get the equivalence: $(\nabla_Z\omega)(X, Y) = g((\nabla_Z J)X, Y)$.

Remark
Usually, the definition of Kähler is given as $N_J = 0$ and $d\omega = 0$. This is just a (rather long) computation, found in most books on Kähler geometry. If we only have $d\omega = 0$ but not necessarily $N_J = 0$, then (M, g, J, ω) is called almost Kähler.

Proposition
Let (M^{2m}, g, J, ω) be almost Kähler. Then ω is a calibration.
Proof:
The fact ω is closed is from almost Kähler. For comass one, let $e_1, e_2 \in T_pM$ be orthonormal. Then

$$\omega(e_1, e_2) = g(Je_1, e_2) \overbrace{\leq}^{CS} |Je_1||e_2| = 1$$

using that J is an isometry. ■

Corollary
If Σ is an oriented real 2-dimensional submanifold of (M, g, J, ω) such that $\omega|_\Sigma = \mu_\Sigma$, then Σ is minimal (in fact, locally volume-minimizing). Moreover, Σ is ω-calibrated if and only if Σ is a J-complex submanifold ($J_p T_p\Sigma = T_p\Sigma$ for all $p \in \Sigma$) and the orientation on Σ induced by J ($e_1 \wedge Je_1 \wedge e_2 \wedge Je_2 \wedge \ldots \wedge e_m \wedge Je_m$) agrees with the given orientation on Σ.
Proof:
Most of that is just the fundamental theorem of calibrated geometry, the moreover is the equality case in Cauchy-Schwarz in the above proof. ■

Remark
One can show if (M, g, J, ω) is a $U(m)$-structure, then $\omega^m/m! = \mu_g$.
You can also show $\alpha_k = \omega^k/k!$ is a calibration for all $k = 1, \ldots m$ (Wirtinger's inequality), and the α_k-calibrated submanifolds are exactly the $2k$-dimensional submanifolds whose tangent spaces are preserved by J.

Corollary
Complex submanifolds of almost Kähler manifolds are minimal. ■

Curvature of Kähler Manifolds

Let (M^{2m}, g, J, ω) be Kähler. Then the Riemann curvature of g has extra symmetries:

$$R(X, Y, JZ, JW) = R(JX, JY, Z, W) = (R, X, Y, Z, W)$$

by definition of R and the fact $\nabla J = 0$. In general, we know $R \in \Gamma(\mathcal{S}^2(\Lambda^2 T^*M))$, but for Kähler manifolds, this says $R \in \Gamma(\mathcal{S}^2(\Lambda^{1,1} T^*M))$. As a consequence, $\mathrm{Ric}(JX, JY) = \mathrm{Ric}(X, Y)$.

Definition
The Ricci form is the $(1,1)$-form $\rho(X, Y) = \mathrm{Ric}(JX, Y)$ (compare with $\omega(X, Y) = g(JX, Y)$). It is always closed. If g is also Einstein with $\mathrm{Ric} = \lambda g$, $\lambda \in \mathbb{R}$, then this is equivalent to saying $\rho = \lambda\omega$, called Kähler-Einstein.

The other question on A2 about curvature of Kähler metrics: if $z^1, \ldots, z^m$ are local holomorphic coordinates, $e_\alpha = \frac{\partial}{\partial z^\alpha}$ and $e_{\overline{\alpha}} = \frac{\partial}{\partial \overline{z}^\alpha}$, we can extend g, ∇, R to complex-valued objects by complex linearity. Writing bars on our indices to correspond to $(0,1)$-coordinates, one can show $g_{\alpha\beta} = g_{\overline{\alpha}\overline{\beta}} = 0$, and from $\nabla J = 0$, one can show

$$0 = \Gamma^\alpha_{\overline{\beta}A} = \Gamma^\alpha_{A\overline{\beta}} = \Gamma^{\overline{\alpha}}_{\beta A} = \Gamma^{\overline{\alpha}}_{A\beta}$$

where A is any-index, so only $\Gamma^\alpha_{\beta\gamma}$ and $\Gamma^{\overline{\alpha}}_{\overline{\beta}\overline{\gamma}}$ can be nonzero. This is equivalent to $\nabla_{e_\alpha} e_{\overline{\beta}} = 0$. Then one can use this to show

$$0 = R^{\overline{\gamma}}_{AB\alpha} = R^\gamma_{AB\overline{\alpha}}$$

implying by lowering an index ($R_{AB\alpha\beta} = R^C_{AB\alpha} g_{C\beta} = R^{\overline{\gamma}}_{AB\alpha} g_{\overline{\gamma}\beta} = 0$) that $R_{AB\alpha\beta} = 0$ and $R_{AB\overline{\alpha}\overline{\beta}} = 0$, implying only $R_{\alpha\overline{\beta}\gamma\overline{\delta}} = -R_{\alpha\overline{\beta}\overline{\delta}\gamma} = -R_{\overline{\beta}\alpha\gamma\overline{\delta}} = R_{\overline{\beta}\alpha\overline{\delta}\gamma}$ can be nonzero.

Riemannian Submersions

Let $u : (M^m, g) \to (N^n, h)$ be a Riemannian submersion. Then
(1) $(u_*)_p : T_pM \to T_{u(p)}N$ is surjective, so we can write $T_pM = \ker((u_*)_p) \oplus [\ker(u_*)_p]^\perp$, where the kernel is called the vertical space H_p at p and its complement is the horizontal space at p.
(2) $(u_*)_p$ is an isometry when restricted to H_p.

Question
Suppose $u : (M^m, g) \to H$ is a surjective submersion. When can we define a metric h on N making this a Riemannian submersion?
Let $q \in N$. Then there exists $p \in M$ such that $u(p) = q$, and we want to define a metric h_q such that

$$h_q((u_*)_p X_p, (u_*)_p Y_p) = g_p(X_p, Y_p)$$

for some unique $X_p \in H_p$. For this to be well-defined, if $q = u(p) = u(\widetilde{p})$ for some $\widetilde{p} \neq p$, we need $g_p(X_p, Y_p) = g_{\widetilde{p}}(X_{\widetilde{p}}, Y_{\widetilde{p}})$ whenever $(u_*)_p X_p = (u_*)_{\widetilde{p}} X_{\widetilde{p}}$ and similarly for $Y_p, Y_{\widetilde{p}}$. If this holds, then the construction works (h as defined is smooth since g is).

Example
Let $\pi : (S^n, g_{round}) \to \mathbb{RP}^n$ be the usual quotient map. This is a smooth submersion, and in fact, π is a local diffeomorphism since S^n and $\mathbb{RP}^n$ are of the same dimension. Let $T : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be the antipodal map $p \mapsto -p$, which is an isometry of $\mathbb{R}^{n+1}$ and maps S^n onto itself, and hence is an isometry of (S^n, g_{round}) when restricted to the sphere. Moreover, since T is linear, $(T_*)_p = T$ when identifying $T_p\mathbb{R}^{n+1} \cong \mathbb{R}^{n+1}$.
Let $X_p, Y_p \in T_p S^n \cong \{v \in \mathbb{R}^{n+1}, \langle v, p \rangle = 0\} \subseteq \mathbb{R}^{n+1} \cong T_p\mathbb{R}^{n+1}$. Since $\pi \circ T = \pi$, differentiating we obtain that

$$(\pi_*)_{-p} \circ \underbrace{(T_*)_p X_p}_{\cong -X_{-p}} = (\pi_*)_p X_p$$

which under our identifications, shows $(\pi_*)_{-p}(-X_{-p}) = (\pi_*)_p X_p$. Therefore,

$$(g_{round})_p(X_p, Y_p) = (g_{round})_{-p}(-X_{-p}, -Y_{-p})$$

so everything holds to say there exists a unique metric $g_{\mathbb{RP}^n}$ making $\pi : S^n \to \mathbb{RP}^n$ a Riemannian submersion, and in fact, a local isometry. In particular, $g_{\mathbb{RP}^n}$ has constant sectional curvature $+1$.

Example
Consider the map $\pi : S^{2n+1} \to \mathbb{CP}^n$ that sends $z = (z^0, \dots, z^n) \in \mathbb{C}^{n+1}$ such that $|z^0|^2 + \dots + |z^n|^2 = 1$ to the homogeneous line $[z^0, z^1, \dots, z^n] = \{\lambda z : \lambda \in S^1\}$. One can check this is a smooth surjective submersion. Now, we mimic the above example. Let $\lambda \in S^1$ be arbitrary, and let T_λ be multiplication by λ, which is a linear isometry of $\mathbb{C}^{n+1}$ and maps S^{n+1} into S^{n+1}, and so T_λ restricted to S^{n+1} is an isometry of (S^{n+1}, g_{round}), and we have $\pi \circ T_\lambda = \pi$. Let $X_z, Y_z \in T_z S^{n+1}$. Then, analogously to above, we have

$$g_z(X_z, Y_z) = g_{T_\lambda z}([(T_\lambda)_*]_z X_z, [(T_\lambda)_*]_z Y_Z)$$

and so we get a unique Riemannian metric $g_{\mathbb{CP}^n}$ on $\mathbb{CP}^n$ such that $\pi : (S^{n+1}, g_{round}) \to (\mathbb{CP}^n, g_{\mathbb{CP}^n})$ is a Riemannian submersion. On assignment 2, we show (up to a positive factor) that $g_{\mathbb{CP}^n}$ is the Fubini-Study metric.

Remarks
(1) The map $\pi : S^{n+1} \to \mathbb{CP}^n$ as above is called the Hopf map and it generalizes the classical Hopf map when $n = 1$, $\pi : S^3 \to \mathbb{CP}^1 \cong S^2$. There is an analogous construction for $\mathbb{HP}^n$ and for $\mathbb{OP}^1$.
(2) One can show that the fibres $\pi^{-1}([z]) = \{\lambda z : \lambda \in S^1\}$ are closed geodesics in S^1.
(3) We'll see soon that the Hopf maps are harmonic maps and then (2) will follow from general results.
(4) There exists a Riemannian submersion analogue of the second fundamental form called the O'Neill tensors, given an analogue of the Gauss equation for Riemannian submersions. One can use these equations to show for $\pi : S^{2n+1} \to \mathbb{CP}^n$ a Riemannian submersion,

$$1 \leq \underbrace{K_{(\mathbb{CP}^n, g_{\mathbb{CP}^n})}(L_p)}_{=1+3\langle X, JY \rangle^2} \leq 4$$

where X, Y is an orthonormal basis of a 2-plane $L_p \subseteq T_p\mathbb{CP}^n$. Using this, we see if L_p is a complex subspace ($JL_p = L_p$) then we can take $Y = JX$ to obtain $K(L_p) = 4$, and if L_p is totally real 2-plane

($JX \perp L_p$ for all $X \in L_p$), then $K(L_p) = 1$. In particular, for $n = 1$, $\mathbb{CP}^1 \cong S^2$ and $L_p = T_p\mathbb{CP}^1$ is a complex subspace, so $g_{\mathbb{CP}^1}$ has constant sectional curvature 4, and so

$$g_{FS,\mathbb{CP}^1} = \frac{1}{4} g_{round,S^2}$$

Harmonic Maps

Let $u : (M^m, g) \to (N^n, h)$ be smooth. We will write $du : TM \to TN$ for the pushforward. Then $(du)(X_p) \in T_{u(p)}N = (u_*TN)_p$, and so we can think of $du \in \Gamma(T^*M \otimes u^*TN)$. The metric g on TM gives a metric we will denote g^{-1} on T^*M, and the metric h on TN gives a metric u^*h on u^*TN. Combined, we get a fiber metric $g^{-1} \otimes u^*h$ on $T^*M \otimes u^*TN$.

Definition
The energy density of u is defined to be the smooth function on M given by

$$|du|^2 = \langle du, du\rangle_{g^{-1}\otimes u^*h}$$

In local coordinates $x^1, \dots, x^m$ on M (indexed by latin) and $y^1, \dots, y^n$ on N (indexed by greek), we have

$$du = \frac{\partial u^\alpha}{\partial x^i} \mathrm{d}x^i \otimes \underbrace{\left(\frac{\partial}{\partial y^\alpha} \circ u\right)}_{u^* \frac{\partial}{\partial y^\alpha}}$$

where one can verify this is correct by evaluating both of these at p and then sticking in a $\frac{\partial}{\partial x^j}|_p$. Then,

$$|du|^2 = \langle du, du\rangle = \langle \frac{\partial u^\alpha}{\partial x^i} \mathrm{d}x^i \otimes u^* \frac{\partial}{\partial y^\alpha}, \frac{\partial u^\beta}{\partial x^j} \mathrm{d}x^j \otimes u^* \frac{\partial}{\partial y^\beta}\rangle$$

$$= \frac{\partial u^\alpha}{\partial x^i} \frac{\partial u^\beta}{\partial x^j} g^{ij} u^* h_{\alpha\beta} = \frac{\partial u^\alpha}{\partial x^i} \frac{\partial u^\beta}{\partial x^j} g^{ij} (h_{\alpha\beta} \circ u)$$

Suppose for now that M is compact and oriented (so we can integrate functions on M).

Definition
We define the energy (or 2-energy) of u to be

$$E(u) = \frac{1}{2} \int_M |du|^2 \mu_g$$

Definition
In this case, u is called a harmonic map (also called a 2-harmonic map) if it is a critical point of E, that is, $\frac{d}{dt}|_{t=0} E(u_t) = 0$ whenever u_t is a smooth 1-parameter family of maps $u_t : M \to N$ with $u_0 = u$.

We'll derive a 2nd order nonlinear PDE for u to be a harmonic map (called the harmonic map equation).

The bundle $T^*M \otimes u^*TN$ inherits a connection $\nabla = \nabla^{T^*M} \otimes (u^*\nabla^{TN})$, where more precisely if $\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \Gamma^k_{ij} \frac{\partial}{\partial x^k}$ and $\nabla_{\frac{\partial}{\partial y^\alpha}} \frac{\partial}{\partial y^\beta} = \Gamma^\gamma_{\alpha\beta} \frac{\partial}{\partial y^\gamma}$ give our Christoffel symbols on M and N, then we know $\nabla_{\frac{\partial}{\partial x^i}} \mathrm{d}x^k = -\Gamma^k_{ij} \mathrm{d}x^j$ for T^*M. For u^*TN, we have

$$\nabla_{\frac{\partial}{\partial x^i}|_p} (u^* \frac{\partial}{\partial y^\beta}) = \nabla_{(u_*)_p \frac{\partial}{\partial x^i}|_p} \frac{\partial}{\partial y^\beta} = \nabla_{\frac{\partial u^\alpha}{\partial x^i}(p) \frac{\partial}{\partial y^\alpha}|_{u(p)}} \frac{\partial}{\partial y^\beta}$$

$$= \frac{\partial u^\alpha}{\partial x^i}(p) \Gamma^\gamma_{\alpha\beta}(u(p)) \frac{\partial}{\partial y^\gamma}|_{u(p)}$$

and so

$$\nabla_{\frac{\partial}{\partial x^i}}(\mathrm{d}x^k \otimes u^*\frac{\partial}{\partial y^\beta}) = (\nabla_{\frac{\partial}{\partial x^i}}\mathrm{d}x^k) \otimes u^*(\frac{\partial}{\partial y^\beta}) + \mathrm{d}x^k \otimes \nabla_{\frac{\partial}{\partial x^i}}(u^*\frac{\partial}{\partial y^\beta})$$
$$= -\Gamma^k_{ij}\mathrm{d}x^j \otimes (u^*\frac{\partial}{\partial y^\beta}) + \frac{\partial u^\alpha}{\partial x^i}(\Gamma^\gamma_{\alpha\beta}\circ u)\mathrm{d}x^k \otimes (u^*\frac{\partial}{\partial y^\gamma})$$

Definition

The tension of u, denoted $\tau(u) \in \Gamma(u^*TN)$ is defined to be

$$\tau(u) = \mathrm{tr}_g(\overbrace{\nabla \mathrm{d}u}^{\in\Gamma(T^*M\otimes T^*M\otimes u^*TN)})$$

Computing this in local coordinates, we have

$$\nabla(du) = \nabla(\frac{\partial u^\alpha}{\partial x^i}\mathrm{d}x^i \otimes u^*\frac{\partial}{\partial y^\alpha}) = \mathrm{d}x^j \otimes \nabla_{\frac{\partial}{\partial x^j}}(\frac{\partial u^\alpha}{\partial x^i}\mathrm{d}x^i \otimes u^*\frac{\partial}{\partial y^\alpha})$$
$$= \mathrm{d}x^j \otimes \left(\frac{\partial^2 u^\alpha}{\partial x^j \partial x^i}\mathrm{d}x^i \otimes u^*\frac{\partial}{\partial y^\alpha} + \frac{\partial u^\alpha}{\partial x^i}(-\Gamma^i_{jk}\mathrm{d}x^k) \otimes u^*\frac{\partial}{\partial y^\alpha} + \frac{\partial u^\alpha}{\partial x^i}\mathrm{d}x^i \otimes \left(\frac{\partial u^\beta}{\partial x^j}\Gamma^\gamma_{\beta\alpha}u^*\frac{\partial}{\partial y^\gamma}\right)\right)$$
$$= \left(\frac{\partial^2 u^\gamma}{\partial x^i \partial x^j} - \frac{\partial u^\gamma}{\partial x^k}\Gamma^k_{ij} + \frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}(\Gamma^\gamma_{\alpha\beta}\circ u)\right)\mathrm{d}x^i \otimes \mathrm{d}x^j \otimes (u^*\frac{\partial}{\partial y^\gamma})$$

where we rename some indices and use that the Christoffel symbols are symmetric. Thus, taking the trace, we have

$$\tau(u) = g^{ij}\left(\frac{\partial^2 u^\gamma}{\partial x^i \partial x^j} - \frac{\partial u^\gamma}{\partial x^k}\Gamma^k_{ij} + \frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}\left(\Gamma^\gamma_{\alpha\beta}\circ u\right)\right)u^*\frac{\partial}{\partial y^\gamma}$$

which we see is second order in derivatives of u and is quasilinear (linear in highest derivatives, nonlinear elsewhere).

Remark

∇du is called the map Hessian of u (or just Hessian of u), and the tension $\tau(u)$ of u is also called the map Laplacian of u (or just Laplacian of u), keeping in mind that we've seen how to define $\mathrm{Hess}(f) = \nabla df$ and $\Delta f = \mathrm{tr}_g(\nabla df)$, so we see these are generalizations: We saw these have coordinate expressions $\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma^k_{ij}\frac{\partial f}{\partial x^k}$ and $g^{ij}(\frac{\partial^2 f}{\partial x^i \partial x^j} - \Gamma^k_{ij}\frac{\partial f}{\partial x^k})$, and so for a map $f : M \to \mathbb{R}$ with the Christoffel symbols of the standard metric on $\mathbb{R}$ vanishing, we see these expressions agree.
If $u : (a, b) \to N$ is a curve, then we see $\tau(u) = 0$ says

$$0 = \frac{d^2 u^\gamma}{dt^2} + \frac{du^\alpha}{dt}\frac{du^\beta}{dt}(\Gamma^\gamma_{\alpha\beta}\circ u)$$

which is the geodesic equation.

Theorem

u is harmonic if and only if $\tau(u) = 0$.

Lemma

Let $v \in \Gamma(u^*TN)$, and let $w \in \Gamma(T^*M \otimes u^*TN)$. Then

$$\int_M \langle \nabla v, w\rangle \mu_g = -\int_M \langle v, \mathrm{div}\, w\rangle \mu_g$$

Proof:

Write $\nabla v = \mathrm{d}x^i \otimes \nabla_i v$ in local coordinates, and $w = \mathrm{d}x^j \otimes w_j$ where $\nabla_i v$ and w_j are local sections of u^*TN. Then,

$$\langle \nabla v, w\rangle = \langle \mathrm{d}x^i \otimes \nabla_i v, \mathrm{d}x^j \otimes w_j\rangle = g^{ij}\langle \nabla_i v, w_j\rangle = g^{ij}\left(\nabla_i\langle v, w_j\rangle - \langle v, \nabla_i w_j\rangle\right)$$

Then $g^{ij}\nabla_i\langle v, w_j\rangle$ is the divergence of some vector field, so it vanishes when integrating by the divergence theorem, and $-\langle v, g^{ij}\nabla_i w_j\rangle = -\langle v, \operatorname{div} w\rangle$, completing the proof. ■

In the above lemma, note that if $w = du \in \Gamma(T^*M \otimes u^*TN)$, then $\operatorname{div} w = \operatorname{tr}_g(\nabla du) = \tau(u)$, and so

$$\int_M \langle \nabla v, du\rangle \mu_g = -\int_M \langle v, \tau(u)\rangle \mu_g$$

Proof of Theorem:
We must show $\frac{d}{dt}|_{t=0}E(u_t) = 0$ for all smooth variations u_t of u_0 if and only if $\tau(u_0) = 0$. Let $u_t, t \in (-\varepsilon, \varepsilon)$ be a smooth family of maps $M \to N$. In particular, we can think of u as a smooth map $(-\varepsilon, \varepsilon) \times M \to N$ where $u_t = u(t, \cdot) : M \to N$ for all $t \in (-\varepsilon, \varepsilon)$. How can we interpret $\frac{d}{dt}|_{t=0}u_t$? Fix $p \in M$. Then $t \mapsto u_t(p)$ is a smooth curve in N passing through $u_0(p)$ at $t = 0$. Thus, this is the velocity at time $t = 0$ of this curve, so $\frac{d}{dt}|_{t=0}u_t(p) \in T_{u_0(p)}N = (u_0^*TN)_p$. In turn, we get $v = \frac{d}{dt}|_{t=0}u_t : M \to TN$ such that $v(p) \in T_{u_0(p)}N = (u_0^*TN)_p$, which is a section of u_0^*TN, where we can think of $\Gamma(u_0^*TN)$ as the tangent space of $\operatorname{Map}(M, N)$ at u_0. Now, differentiating under the integral sign (since μ_g is independent of t),

$$\frac{d}{dt}|_{t=0}E(u_t) = \frac{d}{dt}|_{t=0}\left(\frac{1}{2}\int_M |du_t|^2\mu_g\right) = \frac{1}{2}\int_M \left(\frac{\partial}{\partial t}|_{t=0}\langle du_t, du_t\rangle\right)\mu_g$$

Thinking of $\frac{\partial}{\partial t}$ as a vector field on $(-\varepsilon, \varepsilon) \times M$, by metric compatibility (for the Levi-Civita connection of the product metric),

$$\frac{\partial}{\partial t}\langle u_t, u_t\rangle = 2\langle \nabla_{\frac{\partial}{\partial t}} du_t, du_t\rangle$$

Now, what is $\nabla_{\frac{\partial}{\partial t}} du_t$? If $X \in \Gamma(TM)$, by the Leibniz rule,

$$\left(\nabla_{\frac{\partial}{\partial t}} du_t\right)(X) = \nabla_{\frac{\partial}{\partial t}}((du_t)(X)) - (du_t)(\nabla_{\frac{\partial}{\partial t}} X)$$

Now, $\nabla_{\frac{\partial}{\partial t}} X = 0$ (as can be seen directly looking at the formula for the Christoffel symbols, or since $\frac{\partial}{\partial t}$ is the coordinate on $(-\varepsilon, \varepsilon)$ and X on M doesn't change in that direction). By Assignment 1 Question 1, since $[X, \frac{\partial}{\partial t}] = 0$, $\nabla_{\frac{\partial}{\partial t}}((du_t)(X)) = \nabla_X((du_t)(\frac{\partial}{\partial t}))$, and so without the X we see $\nabla_{\frac{\partial}{\partial t}} du_t = \nabla(du_t)(\frac{\partial}{\partial t}) = \nabla\frac{\partial u_t}{\partial t}$. Plugging this all into the integral, we obtain

$$\frac{d}{dt}|_{t=0}E(u_t) = \int_M \langle \nabla\frac{\partial u_t}{\partial t}, du_t\rangle|_{t=0}\mu_g \overbrace{=}^{\text{construction of } v} \int_M \langle \nabla v, du_0\rangle \mu_g \overbrace{=}^{\text{Lemma}} -\int_M \langle v, \tau(u_0)\rangle \mu_g$$

where this is zero for all $v \in \Gamma(u_0^*TN)$ if and only if $\tau(u_0) = 0$. ■

Remark
In fact, this shows that the gradient (with respect to the L^2 metric on $\operatorname{Map}(M, N)$ of the energy functional is $-$tension, in the sense that if $E : \operatorname{Map}(M, N) \to \mathbb{R}$ is the energy, then $\operatorname{grad} E$ is the vector field on $\operatorname{Map}(M, N)$ such that $\langle \operatorname{grad}_{u_0} E, v\rangle_{L^2} = \frac{d}{dt}|_{t=0}E(u_t)$ where u_t is such that $\frac{d}{dt}|_{t=0}u_t = v$ for $v \in T_{u_0}\operatorname{Map}(M, N) = \Gamma(u_0^*TN)$.
Later, we'll consider the "harmonic map heat flow" where $\frac{\partial}{\partial t}u_t = -\tau(u_t)$, which is a flow for smooth maps $M \to N$ which could possibly flow to a harmonic map.

Examples of Harmonic Maps

We've already seen:
(1) harmonic functions $(M, g) \to \mathbb{R}$.
(2) geodesics $(a, b) \to (N, h)$.
(3) (locally) constant maps, which are maps $u : M \to N$ with $du \equiv 0$.

(4) The identity map: $u = I : (M, g) \to (M, g)$ is harmonic. Indeed, take $y^i = x^i$ so then $\frac{\partial u^\alpha}{\partial x^j} = \frac{\partial y^\alpha}{\partial x^j} = \frac{\partial x^\alpha}{\partial x^j} = \delta^\alpha_j$, and so in the harmonic map equation, we have

$$g^{ij}(-\Gamma^\gamma_{ij} + \Gamma^\gamma_{ij}) = 0$$

since the second derivatives of I vanish. Even though the identity map seems like a trivial example, its property of being a local minimum/maximum/saddle of energy depends on the geometry of (M, g).

Proposition

Let $u : (M, g) \to (N, h)$ be smooth. The map Hessian $\nabla du \in \Gamma(T^*M \otimes T^*M \otimes u^*TN)$ is symmetric: $(\nabla du)(X, Y) = (\nabla du)(Y, X)$, and thus is a u^*TN-valued symmetric bilinear form on M.

Proof:

$$(\nabla du)(X, Y) = (\nabla_X du)(Y) = \overbrace{\nabla_X((du)Y)}^{(u^*\nabla^T N)_X(u_*Y) = \nabla^{TN}_{u_*X}(u_*Y)} - du(\nabla_X Y)$$

Thus, by Assignment 1,

$$(\nabla du)(X, Y) - (\nabla du)(Y, X) = (u^*\nabla)_X(u_*Y) - u_*(\nabla_X Y) - (u^*\nabla)_Y(u_*X) + u_*(\nabla_Y X)$$

$$= u_*[X, Y] - u_*[X, Y] = 0 \quad \blacksquare$$

Corollary (Example 5)

Let $u : (M, g) \to (N, h)$ be a Riemannian immersion. Then u is harmonic if and only if u is minimal.

Proof:

We have

$$(\nabla du)(X, Y) = \nabla_X(u_*Y) - u_*(\nabla_X Y) \overbrace{=}^{\text{Riemannian immersion}} \nabla^{TN}_X Y - \nabla^{TM}_X Y = A(X, Y)$$

Thus, we see ∇du is the second fundamental form of u, and so taking traces, we see the tension coincides with the mean curvature. $\blacksquare$

Remark

The above corollary explains why the map Hessian is sometimes called the second fundamental form of u.

Continuing this analogy,

Definition

Let $u : (M, g) \to (N, h)$ be smooth. We say u is totally geodesic if $\nabla du = 0$. So totally geodesic implies harmonic, but the converse need not hold.

Theorem (Composition formulas)

Let $u : (M, g) \to (N, h)$ and $v : (N, h) \to (P, k)$ be smooth maps. Then

(1) $\nabla d(v \circ u) = dv \circ \nabla du + (\nabla dv)(du, du)$.

(2) $\tau(v \circ u) = dv \circ \tau(u) + \mathrm{tr}((\nabla dv)(du, du))$

Proof:

(2) is just the trace of (1), so it suffices to prove (1).

By the Leibniz rule,

$$(\nabla d(v \circ u))(X, Y) = \nabla_X(d(v \circ u)(Y)) - d(v \circ u)(\nabla_X Y)$$

$$= \nabla_X((dv \circ du)(Y)) - dv \circ du \nabla_X Y$$

$$= \underbrace{(\nabla_X dv)(duY)}_{(\nabla dv)(duX, duY)} + \underbrace{dv(\nabla_X(duY)) - dv \circ du \nabla_X Y}_{= dv((\nabla du)(X,Y))}$$

and so
$$\nabla d(v\circ u)(\cdot,\cdot) = dv(\nabla du(\cdot,\cdot)) + (\nabla dv)(du(\cdot), dv(\cdot))$$
as desired. ■

Corollary
(1) The composition of totally geodesic maps is totally geodesic.
(2) if u is harmonic and v is totally geodesic, then $v\circ u$ is harmonic. ■

Corollary
Let $u:(M,g)\to(N,h)$ be smooth. The following are equivalent:
(a) u is totally geodesic.
(b) u "preserves connections" in the sense that $(u^*\nabla)_X(u_*Y) = u_*(\nabla_X Y)$.
(c) u maps geodesics of M to geodesics of N.
Proof:
(a) if and only if (b) comes from
$$(\nabla du)(X,Y) = (\nabla_X du)(Y) = \nabla_X(duY) - du\nabla_X Y = (u^*\nabla)_X(u_*Y) - u_*(\nabla_X Y)$$
For (a) if and only if (c), consider the above theorem with $\gamma = u, v = u$ for γ a geodesic, where we then compute
$$\tau(u\circ\gamma) = du\circ\tau(\gamma) + \underbrace{\mathrm{tr}}_{=1 \text{ on } ((-\varepsilon,\varepsilon),dt^2)}((\nabla du)(d\gamma,d\gamma)) = du\circ\tau(\gamma) + (\nabla du)(d\gamma,d\gamma)$$
from which (a) implies (c) is immediate, and if (c) holds then we see $(\nabla du)(\gamma'(t),\gamma'(t)) = 0$ for a geodesic γ, and we know any $X_p\in T_pM$ is $\gamma'(0)$ for some geodesic γ, which shows $\nabla du = 0$. ■

What about submersions? Which Riemannian submersions are harmonic maps?
Let $u:(M^m,g)\to(N^n,h)$ be a surjective Riemannian submersion. Then we have an orthogonal decomposition $T_pM = V_p\oplus H_p$ where $V_p = \ker(u_*)_p$ and $H_p\cong T_{u(p)}N$.
By the implicit function theorem, for all $q\in N$, $u^{-1}(q)$ is a codimension n submanifold of M, where $V_p = T_p(u^{-1}(u(p)))$.

Theorem
A Riemannian submersion u is harmonic if and only if all of its fibres are minimal.

To prove this, we need some preliminary results.

Definition
Based on the splitting of T_pM above, we have an orthogonal decomposition of TM into subbundles $TM = V\oplus H$. A vector field X on M is called horizontal if it is a section of H, that is, if $X_p\in H_p$ for all p, and vertical it's a section of V, that is, if $X_p\in V_p$ for all p. If $X\in\Gamma(TN)$ is an arbitrary vector field, then its horizontal lift $X^*\in\Gamma(H)$ is the horizontal vector field on M determined by
$$(u_*)_pX^*_p = X_{u(p)}$$
where we think of $(u_*)_p$ as a map $(u_*)_p : H_p\xrightarrow{\cong} T_{u(p)}N$. It's not hard to check that X^* is smooth.

Remarks
(1) $W\in\Gamma(TM)$ is vertical if and only if it is u-related to 0.
(2) $X^*\in\Gamma(H)$ is the horizontal lift of X if and only if X^* is u-related to X.
(3) Not all horizontal vector fields are horizontal lifts. For example, let X^* be the horizontal lift of X. Let $W = fX^*$ for some $f\in\mathcal{C}^\infty(M)$ which is not constant on fibres. Then $(u_*)_pW_p = f(p)(u_*)_pX^*_p = f(p)X_{u(p)}$, so if there is some $\widetilde{p}\neq p$ such that $q = u(p) = u(\widetilde{p})$, then we see $(u_*)_{\widetilde{p}}W_{\widetilde{p}} = f(\widetilde{p})X_{u(\widetilde{p})}$, so W

is not u-related to any vector field on N.
(4) Let $X \in \Gamma(TN)$ and $f \in \mathcal{C}^\infty(N)$. Then

$$X^*(f \circ u) = (Xf) \circ u$$

Indeed, at $p \in M$,

$$X_p^*(f \circ u) = d(f \circ u)_p(X_p^*) = (df)_{u(p)}((u_*)_p(X_p^*)) = (df)_{u(p)} X_{u(p)} = (Xf)_{u(p)} = ((Xf) \circ u)_p$$

Lemma (O'Neill formulas)
Let $u : (M, g) \to (N, h)$ be a Riemannian submersion. Let $X, Y \in \Gamma(TN)$. Then:
(i) $g_p(X_p^*, Y_p^*) = h_{u(p)}(X_{u(p)}, Y_{u(p)})$ for all $p \in M$, written more compactly as $g(X^*, Y^*) = h(X, Y) \circ u$.
(ii) $u_*([X^*, Y^*]) = [X, Y]$.
(iii) $u_*(\nabla^M_{X^*} Y^*) = \nabla^N_X Y$.
Proof:
(i) is immediate by definition of a Riemannian submersion and horizontal vector field.
(ii) is naturality of the Lie bracket: X^*, X and Y^*, Y are u-related by definition.
(iii) By the Koszul formula and (i),

$$2h(u_* \nabla^M_{X^*} Y^*, Z^*) \circ u = 2g(\nabla^M_{X^*} Y^*, Z^*)$$

$$= X^*(g(Y^*, Z^*)) + Y^*(g(X^*, Z^*)) - Z^*(g(X^*, Y^*)) - g(X^*, [Y^*, Z^*]) - g(Y^*, [X^*, Z^*]) + g(Z^*, [X^*, Y^*])$$

By (i) and Remark (4) above, $X^*(g(Y^*, Z^*)) = X^*(h(Y, Z) \circ u) = X(h(Y, Z)) \circ u$. By (i) and (ii), we have $g(X^*, [Y^*, Z^*]) = h(X, u_*[Y^*, Z^*]) \circ u = h(X, [Y, Z]) \circ u$, and so using these, we simplify the above with the Koszul formula for h to

$$2h(u_* \nabla^M_{X^*} Y^*, u_* Z^*) \circ u = 2h(\nabla^N_X Y, Z) \circ u$$

Since u is surjective, we get $h(u^* \nabla^M_{X^*} Y^*, u_* Z^*) = h(\nabla^N_X Y, Z)$ for all Z, implying $u_* \nabla^M_{X^*} Y^* = \nabla^N_X Y$. ■

Proof of the Theorem
Let $\{e_1', \dots, e_n'\}$ be a local orthonormal frame for TN with respect to h, and let $\{e_1, \dots, e_n\}$ be their horizontal lifts. This is a local orthonormal frame for H with respect to g by (i) of the Lemma. We can extend this to a local orthonormal frame $\{e_1, \dots, e_m\}$ for TM with respect to g. Computing the tension, since our frame is orthonormal, we get

$$\tau(u) = \mathrm{tr}_g(\nabla du) = \sum_{i=1}^m (\nabla_{e_i}(u_* e_i) - u_*(\nabla_{e_i} e_i))$$

For $1 \le i \le n$, by (iii) of the previous lemma, we have

$$\nabla_{e_i}(u_* e_i) = \nabla^N_{e_i'} e_i' = u_* \nabla^M_{e_i} e_i$$

and so our sum becomes

$$\tau(u) = \sum_{i=n+1}^m (\nabla_{e_i}(u_* e_i) - u_* \nabla_{e_i} e_i)$$

For $n + 1 \le i \le m$, e_i is vertical, and so $u_* e_i = 0$, so that

$$\tau(u) = -u_* \sum_{i=n+1}^m \nabla_{e_i} e_i$$

Thus we see u is harmonic if and only if $\sum_{i=n+1}^m \nabla_{e_i} e_i$ is vertical, which is to say its normal part is zero. But we know H_p is the normal space to the fibre $u^{-1}(u(p))$, and the normal part of $\sum_{i=n+1}^m \nabla_{e_i} e_i$ is the mean curvature of the fibre $u^{-1}(u(p))$, so putting these together, u is harmonic if and only if the mean curvature of the fibre $u^{-1}(u(p))$ vanishes. ■

Corollary

Let (M, g) be Riemannian. Then an isometry $u : (M, g) \overset{\cong}{\to} (M, g)$ is harmonic.

Proof 1:

u is a Riemannian immersion, so u is harmonic if and only if its minimal, but νM in M is rank zero, so its second fundamental form vanishes identically. ■

Proof 2:

u is a Riemannian submersion, so u is harmonic by the previous theorem if and only if its fibres are minimal. But the fibres of an isometry are points, which are certainly minimal. ■

Our final example of harmonic maps:

Definition

Let (M^{2m}, g, J_M) and (N^{2n}, h, J_N) be Kähler manifolds.

A smooth map $u : M \to N$ is called holomorphic if and only if $u_* J_M = J_N u_*$. (In local holomorphic coordinates, this is the Cauchy-Riemann equations). u is called antiholomorphic if $u_* J_M = -J_N u_*$.

Proposition

If u is holomorphic or antiholomorphic, then u is harmonic.

Proof:

We can choose a local orthonormal frame for TM of the form $\{e_1, \dots, e_m, f_1, \dots, f_m\}$ where $f_i = Je_i$ and $Jf_i = -e_i$. Compute

$$\nabla_{f_i} u_* f_i = \nabla_{f_i} u_* J e_i = \overbrace{\pm}^{\text{hol}} \nabla_{f_i} J u_* e_i \overbrace{=}^{N \text{ Kähler}} \pm J \nabla_{f_i} u_* e_i \overbrace{=}^{A1} \pm J(\nabla_{e_i} u_* f_i + u_*[f_i, e_i])$$

$$\overbrace{=}^{N \text{ Kähler}} \pm \nabla_{e_i} J u_* f_i \pm J u_*[f_i, e_i] = \nabla_{e_i} u_* J f_i \pm J u_*[f_i, e_i] = -\nabla_{e_i} u_* e_i \pm J u_*[f_i, e_i]$$

We also have

$$u_* \nabla_{f_i} f_i = u_* \nabla_{f_i} J e_i \overbrace{=}^{M \text{ Kähler}} u_* J \nabla_{f_i} e_i = u_* J(\nabla_{e_i} f_i + [f_i, e_i]) \overbrace{=}^{M \text{ Kähler}} u_* \nabla_{e_i}(J f_i) \pm J u_*[f_i, e_i]$$

Combining these identifies, we obtain

$$\tau(u) = \sum_{i=1}^{m} (\nabla_{e_i}(u_* e_i) - u_*(\nabla_{e_i} e_i)) + \sum_{i=1}^{m} (\nabla_{f_i}(u_* f_i) - u_*(\nabla_{f_i} f_i)) = 0$$

where everything in the above two identities cancel. ■

Weitzenböck Formula

Let (M, g) be Riemannian, and let $E \to M$ be a real vector bundle with connection ∇ and compatible fibre metric h^E. The bundle of E-valued k-forms on M is $\Lambda^k(T^*M) \otimes E$, and its sections are Ω^k_E. Given $\xi \in \Omega^k_E$, locally we can write $\xi = \alpha^a \otimes \sigma_a$ if $\sigma_1, \dots, \sigma_r$ is a local frame for E and α^a is a (local) k-form on M.

Definition

Given the connection ∇ on E, we define a map $d^\nabla : \Omega^k_E \to \Omega^{k+1}_E$, called the exterior covariant derivative associated to ∇, as follows:

$$(d^\nabla \xi)(X_1, \dots, X_{k+1}) = \sum_{i=1}^{k+1} (-1)^{i+1} \nabla^E_{X_i}(\xi(X_1, \dots, \widehat{X_i}, \dots, X_{k+1}))$$

$$+\sum_{i<j}(-1)^{i+j}\xi([X_i,X_j],X_1,\dots,\widehat{X_i},\dots,\widehat{X_j},\dots,X_{k+1})$$

This is a generalization of the coordinate-free formula for the ordinary exterior derivative with $E=\underline{\mathbb{R}}_M$ the trivial real line bundle and $\nabla=d$ is the trivial connection.
We need to show $d^\nabla\xi$ is $\mathcal{C}^\infty(M)$-linear and is totally skew-symmetric in its arguments.
Indeed, by the Leibniz rule, for ∇ the connection on $\Lambda^k(T^*M)\otimes E$ induced by the Levi-Civita connection and ∇^E,

$$(\nabla_Y\xi)(X_1,\dots,X_k)=\nabla^E_Y(\xi(X_1,\dots,X_k))-\sum_{j=1}^k\xi(X_1,\dots,\nabla^{TM}_YX_j,\dots,X_k)$$

Hence by the Leibniz rule

$$(d^\nabla\xi)(X_1,\dots,X_{k+1})=\sum_{i=1}^{k+1}(-1)^{i+1}(\nabla_{X_i}\xi)(X_1,\dots,\widehat{X_i},\dots,X_{k+1})$$

$$+\sum_{i=1}^{k+1}(-1)^{i+1}\left(\sum_{j=1}^{i-1}\xi(X_1,\dots,\nabla_{X_i}X_j,\dots,\widehat{X_i},X_{k+1})+\sum_{j=i+1}^{k+1}\xi(X_1,\dots,\widehat{X_i},\dots,\nabla_{X_i}X_j,\dots,X_{k+1})\right)$$

$$+\sum_{i<j}(-1)^{i+j}\xi([X_i,X_j],X_1,\dots,\widehat{X_i},\dots,\widehat{X_j},X_{k+1})$$

For the middle term, swapping the roles of i and j, we have (same calculation as on A1)

$$\sum_{j<i}(-1)^{i+1}\xi(X_1,\dots,\nabla_{X_i}X_j,\dots,\widehat{X_i},\dots,X_{k+1})+\sum_{i<j}(-1)^{i+1}\xi(X_1,\dots,\widehat{X_i},\dots,\nabla_{X_i}X_j,\dots,X_{k+1})$$

$$=\sum_{i<j}(-1)^{i+j}\xi(\nabla_{X_j}X_i,X_1,\dots,\widehat{X_i},\widehat{X_j},\dots,X_{k+1})+\sum_{i<j}(-1)^{i+j+1}\xi(\nabla_{X_i}X_j,X_1,\dots,\widehat{X_i},\widehat{X_j},\dots,X_{k+1})$$

$$=-\sum_{i<j}(-1)^{i+j}\xi([X_i,X_j],X_1,\dots,\widehat{X_i},\widehat{X_j},\dots,X_{k+1})$$

and thus the middle term and last term in $d^\nabla\xi(X_1,\dots,X_{k+1})$ cancel.
This shows

$$(d^\nabla\xi)(X_1,\dots,X_{k+1})=(\nabla_{X_1}\xi)(X_2,\dots,X_{k+1})-(\nabla_{X_2}\xi)(X_1,X_3,\dots,X_k)+\dots+(-1)^k(\nabla_{X_{k+1}}\xi)(X_1,\dots,X_k)$$

is the skew-symmetrization of the covariant derivative, giving the result.

Remark
In general, $(d^\nabla)^2\neq0$. In fact, if $\xi\in\Omega^0_E=\Gamma(E)$, then $(d^\nabla)^2\xi\in\Omega^2_E$, and we get

$$((d^\nabla)^2\xi)(X,Y)=\underbrace{R^\nabla(X,Y)}_{\in\Gamma(\mathrm{End}(E))}\xi$$

Now, using the fibre metric h^E on E, we can define the formal adjoint $(d^\nabla)^*:\Omega^k_E\to\Omega^{k-1}_E$ by

$$\int_M\langle d^\nabla\xi,\eta\rangle\mu_g=\int_M\langle\xi,(d^\nabla)^*\eta\rangle\mu_g$$

if M is compact and oriented.

Proposition
If $\xi \in \Omega^k_E$, then

$$((d^\nabla)^*\xi)(X_1,\dots,X_{k-1}) = -g^{ij}(\nabla_{e_i}\xi)(e_j,X_1,\dots,X_{k-1}) = -\operatorname{tr}_g(\nabla\xi)(X_1,\dots,X_{k-1})$$

Proof:
Fix $p \in M$ and choose geodesic normal coordinates centered at p, so that $\nabla_{\partial_i}\partial_j = 0$ at p and $\nabla_i g^{ab} = 0$. Let $\xi \in \Omega^{k-1}_E$ and $\eta \in \Omega^k_E$. Then at p,

$$\langle d^\nabla\xi,\eta\rangle = \frac{1}{k!}\underbrace{\langle (d^\nabla\xi)_{i_1\dots i_k},\eta_{j_1\dots j_k}\rangle}_{=h^E \text{ applied to } E\text{-parts}} g^{i_1j_1}\dots g^{i_kj_k}$$

$$\overbrace{=}^{\text{result above}} \frac{1}{k!}\sum_{j=1}^k (-1)^{j+1}\langle (\nabla_{i_j}\xi)_{i_1\dots\widehat{i_j}\dots i_k},\eta_{j_1\dots j_k}\rangle g^{i_1j_1}\dots g^{i_kj_k}$$

$$= \frac{1}{k!}k\langle (\nabla_{i_1}\xi)_{i_2\dots i_k},\eta_{j_1\dots j_k}\rangle g^{i_1j_1}\dots g^{i_kj_k} = \frac{1}{(k-1)!}\langle (\nabla_{i_1}\xi)_{i_2\dots i_k},\eta_{j_1\dots j_k}\rangle g^{i_1j_1}\dots g^{i_kj_k}$$

On the other hand, using the definition stated in the proposition,

$$\langle \xi,(d^\nabla)^*\eta\rangle = \frac{1}{(k-1)!}\langle \xi_{i_2\dots i_k}, -g^{i_1j_1}\nabla_{i_1}\eta_{j_1\dots j_k}\rangle g^{i_2j_2}\dots g^{i_kj_k}$$

$$= -\frac{1}{(k-1)!}\langle \xi_{i_2\dots i_k},\nabla_{i_1}\eta_{j_1\dots j_k}\rangle g^{i_1j_1}\dots g^{i_kj_k}$$

and so

$$\langle d^\nabla\xi,\eta\rangle - \langle \xi,(d^\nabla)^*\eta\rangle = \frac{1}{(k-1)!}\nabla_{i_1}(\langle \xi_{i_2\dots i_k},\eta_{j_1\dots j_k}\rangle) g^{i_1j_1}\dots g^{i_kj_k}$$

$$\overbrace{=}^{\text{metric compatible}} \frac{1}{(k-1)!}\nabla_{i_1}(\langle \xi_{i_2\dots i_k},\eta_{j_1\dots j_k}\rangle g^{i_1j_1}\dots g^{i_kj_k}) = \frac{1}{(k-1)!}\operatorname{div}(\text{some vector field})$$

which integrates to 0 by the divergence theorem, giving the result. ■

Given the above result, we take the definition of $(d^\nabla)^*$ to be the above formula.

Definition
The d^∇-Laplacian is defined to be $\Delta_{d^\nabla} : \Omega^k_E \to \Omega^k_E$, $\Delta_{d^\nabla} = d^\nabla(d^\nabla)^* + (d^\nabla)^*d^\nabla$ (we note this is self-adjoint).

Remark
This generalizes the Hodge Laplacian $\Delta_d = dd^* + d^*d$ on $E = \underline{\mathbb{R}}_M$, where $d^\nabla = d$.

Lemma
If M is compact and oriented, then $\Delta_{d^\nabla}\xi = 0$ if and only if $d^\nabla\xi = 0$ and $(d^\nabla)^*\xi = 0$ (these conditions are called d^∇-harmonic, d^∇-closed, and d^∇-coclosed respectively).
Proof:
The reverse direction is immediate. For the forward direction, we have (letting $\langle\langle\cdot,\cdot\rangle\rangle$ be the L^2-inner product),

$$0 = \langle\langle \Delta_{d^\nabla}\xi,\xi\rangle\rangle = \langle\langle (d^\nabla)(d^\nabla)^*\xi,\xi\rangle\rangle + \langle\langle (d^\nabla)^*(d^\nabla)\xi,\xi\rangle\rangle = ||(d^\nabla)^*\xi||^2 + ||(d^\nabla)\xi||^2$$

giving the result. ■

We will soon see that Δ_{d^∇} is elliptic.

Definition
Let $V \to M$ be a real vector bundle with connection ∇ and compatible fibre metric h^V.
Similarly to before, for $\varphi \in \Gamma(T^*M \otimes V)$, we have

$$(\nabla^*\varphi)(\cdot) = -g^{ij}(\nabla_{e_i}\varphi)(e_j, \cdot) = -\operatorname{tr}_g(\nabla\varphi)$$

The rough Laplacian (or connection Laplacian) is the map

$$\nabla^*\nabla : \Gamma(V) \to \Gamma(V)$$

given locally by

$$(\nabla^*\nabla\xi) = -g^{ij}(\nabla_{e_i}(\nabla\xi))(e_j) = -g^{ij}\left(\nabla_{e_i}((\nabla\xi)(e_j)) - (\nabla\xi)(\nabla_{e_j}e_j)\right)$$
$$= -g^{ij}(\nabla_{e_i}(\nabla_{e_j}\xi) - \nabla_{\nabla_{e_i}e_j}\xi) =: -g^{ij}(\nabla\nabla\xi)(e_i, e_j) \underbrace{=:}_{\text{careful, not equal to } -g^{ij}\nabla_i(\nabla_j\xi)} -g^{ij}\nabla_i\nabla_j\xi$$

"Definition"
A self-adjoint 2nd-order linear differential operator $P : \Gamma(V) \to \Gamma(V)$ is called elliptic if $P = \nabla^*\nabla +$ first and zero-order terms.

Fact
If P is an elliptic operator on $\Gamma(V)$ and M is compact, then:
(1) The eigenvalues of P are real by the spectral theorem, and form a countable discrete set accumulating only at $+\infty$.
(2) Each eigenspace of P is finite-dimensional.
(3) P is called Fredholm, implying:
$\ker P$ is finite-dimensional, and $\operatorname{im} P$ is closed (with respect to the L^2-inner product), and $\operatorname{coker} P = \Gamma(V)/\operatorname{im} P$ is finite dimensional.
For references, see Warner or Griffiths and Harris.

Example
$P = \nabla^*\nabla$ is elliptic, and in this case all eigenvalues are non-negative: If $P\xi = \lambda\xi$ with $\xi \neq 0$, then

$$||\nabla\xi||^2 = \langle\langle \nabla^*\nabla\xi, \xi\rangle\rangle = \langle\langle \lambda\xi, \xi\rangle\rangle = \lambda||\xi||^2$$

implying $\lambda \geq 0$. Here, $\ker(\nabla^*\nabla)$ is the parallel sections, i.e. ξ satisfying $\nabla\xi = 0$.

Now, let $V = \Lambda^K(T^*M) \otimes E$, $\xi \in \Gamma(V) = \Omega^k_E$. By the Leibniz rule for covariant differentiation (do it twice then subtract to get the definition of curvature),

$$(R^V(X,Y)\xi)(X_1, \ldots, X_k) = R^E(X,Y)(\xi(X_1, \ldots, X_k)) - \sum_{i=1}^{k} \xi(X_1, \ldots, R^{TM}(X,Y)X_i, \ldots, X_k)$$

Definition
The Ricci operator $\rho \in \Gamma(\operatorname{End}(V))$ is defined as follows: if $k = 0$, $\rho = 0$, and if $k \geq 1$,

$$(\rho\xi)(X_1, \ldots, X_k) = g^{ij}\left(\sum_{\ell=1}^{k}(-1)^{\ell+1}(R^V(e_i, X_\ell)\xi)(e_j, X_1, \ldots, \widehat{X_\ell}, \ldots, X_k)\right)$$

Remark
If $k = 1$, and $E = \underline{\mathbb{R}}_M$, then $\Omega^1_E = \Omega^1 \cong \Gamma(TM)$ using the metric, so taking $\xi = Y \in \Omega^1$, then

$$(\rho Y)(X) = g^{ij}(R(e_i, X)Y)(e_j) = g^{ij}R(e_i, X, Y, e_j) = \operatorname{Ric}(X,Y) = \operatorname{Ric}(Y)(X)$$

and so $\rho = \mathrm{Ric}$ under our identifications, so ρ generalizes Ric in this sense.

Theorem (Weitzenböck Formula)
Let $\xi \in \Omega^k_E$. Then

$$\Delta_{d^\nabla}\xi = \nabla^*\nabla\xi + \rho(\xi)$$

Proof:
If $k = 0$, then $\rho(\xi) = 0$ and $\Delta_{d^\nabla}\xi = (d^\nabla)^*(d^\nabla)\xi = \nabla^*\nabla\xi$ because $d^\nabla = \nabla$ on Ω^0_E. Now let $k \geq 1$. We will compute $d^\nabla(d^\nabla)^*\xi$ and $(d^\nabla)^* d^\nabla\xi$. Fix $p \in M$, and let $x^1, \ldots, x^m$ be geodesic normal coordinates centered at p. Thus $\partial_i|_p$ are orthonormal and $\nabla_i\partial_j = 0$ at p. Then

$$(d^\nabla(d^\nabla)^*\xi)(\partial_{i_1}, \ldots, \partial_{i_k}) = \sum_{j=1}^k (-1)^{j+1}\nabla_{i_j}((d^\nabla)^*\xi)(\partial_{i_1}, \ldots, \widehat{\partial_{i_j}}, \ldots, \partial_{i_k})$$

$$\overbrace{=}^{\text{at } p\text{, since } \nabla_i\partial_j = 0} \sum_{j=1}^k (-1)^{j+1}\nabla_{i_j}(((d^\nabla)^*\xi)(\partial_{i_1}, \ldots, \widehat{\partial_{i_j}}, \ldots, \partial_{i_k}))$$

$$\overbrace{=}^{\text{at } p} -\sum_{j=1}^k (-1)^{j+1}\nabla_{i_j}(g^{ab}(\nabla_a\xi)(\partial_b, \partial_{i_1}, \ldots, \widehat{\partial}_{i_k}, \ldots, \partial_{i_k}))$$

On the other hand,

$$((d^\nabla)^* d^\nabla\xi)(\partial_{i_1}, \ldots, \partial_{i_k}) = -g^{ab}\nabla_a((d^\nabla\xi))(\partial_b, \partial_{i_1}, \ldots, \partial_{i_k})$$

$$\overbrace{=}^{\text{at } p\text{, since } \nabla_i\partial_j = 0} -g^{ab}\nabla_a((d^\nabla\xi)(\partial_b, \partial_{i_1}, \ldots, \partial_{i_k})) = \overbrace{-g^{ab}\nabla_a((\nabla_b\xi)(\partial_{i_1}, \ldots, \partial_{i_k}))}^{=\nabla^*\nabla\xi(\partial_{i_1}, \ldots, \partial_{i_k})} - g^{ab}\nabla_a\left(\sum_{j=1}^k (\nabla_{i_j}\xi)(\partial_b, \partial_{i_1}, \ldots, \widehat{\partial_{i_j}}, \ldots, \partial_{i_k})\right)$$

TODO SIGN WRONG Adding these together, we obtain

$$(\Delta_{d^\nabla}\xi)(\partial_{i_1}, \ldots, \partial_{i_k}) = (\nabla^*\nabla\xi)(\partial_{i_1}, \ldots, \partial_{i_k}) + g^{ab}\sum_{j=1}^k (-1)^{j+1}(\underbrace{(\nabla_a\nabla_{i_j} - \nabla_{i_j}\nabla_a)}_{=R(e_a, e_{i_j})}\xi)(\partial_b, \partial_{i_1}, \ldots, \widehat{\partial_{i_j}}, \ldots, \partial_{i_k})$$

$$= (\nabla^*\nabla\xi)(\partial_{i_1}, \ldots, \partial_{i_k}) + g^{ab}\sum_{j=1}^k (-1)^{j+1}(R(e_a, e_{i_j})\xi)(e_b, \partial_{i_1}, \ldots, \widehat{\partial_{i_j}}, \ldots, \partial_{i_k}) = (\nabla^*\nabla\xi)(\partial_{i_1}, \ldots, \partial_{i_k}) + \rho(\xi)(\partial_{i_1}, \ldots, \partial_{i_k})$$

as desired. ■

Corollary
Δ_{d^∇} is elliptic. ■

Harmonic Maps and the Exterior Covariant Derivative

Let $u : (M^m, g) \to (N^n, h)$ be a smooth map between Riemannian manifolds, and consider $du = u_* \in \Gamma(T^*M \otimes u^*TN) = \Omega^1_E$ when $E = u^*TN$.

Lemma
du is always d^∇-closed.
Proof:
Since d^∇ is the skew-symmetrization of $\nabla^E = u^*\nabla^{TN}$,

$$(d^\nabla du)(X, Y) = (\nabla du)(X, Y) - (\nabla du)(Y, X)$$

Since the map Hessian ∇du is symmetric, this is zero. ■

Lemma
u is harmonic if and only if $(d^\nabla)^* = 0$.
Proof:
We have

$$\tau(u) = \mathrm{tr}_g(\nabla du) = -(d^\nabla)^*(du) \quad ■$$

Corollary
If M is compact and oriented, then u is a harmonic map if and only if du is a d^∇-harmonic u^*TN-valued 1-form. ■

Second Variation Formula for Harmonic Maps

For now, assume M is compact and oriented, so its energy functional E is well-defined, and u is harmonic if and only if it is a critical point of E.

Definition
Suppose u is harmonic. The Hessian of E at u is the symmetric bilinear form on $\Gamma(u^*TN) = T_u(\mathrm{Map}(M,N))$ defined by, for $v = \frac{d}{ds}\Big|_{s=0} u_{s,0}$, and $w = \frac{d}{dt}\Big|_{t=0} u_{0,t}$, where $u_{s,t}$ is a smooth 2-parameter family of maps $M \to N$ with $u_{0,0}$,

$$(\mathrm{Hess}\, E)_u(v,w) = \frac{\partial^2}{\partial s \partial t}\Big|_{(s,t)=(0,0)} E(u_{s,t})$$

Theorem
Let u, v, w be as above. Then,

$$(\mathrm{Hess}\, E)_u(v,w) = \int_M \langle \nabla^*\nabla v + \underbrace{\mathrm{tr}_g(R^{u^*\nabla^{TN}}(\cdot, v)\cdot)}_{=\sum_{i=1}^m R^{\nabla^{TN}}(u_*e_i, v)u_*e_i,\ e_1,\dots,e_m \text{ o.n.f.}}, w\rangle \mu_g$$

Notice that this formula involves 2 derivatives of v and the curvature of N.

Definition
Define $J_u : \Gamma(u^*TN) \to \Gamma(u^*TN)$ to be

$$J_u(v) = \nabla^*\nabla v + \mathrm{tr}_g(R^{u^*\nabla^{TN}}(\cdot, v)\cdot)$$

This is called the Jacobi operator of the harmonic map u, which is linear. So the above theorem says $(\mathrm{Hess}\, E)_u(v,w) = \int_M \langle J_u(v), w\rangle \mu_g = \langle\langle J_u(v), w\rangle\rangle$. Then J_u is formally self-adjoint with respect to the L^2 metric, which is equivalent to Hess_u being symmetric by the above theorem:

$$(\mathrm{Hess}\, E)_u(v,w) = \langle\langle \nabla v, \nabla w\rangle\rangle + \int_M \left(\sum_{i=1}^m R^N(u_*e_i, v, u_*e_i, w)\right)\mu_g$$

which we see is symmetric in v, w by symmetries of the Riemann curvature tensor.
Moreover, J_u is elliptic.

Proof of Theorem:
Define $E(s,t) = E(u_{s,t}) = \frac{1}{2}\int_M |du_{s,t}|^2 \mu_g$, so that because the metric and volume form are parallel,

$$\frac{\partial E}{\partial s} = \int_M \langle \nabla_{\frac{\partial}{\partial s}} du, du\rangle \mu_g$$

where we write $u = u_{s,t}$. Now, when we derived the first variation formula, we saw

$$(\nabla_{\frac{\partial}{\partial s}} du)(X) = \nabla_{\frac{\partial}{\partial s}}(du \cdot X) - du(\cancelto{0}{\nabla_{\frac{\partial}{\partial s}} X}) = \nabla_X (du \cdot \frac{\partial}{\partial s})$$

where the last equality is because of what we proved on Assignment 1, since $[X, \frac{\partial}{\partial s}]$, so that $\nabla_{\frac{\partial}{\partial s}} du = \nabla \frac{\partial u}{\partial s}$. Thus

$$\frac{\partial E}{\partial s} = \int_M \langle \nabla \frac{\partial u}{\partial s}, du \rangle \mu_g$$

Thus, since $\nabla_{\frac{\partial}{\partial t}} du = \nabla \frac{\partial u}{\partial t}$ in similar fashion, we have

$$\frac{\partial^2 E}{\partial t \partial s} = \int_M \left(\langle \nabla_{\frac{\partial}{\partial t}} \nabla \frac{\partial u}{\partial s}, du \rangle + \langle \nabla \frac{\partial u}{\partial s}, \nabla \frac{\partial u}{\partial t} \rangle \right) \mu_g$$

At $(s,t) = (0,0)$, the right term becomes $\langle\langle \nabla v, \nabla w \rangle\rangle$. For the left term, $\int_M \langle \nabla_{\frac{\partial}{\partial t}} \nabla \frac{\partial u}{\partial s}, du \rangle \mu_g$, consider $\nabla_{\frac{\partial}{\partial t}} \nabla \frac{\partial u}{\partial s}$: for X a vector field,

$$\nabla_{\frac{\partial}{\partial t}} \nabla_X = \nabla_X \nabla_{\frac{\partial}{\partial t}} + R(\frac{\partial}{\partial t}, X) + \cancel{\nabla_{[\frac{\partial}{\partial t}, X]}}$$

so since

$$R(\frac{\partial}{\partial t}, X) = R^N(u_* \frac{\partial}{\partial t}, u_* X) = R^N(\frac{\partial u}{\partial t}, u_* X)$$

applying this to our integrand, we have

$$\nabla_{\frac{\partial}{\partial t}} \nabla \frac{\partial u}{\partial s} = \nabla \nabla_{\frac{\partial}{\partial t}} \frac{\partial u}{\partial s} + R^N(\frac{\partial u}{\partial t}, u_* \cdot) \frac{\partial u}{\partial s}$$

so the first term becomes

$$\int_M \left(\langle \nabla \underbrace{\nabla_{\frac{\partial}{\partial t}} \frac{\partial u}{\partial s}}_{\in \Gamma(u^*TN) = \Omega^0_E}, du \rangle + \langle R^n(\frac{\partial u}{\partial t}, u_* \cdot) \frac{\partial u}{\partial s}, du \cdot \rangle \right) \mu_g$$

Since $\nabla = d^\nabla$ on vector-valued 0-forms, taking formal adjoints, the first term is

$$\langle \ldots, (d^\nabla)^* du \rangle \mu_g$$

which is 0 at $(s,t) = (0,0)$ since $u_{0,0}$ is harmonic. For the second term at $(s,t) = (0,0)$, we have, by the symmetries of the curvature tensor,

$$\int_M \langle R^N(w, u_* \cdot) v, u_* \cdot \rangle \mu_g = \int_M g^{ab} \langle R^N(w, u_* e_a) v, u_* e_b \rangle \mu_g$$

so by symmetries of the curvature tensor, we get the second term of the desired integrand, as desired. ■

Stability and Instability

Since J_u is second-order linear self-adjoint elliptic on $\Gamma_{L^2}(u^*TN)$, by the spectral theorem $\Gamma(u^*TN)$ splits as the L^2-orthogonal direct sum of the eigenspaces J_u, where each eigenspace is finite-dimensional and consists of smooth sections (by elliptic regularity), and J_u has real, countable point-spectrum that only accumulates at $+\infty$. This allows us to define:

Definition
Let u be a harmonic map. The index of u is the dimension of the largest subspace of $\Gamma(u^*TN)$ on which $(\operatorname{Hess} E)_u$ is negative definite.
This is well-defined, because seeing smooth sections as contained in $\Gamma_{L^2}(u^*TN)$, with respect to the above direct sum, an eigenspace on which it is negative definite is one where J_u has a negative eigenvalue, of which there are only finitely many.
The nullity of u is $\dim\ker J_u$.

Definition
A harmonic map u is called stable (up to second order) if $\operatorname{ind}(u) = 0$, if and only if $(\operatorname{Hess} E)_u(v, v) \geq 0$ for all $v, w \in \Gamma(T^*N)$.

Remark
This could be misleading. The map u could be stable, but it's possible that for some variation $v = \frac{\partial u}{\partial s}|_{s=0}$, we have $\frac{\partial^k u_s}{\partial s^k}|_{s=0} < 0$ for some $k \geq 3$, which would prevent u from being a "local minimum".

Example
Let $u : (S^1, g_{round}) \to (N, h)$ be a geodesic (and thus harmonic). Then $J_u v = 0$ if and only if

$$-\nabla^N_{u'}\nabla^N_{u'}v + R(u', v)u' = 0$$

which is the classical Jacobi field equation from Riemannian geometry (v is a Jacobi field of u if and only if it satisfies this).

Proposition
Let $u : M \to N$ be harmonic and suppose $R^N \leq 0$ has nonpositive sectional curvature (i.e. $R^N(V, W, W, V) \leq 0$ for all $V, W \in \Gamma(TN)$). Then u is stable.
Proof:
Integrating the second variation formula by parts, we obtain

$$(\operatorname{Hess} E)_u(v, v) = ||\nabla v||^2_{L^2} + \sum_{i=1}^m \int_M \underbrace{R^N(u_*e_i, v, u_*e_i, v)}_{=-R^N(u_*e_i,v,v,u_*e_i)\geq 0} \mu_g \geq 0$$

and so $\operatorname{ind}(u) = 0$. ■

Example
Consider a constant map $u : M \to N$, $u(p) = q$ for all $p \in M$. Then u is harmonic. This is an absolute minimum of the energy, so we expect it to be stable. We'll verify this and compute the nullity.
First note that $\Gamma(u * TN) = \{V : M \to TN, V_p \in T_{u(p)}N = T_qN\}$. In particular, letting $v_1, \ldots, v_n$ be a orthonormal basis of T_qN, extend them to a local coordinate frame $\widetilde{v}_1, \ldots, \widetilde{v}_n$ in a neighbourhood W of q for geodesic normal coordinates centered at q.
Now, define $V_1, \ldots, V_n \in \Gamma(u^*TN)$ by $V_i|_p = v_i = \widetilde{v}_i|_{u(p)}$ for all $p \in M$, so that $V_i = u^*\widetilde{v}_i$ on $u^{-1}(W)$. But the V_i's are a global frame for u^*TN, giving an isomorphism $u^*TN \cong M \times T_qN$, so that we can write $\Gamma(u^*TN) = \{\sum_{i=1}^n f_i v_i : f_i \in \mathcal{C}^\infty(M)\}$. We can construct a smooth variation u_t of u corresponding to $V = \sum_i f_i v_i$ by $u_t(p) = \exp_q(tV_p)$, where exp is the Riemannian exponential map of (N, h) at q. Then $u_0(p) = q$ for all p, so that $u_0 = u$. Then this is a variation of u and $\frac{d}{dt}\Big|_{t=0} u_t = V$.
Let's compute $J_u(V)$. Let $e_1, \ldots, e_m$ be a local orthonormal frame for TM. Then $(u_*)_p : T_pM \to T_qN$ is the zero map, so that the second term in the Jacobi operator is zero, $\sum_{i=1}^n R^N(u_*e_i, V)u_*e_i = 0$ so that

$$\langle\langle J_u(V), V\rangle\rangle = ||\nabla V||^2_{L^2} \geq 0$$

showing u is stable as expected.

Because $\widetilde{v}_1, \ldots, \widetilde{v}_n$ of TN are normal coordinates, $\nabla v_j = 0$ at q for all j, so that

$$\nabla_X V = \sum_{i=1}^n \nabla_X(f_i V_i) = \sum_{i=1}^n ((Xf_i)V_i + f_i \nabla_X V_i)$$

But unentangling our notation, by normality we obtain

$$\nabla_X V_i = (u^*\nabla^{TN})_X(u^*\widetilde{v}_i) = \nabla_{u_* X} v_i = 0$$

so that

$$\nabla_X V = \sum_{i=1}^n (Xf_i)V_i$$

However

$$\nabla^*\nabla V = -g^{ab}\nabla_a\nabla_b V = -\sum_i g^{ab}(\nabla_a\nabla_b f_i)V_i = \sum_i (\Delta_g f_i)V_i$$

Thus $J_u(V) = 0$ if and only if $\Delta_g f_i = 0$ for all i so (if M connected), since the only harmonic functions on a compact connected manifold are constants, we get $\operatorname{nullity}(u) = n = \dim N$.

Example
Let $I : M \to M$ be the identity. Then in our above setup, $\Gamma(I^*TN) = \Gamma(TM)$. Then given $X \in \Gamma(TM)$,

$$J_I(X) = \nabla^*\nabla X + \sum_{i=1}^m R^M(e_i, X)e_i = \nabla^*\nabla X - \operatorname{Ric} X$$

But by the Weitzenböck formula,

$$\Delta_d X = \nabla^*\nabla X + \operatorname{Ric} X$$

In particular,

$$J_I(X) = \Delta_d X - 2\operatorname{Ric} X$$

This implies:

Theorem (on Assignment 4)
Suppose (M^m, g) is compact, oriented, and Einstein with Einstein constant C. Then:
(1) The identity map $I : M \to M$ is stable if and only if the first positive eigenvalue λ_1 of Δ_d on functions satisfies $\lambda_1 \geq 2C$.
(2) The nullity of I is

$$\dim(\operatorname{Isom}(M, g)) + \dim(\{f \in \mathcal{C}^\infty(M) : \Delta_d f = 2Cf\})$$

Continuing with our study of the identity map,

Lemma
For $X \in \Gamma(TM)$,

$$\int_M \langle J_I X, X\rangle \mu_g = \int_M \left(\frac{1}{2}|\mathcal{L}_X g|^2 - (\operatorname{div} X)^2\right)\mu_g$$

Proof:
By the formula $J_I(X) = \Delta_d X - 2\operatorname{Ric} X$, we have

$$\text{LHS} = \int_M \langle \nabla^*\nabla X - \operatorname{Ric} X, X\rangle \mu_g \underbrace{=}_{A3} \int_M \langle 2\operatorname{div}\operatorname{div}^* X - dd^* X, X\rangle \mu_g$$

$$= \int_M \left(2|\operatorname{div}^* X|^2 - |d^* X|^2\right)\mu_g = \int_M \left(\frac{1}{2}|\mathcal{L}_X g|^2 - (\operatorname{div} X)^2\right) \quad \blacksquare$$

A quick digression on Killing vector fields:

Recall
$X \in \Gamma(TM)$ is called a Killing vector field if $\mathcal{L}_X g = 0$ if and only if its flow Φ_t acts by isometries. More generally, we can consider a "conformal Killing" vector field X such that its flow Φ_t acts by conformal isometries: $\Phi_t^* g = e^{2f_t} g$ for $f_t \in \mathcal{C}^\infty(M), f_0 = 0$. Differentiating this equality, we get

$$\mathcal{L}_X g = \frac{d}{dt}\Big|_{t=0} (\Phi_t^* g) = 2\frac{df_t}{dt} e^{2f_t} g\Big|_{t=0} = 2Fg$$

where $F = \frac{df_t}{dt}(0)$. Suppose $\mathcal{L}_X g = 2Fg$. Then in a local frame,

$$\nabla_i X_j + \nabla_j X_i = 2F g_{ij}$$

taking the trace, we obtain

$$2 \operatorname{div} X = 2Fm$$

so $F = \frac{1}{m} \operatorname{div} X$. Thus, $X \in \Gamma(TM)$ is "conformal Killing" if and only if $\mathcal{L}_X g = \frac{2}{m}(\operatorname{div} X) g$.

Proposition
If $m = \dim M \geq 3$, then $\operatorname{ind}(I) \geq \dim \left(\frac{\text{conformal Killing vector fields}}{\text{Killing vector fields}} \right)$.
Proof:
Let X be a conformal Killing, but not Killing, vector field. So $\mathcal{L}_X g = \frac{2}{m}(\operatorname{div} X) g$ but $\operatorname{div} X \neq 0$. Thus, since $|g|^2 = m$,

$$|\mathcal{L}_X g|^2 = \frac{4}{m^2}(\operatorname{div} X)^2 m = \frac{4}{m}(\operatorname{div} X)^2$$

so by the Lemma,

$$\int_M \langle J_I X, X \rangle \mu_g = \int_M \left(\frac{2}{m}(\operatorname{div} X)^2 - (\operatorname{div} X)^2 \right) \mu_g = \frac{2-m}{m} \int_M (\operatorname{div} X)^2 \mu_g \overbrace{<}^{m>2} 0$$

since $\operatorname{div} X \neq 0$. Thus, if $X_1, \ldots, X_d$ are linearly independent in conformal Killing/Killing, we see $(\operatorname{Hess} E)_I$ is negative definite on their span, which shows the desired inequality. ■

Remark
The same argument shows $\operatorname{nullity}(I) \geq \dim(\text{Killing vector fields})$.

Example
Let (L, g) be any compact oriented Riemannian manifold. Define $M = S^1 \times L$, $g = f^2(d\theta^2 + g_L)$, $f \in \mathcal{C}^\infty(S^1)$ is nonconstant. Consider $X = \frac{\partial}{\partial \theta}$. Then $\Phi_t(\theta, p) = (\theta + t, p)$, so that $\Phi_t^* d\theta = d\theta$ and $\Phi_t^* g_L = g_L$, so that

$$\Phi_t^*(g) = \Phi_t^*(f(\theta)^2(d\theta^2 + g_L)) = f^2(\theta + t)(d\theta^2 + g_L)$$

so since f is non-constant, $\Theta_t^* \neq \operatorname{Id}$ for some t's. However, since

$$f^2(\theta + t)(d\theta^2 + g_L) = \frac{f^2(\theta + t)}{f^2(\theta)} f^2(\theta)(d\theta^2 + g_L)$$

we see X is conformal Killing but not Killing. Thus $\operatorname{ind}(I) \geq 1$, so $I : S^1 \times M \to S^1 \times M$ is not stable.

Here we will have two instability theorems for harmonic maps from/to spheres:

Theorem A
Let $u : (S^m, g_{round}) \to (N, h)$ be a non-constant harmonic map with $m \geq 3$. Then $\operatorname{ind}(u) > 0$, so u is

not stable.

Theorem B
Let $u : (M^m, g) \to (S^n, h_{round})$ with M compact, oriented, be a non-constant harmonic map with $n \geq 3$. Then $\operatorname{ind}(u) > 0$.

Lemma
Let $f : \mathbb{R}^{m+1} \to \mathbb{R}$ be linear, say $f(x) = a_i x^i$ for some $a_1, \ldots, a_{m+1} \in \mathbb{R}$. Define $X_f = (df)^\sharp = \sum_{i=1}^{m+1} a_i \frac{\partial}{\partial x^i}$. Let X_f^T, $X_f^\perp$ be the tangent and normal parts of X_f on S^m. Let $\nu = x^i \frac{\partial}{\partial x^i}$ be the outward unit normal vector field on S^m. Then, for any $Y \in \Gamma(TS^m)$,

$$\nabla_Y^{S^m}(X_f^T) = -\langle X_f, \nu \rangle Y = -fY$$

Proof:
We calculate

$$\nabla_Y^{S^m}(X_f^T) = \left(\nabla_Y^{\mathbb{R}^{m+1}} X_f^T\right)^T = \left(\nabla_Y^{\mathbb{R}^{m+1}}(X_f - \langle X_f, \nu\rangle \nu)\right)^T$$

$$= \left(\overbrace{\nabla_Y^{\mathbb{R}^{m+1}} X_f}^{=0 \text{ since } X_f \text{ constant}} - \overbrace{Y(\langle X_f, \nu\rangle)\nu}^{=0 \text{ when tangential projecting}} - \langle X_f, \nu\rangle \nabla_Y^{\mathbb{R}^{m+1}} \nu \right)^T = -\left(\langle X_f, \nu\rangle \nabla_Y^{\mathbb{R}^{m+1}} \nu\right)^T$$

However, we know

$$\nabla_Y^{\mathbb{R}^{m+1}} \nu = Y^i \frac{\partial}{\partial x^i}(x^j) \frac{\partial}{\partial x^j} = Y^j \frac{\partial}{\partial x^j} = Y$$

so that we get

$$\nabla_Y^{S^m} X_f^T = -\langle X_f, \nu\rangle Y = -fY \quad \blacksquare$$

Proof of Theorem A:
We want to find $v \in \Gamma(u^*TN)$ such that $(\operatorname{Hess} E)_u(v, v) < 0$. Let f, X_f^T be as in the lemma. Let $v = u_*(X_f^T) \in \Gamma(u^*TN)$. We compute

$$J_u(v) = \nabla^*\nabla v + \operatorname{tr}_g(R^N(u_*\cdot, v)u_*\cdot)$$

Fix $p \in S^m$ and pick geodesic normal coordinates $x^1, \ldots, x^m$ centered at p. Using the lemma, at p,

$$\nabla^*\nabla v = -\sum_{i=1}^m \nabla_i \nabla_i (u_* X_f^T) = -\sum_{i=1}^m \nabla_i \left((\nabla_i u_*) X_f^T + u_* \underbrace{\nabla_i X_f^T}_{=-f\partial_i \text{ by lemma}} \right)$$

so that

$$\nabla^*\nabla v = -\sum_{i=1}^m \left(\nabla_{\partial_i}\nabla_{\partial_i} u_*\right) X_f^T - \sum_{i=1}^m (\nabla_{\partial_i} u_*)(\nabla_{\partial_i} X_f^T) + \sum_{i=1}^m \nabla_{\partial_i}(u_*(f\partial_i)) \quad (\star)$$

Since u is harmonic, at p we have

$$\sum_{i=1}^m (\nabla du)(\partial_i, \partial_i) = 0 \quad (\sharp)$$

Computing the third term of $(\star)$, we get

$$\sum_i \nabla_{\partial_i}(u_* f \partial_i) = \sum_i (\nabla_{\partial_i} f)(u_* \partial_i) + \underbrace{\sum_i f \nabla_{\partial_i}(u_* \partial_i)}_{=0 \text{ by } (\sharp)}$$

$$= \sum_i u_*((\nabla_{\partial_i} f)\partial_i) = u_*((df)^\sharp)^T = u_*(X_f^T)$$

For the second term of $(\star)$, by the lemma,

$$-\sum_i (\nabla_{\partial_i} u_*)(\underbrace{\nabla_{\partial_i} X_f^T}_{=-f\partial_i}) = \sum_i (\nabla_{\partial_i} u_*)(f\partial_i) = \sum_i \nabla_{\partial_i}(u_* f\partial_i) + u_*(\underbrace{\nabla_{\partial_i}(f\partial_i)}_{=(\nabla_{\partial_i} f)\partial_i = X_f^T \text{ by normal coords and previous calculation}})$$

$$= -\sum_i u_*((\nabla_{\partial_i} f)\partial_i) + u_* X_f^T = -u_* X_f^T + u_* X_f^T = 0$$

so the second term vanishes. Thus,

$$\nabla^*\nabla v = -\sum_i (\nabla_{\partial_i}\nabla_{\partial_i} u_*) X_f^T + u_* X_f^T$$

By the Weitzenböck formula and the harmonicity of u, we know $\nabla^*\nabla(u_*) = \Delta_{d^\nabla} u_* - \rho(u_*) = -\rho(u_*)$, so that

$$-\sum_i (\nabla_{\partial_i}\nabla_{\partial_i} u_*)(X_f^T) = -\rho(u_*)(X_f^T) = -g^{ij}((R(e_i, X_f^T)u_*)(e_i))$$

$$= -g^{ij}(R^{u^*TN}(e_i, X_f^T)(u_* e_i) - u_*(R^{TM}(e_i, X_f^T)e_i)) = -g^{ij} R^N(u_* e_i, u_* X_f^T)(u_* e_i) + g^{ij} u_* R^M(e_i, X_f^T) e_i$$

Substituting with $v = u_* X_f^T$,

$$J_u(v) = \nabla^*\nabla v + \sum_{i=1}^m R^N(u_* e_i, v) u_* e_i$$

$$= g^{ij} R^N(u_* e_i, u_* X_f^T)(u_* e_i) - u_* \operatorname{Ric}(v) + u_* X_f^T + \sum_{i=1}^m R^N(u_* e_i, v) u_* e_i = -u_* \operatorname{Ric}(v) + u_* X_f^T$$

For the round sphere, we know $\operatorname{Ric} = (m-1) g_{round}$, so that

$$J_u(v) = (2-m)v$$

Since u is non-constant, there exists $p \in S^m$, $(u_*)_p \neq 0$ so that there exists $Y_p \in T_p S^m$ such that $(u_*)_p Y_p \neq 0$, so there exists a linear functional f such that $Y_p = (X_f^T)_p$, so in a neighbourhood of p, $u_* X_f^T \neq 0$. Thus

$$(\operatorname{Hess} E)_u(v,v) = \langle\langle J_u(v), v\rangle\rangle = (2-m)||u_* X_f^T||^2 < 0$$

as desired. TODO THERE ARE A BILLION TYPOS ■

Proof of Theorem B:
Let $X_f^T \in \Gamma(TS^n)$ be as in the lemma. Let $v = u^*(X_f^T) \in \Gamma(u^*TN)$. By the second variation formula,

$$(\operatorname{Hess} E)_u(v,v) = \int_M \left(|\nabla v|^2 + \sum_{i=1}^m \langle R^{S^n}(u_* e_i, X_f^T), u_* e_i, X_f^T\rangle \right) \mu_g$$

However,

$$|\nabla v|^2 = \sum_{i=1}^m |\nabla_{e_i}^{u^*TS^n}(u^* X_f^T)|^2 = \sum_{i=1}^m |\nabla_{u_* e_i}^{S^n} X_f^T|^2$$

By the lemma, with $Y = u_* e_i$, we have

$$\nabla_{u_* e_i}^{S^n} X_f^T = -f u_* e_i = -\langle X_f, \nu\rangle u_* e_i$$

so that

$$|\nabla v|^2 = \sum_{i=1}^n \langle X_f, \nu\rangle^2 |u_* e_i|^2$$

For the curvature term, since the sphere has constant sectional curvature $+1$,

$$\langle R^{S^n}(u_*e_i, X_f^T)u_*e_i, X_f^T\rangle = \langle u_*e_i, X_f^T\rangle^2 - |u_*e_i|^2|X_f^T|^2$$

Thus,

$$(\mathrm{Hess})_u(v,v) = \int_M \left(\sum_{i=1}^m \left(|u_*e_i|^2\left(\langle X_f,\nu\rangle^2 - |X_f^T|^2\right) + \langle u_*e_i, X_f^T\rangle^2\right)\right)\mu_g$$

Consider $n+1$ functions $f_j(x) = x^j$, so that $X_j := X_{f_j} = \frac{\partial}{\partial x^j}$, which collectively form a global orthonormal frame for $\mathbb{R}^{n+1}$. Let $v_j = u^*X_j^T$. We want to compute

$$\sum_{j=1}^{n+1}(\mathrm{Hess}\, E)_u(v_j,v_j)$$

Since u_*e_i is tangent to S^n, we have

$$\sum_{j=1}^{n+1}\langle u_*e_i, X_j^T\rangle^2 = \sum_{j=1}^{n+1}\langle u_*e_i, X_j\rangle^2 = |u_*e_i|^2$$

since $X_1,\ldots,X_{n+1}$ are orthonormal.
We can also look at

$$\sum_{j=1}^{n+1}(\langle X_j,\nu\rangle^2 - |X_j^T|^2) = \sum_{j=1}^{n+1}(|X_j^\perp|^2 - (|X_j|^2 - |X_j^\perp|^2))$$

$$= 2\sum_{j=1}\langle X_j,\nu\rangle^2 - \sum_{j=1}^{n+1}|X_j|^2 = 2|\nu|^2 - (n+1) = 2-(n+1) = 1-n$$

Hence, putting all the above together

$$\sum_{j=1}^{n+1}(\mathrm{Hess}\, E)_u(v_j,v_j) = \sum_{i=1}^m\int_M (2-n)|u_*e_i|^2\mu_g = (2-n)\int_M\sum_{i=1}^m|u_*e_i|^2\mu_g$$

$$= (2-n)\int_M |du|^2\mu_g = (2-n)2E(u)$$

However, $E(u)>0$ since u is non-constant, and since $n>2$, we get $\sum_{j=1}^{n+1}(\mathrm{Hess}\, E)_u(v_j,v_j) < 0$, so at least one term is negative, so that desired v_j is the one we need. Thus $\mathrm{ind}(u)>0$. ■

Stress-Energy Tensor

Definition
Let $u:(M^m,g)\to(N^n,h)$ be a smooth map between Riemannian manifolds. Let $e = e(u) = \frac{1}{2}|du|^2$ denote the energy density, so that $E(u) = \int_M e(u)\mu_g$ when M is compact oriented. The stress-energy tensor of u, denoted S_u, is a section of $\mathcal{S}^2(T^*M)$, and is defined to be

$$S_u = e(u)g - u^*h$$

(notice this is quadratic in derivatives of u).

Theorem
Let M be compact and oriented.

$$\underbrace{\mathrm{div}(S_u)}_{\in\Gamma(T^*M)} = -\langle\underbrace{\tau(u)}_{\in\Gamma(u^*TN)}, \overbrace{du}^{\in\Gamma(T^*M\otimes u^*TN)}\rangle$$

where the inner product means we take the inner product on the u^*TN part and leave the T^*M part left-over.

To prove this, recall

$$\mathcal{L}_X\mu_g = (\operatorname{div} X)\mu_g = \frac{1}{2}\operatorname{tr}_g(\nabla_i X_j + \nabla_j X_i)\mu_g = \frac{1}{2}\langle \mathcal{L}_X g, g\rangle\mu_g$$

Lemma

$$\mathcal{L}_X e = \langle du, \nabla(du \cdot X)\rangle - \frac{1}{2}\langle \mathcal{L}_X g, u^*h\rangle$$

Proof:
We calculate

$$\mathcal{L}_X e = X(e) = X\left(\frac{1}{2}\langle du, du\rangle\right) = \langle \nabla_X du, du\rangle$$

$$= \sum_{i=1}^m \langle (\nabla_X du)(e_i), du(e_i)\rangle \underbrace{=}_{\nabla du \text{ symmetric}} \sum_{i=1}^m \langle (\nabla_{e_i} du)(X), du(e_i)\rangle = \langle (\nabla du)(X), du\rangle$$

$$= \langle \nabla(du \cdot X) - du(\nabla X), du\rangle$$

On the other hand,

$$\frac{1}{2}\langle \mathcal{L}_X g, u^*h\rangle = \frac{1}{2}\sum_{i,j}\big(\langle \nabla_{e_i} X, e_j\rangle + \langle \nabla_{e_j} X, e_i\rangle\big)\langle du\cdot e_i, du\cdot e_j\rangle = \sum_{i,j}\langle \nabla_{e_i}X, e_j\rangle\langle du\cdot e_i, du\cdot e_j\rangle$$

$$= \sum_i \langle du\cdot e_i, du\cdot (\sum_j \langle \nabla_{e_i}X, e_j\rangle e_j)\rangle = \sum_i \langle du\cdot e_i, du\cdot \nabla_{e_i}X\rangle = \langle du, du(\nabla X)\rangle$$

and so putting things together we obtain the lemma. ■

Lemma

$$\mathcal{L}_X(e\mu_g) = \langle du, \nabla(du\cdot X)\rangle\mu_g + \frac{1}{2}\langle \mathcal{L}_X g, S_u\rangle\mu_g$$

Proof:
By the identity above and the previous lemma,

$$\mathcal{L}_X(e\mu_g) = (\mathcal{L}_X e)\mu_g + e\mathcal{L}_X(\mu_g) = \langle du, \nabla(du\cdot X)\rangle\mu_g - \frac{1}{2}\langle \mathcal{L}_X g, u^*h\rangle\mu_g + e\frac{1}{2}\langle \mathcal{L}_X g, g\rangle\mu_g$$

$$= \langle du, \nabla(du\cdot X)\rangle\mu_g + \frac{1}{2}\langle \mathcal{L}_X g, \underbrace{eg - u^*h}_{=S_u}\rangle\mu_g$$

as desired. ■

Proof of Theorem:
We calculate

$$\frac{1}{2}\langle \mathcal{L}_X g, S_u\rangle = \sum_{i,j}\frac{1}{2}(\nabla_i X_j + \nabla_j X_i)(S_u)_{ij} = \sum_{i,j}\frac{1}{2}\nabla_i X_j (Su)_{ij} = \sum_{i,j}\langle \nabla X, S_u\rangle$$

On the other hand, by Cartan's formula,

$$\mathcal{L}_X(e\mu_g) = d(X \lrcorner (e\mu_g))$$

is exact, so by Stokes' theorem,

$$0 = \int_M \mathcal{L}_X(e\mu_g) = \int_M \langle du, \nabla(du \cdot X)\rangle \mu_g + \frac{1}{2}\int_M \langle \mathcal{L}_X g, S_u\rangle \mu_g$$
$$= \int_M \langle du, \nabla(du \cdot X)\rangle \mu_g + \int_M \langle \nabla X, S_u\rangle \mu_g$$

so since $\nabla^* = -\operatorname{div}$ from earlier,

$$0 = \int_M \langle -\operatorname{div}(du), du \cdot X\rangle \mu_g - \int_M \langle X, \operatorname{div}(S_u)\rangle \mu_g = -\int_M (\langle \tau(u), du \cdot X\rangle + \langle \operatorname{div} S_u, X\rangle)\mu_g$$
$$= \int_M \langle \langle \tau(u), du\rangle + \operatorname{div} S_u, X\rangle \mu_g$$

for all vector fields X, implying $\langle \tau(u), du\rangle + \operatorname{div}(S_u) = 0$. ■

Corollary

If u is harmonic, then its stress-energy tensor is divergence-free, $\operatorname{div}(S_u) = 0$ (sometimes called a "conservative tensor").

Corollary

Conversely, suppose $\operatorname{div}(S_u) = 0$ and that u is a smooth submersion almost everywhere (e.g. whose complement is dense). Then u is harmonic.

Proof:

Since $\operatorname{div}(S_u) = 0$, we have $\langle \tau(u), du \cdot X\rangle = 0$ for all X. At points p in M where u is a submersion, du at p is surjective, so that $\tau(u) = 0$ at p, so that $\tau(u) = 0$ on M by continuity. ■

Remark

The submersion almost everywhere condition in the above corollary is essential. For example, let $u : M \to N$ be any Riemannian immersion (not necessarily harmonic). Then, by Assignment 3,

$$e = \frac{1}{2}|du|^2 = \frac{1}{2}\langle u^* h, g\rangle \overbrace{=}^{\text{Riem. immersion}} \frac{1}{2}\langle g, g\rangle = \frac{1}{2}m$$

But $S_u = eg - u^*h = \frac{1}{2}mg - g = \left(\frac{m-2}{2}\right) g$ so that $\operatorname{div}(S_u) = 0$, but u need not be harmonic.

Remark

This observation is consistent with the following: suppose u is an isometry. Then u is harmonic, and a Riemannian immersion, so $\operatorname{div}(S_u) = 0$ by the example, and u is a Riemannian submersion, so by the corollary, u is harmonic.

Example

Let $M = S^1$, and let $u : (S^1, g_{round}) \to (N^n, h)$. Then we know u is harmonic if and only if it is a geodesic. Let's compute S_u. Then

$$(du)\left(\frac{\partial}{\partial \theta}\right) = u'(\theta) \in T_{u(\theta)}N = (u^*TN)_\theta$$

so that $|du|^2 = |u'|^2$, and $u^*h = |u'|^2 g$, so that

$$S_u = \frac{1}{2}|du|^2 g - u^*h$$

implies $S_u = -|u'|^2 g$. Thus

$$(\operatorname{div} S_u)_k = g^{ij}\nabla_i (S_u)_{jk} = -\nabla_i |u'|^2 g_{jk} g^{ij} = -\nabla_k |u'|^2$$

so $\operatorname{div}(S_u) = 0$ if and only if $|u'|^2$ is constant (i.e. constant-speed parameterization).
In particular, u is harmonic if and only if u is a geodesic, implying $|u'|^2$ is constant, so that $\operatorname{div}(S_u) = 0$, which is consistent with the first corollary.
Moreover, u is a Riemannian immersion if and only if $u^*h = g$ if and only if $|u'|^2 = 1$ (unit-speed parameterization), so $\operatorname{div}(S_u) = 0$ in this case (Riemannian immersion) regardless of whether u is a geodesic.

Weakly Conformal Maps and Harmonic Morphisms

Definition
A smooth map $u : (M, g) \to (N, h)$ is (weakly) conformal if

$$u^*h = fg$$

for some $f \in \mathcal{C}^\infty(M)$ (if $f = 1$, u is a Riemannian immersion).

Lemma
If u is weakly conformal, $f = \frac{1}{m}|du|^2 \geq 0$, so we write $f = \lambda^2$ with $\lambda = \frac{1}{\sqrt{m}}|du|$.
Proof:
$u^*h = fg$ in coordinates is

$$\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}h_{\alpha\beta} = fg_{ij}$$

Tracing both sides, using that $g^{ij}\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}h_{\alpha\beta} = |du|^2$ and $g^{ij}fg_{ij} = mf$, we obtain the result. ■

Corollary
If u is weakly conformal, then at every $p \in U$, either $(du)_p = 0$ or else $(du)_p : T_pM \to T_{u(p)}N$ is conformally injective, that is,

$$h_{u(p)}((du)_pX_p, (du)_pY_p) = \lambda^2(p)g_p(X_p, Y_p) \quad ■$$

Proposition
Let $u : (M^m, g) \to (N^n, h)$ be non-constant. Then $S_u = 0$ if and only if $m = 2$ and u is weakly conformal.
Proof:
Suppose $S_u = 0$. Then $u^*h = \frac{1}{2}|du|^2 g$, so by the lemma, $f = \frac{|du|^2}{m}$, so $m = 2$, and $u^*h = fg$, so u is weakly conformal.
Conversely, if u is weakly conformal, $u^*h = \frac{1}{2}|du|^2 g$, so $S_u = 0$. ■

Proposition
Let $m \neq 2$ and $u : (M^m, g) \to (N^n, h)$ be non-constant, harmonic, and weakly conformal. Then u is a homothety, that is, $u^*h = \lambda g$ for a positive constant λ.
Proof:

$$0 = \operatorname{div}(S_u) = \operatorname{div}\left(\left(\frac{1}{2} - \frac{1}{m}\right)|du|^2 g\right) = \left(\frac{1}{2} - \frac{1}{m}\right)d(|du|^2)$$

so that $|du|^2$ is constant if $m \neq 2$, and thus so is $\frac{|du|^2}{m}$. ■

In the $m = 1$ case, then $g = k\ \mathrm{d}x^2$ for some function k in local coordinates, so that $u^*h =$ (non-negative) $\mathrm{d}x^2 =$ (positive)g, so when $m = 1$, any u is weakly conformal.
Thus, in the $m = 1$ case, u is a geodesic if and only if u is harmonic, where the former implies u is constant-speed, and the latter implies u is a homothety by the above proposition, but we see in this case, constant-speed and homothety are the same thing ($|du|^2$ is constant).
Definition
Let $u : (M^m, g) \to (N^n, h)$ be smooth. We say u is a harmonic morphism if and only if whenever $V \subseteq N$

is open, $f \in \mathcal{C}^\infty(N)$ is harmonic (with respect to $h|_V$), then $u^*f = f \circ u \in \mathcal{C}^\infty(u^{-1}(V))$ is harmonic with respect to $g|_{u^{-1}(V)}$. Thus u is a harmonic morphism if it (locally) pulls back harmonic functions to harmonic functions (which is a local property).

Now, recall the decomposition $T_pM = V_p \oplus H_p$, where by definition of the horizontal space (orthogonal complement to the kernel), $(du)_p|_{H_p} : H_p \to T_{u(p)}N$ is injective.

Definition
We say u is weakly horizontally conformal if for all $p \in M$ such that $(du)_p \neq 0$, $(du)_p|_{H_p} : H_p \to T_{u(p)}N$ is a conformal isomorphism.

Remark
Being weakly horizontally conformal is equivalent to, for all $p \in M$ with $(du)_p \neq 0$, then $(du)_p$ is surjective onto $T_{u(p)}N$ (so u is a submersion at p) and $(du)_p : (H_p, g_p|_{H_p}) \to (T_{u(p)}N, h_{u(p)})$ is a conformal isomorphism (implying $m \geq n$).

As a special case, if $V_p = 0$ for all p with $(du)_p \neq 0$, then $H_p = T_pM$, and this says that $u^*h = \lambda^2 g$, in which case a weakly horizontally conformal map is weakly conformal.

When $m > n$, given the decompositions $T_pM = V_p \oplus H_p$, $T_p^*M = V_p^* \oplus H_p^*$, we have a decomposition

$$\mathcal{S}^2(T_p^*M) = \underbrace{\mathcal{S}^2(V_p^*)}_{(2,0)} \oplus \underbrace{(V_p^* \otimes H_p^*)}_{(1,1)} \oplus \underbrace{\mathcal{S}^2(H_p^*)}_{(0,2)}$$

and so we can write

$$u^*h = (u^*h)^{2,0} + (u^*h)^{1,1} + (u^*h)^{0,2}$$

With this in mind, a linear algebra aside:
Let V^m, W^n be real inner product spaces, $A : V \to W$ a nonzero linear map, and $A^* : W \to V$ the adjoint map (using the inner product), so that $\langle v, A^*w\rangle = \langle Av, w\rangle$ for all $v \in V, w \in W$. We say A is horizontally conformal if $A : \ker(A)^\perp \to W$ is a conformal isomorphism, i.e. there exists $\lambda > 0$ such that $\langle Ax, Ay\rangle = \lambda^2\langle x, y\rangle$ for all $x, y \in \ker(A)^\perp$.

Lemma
(1) A is horizontally conformal if and only if A^* embeds W conformally onto $\ker A$.
(2) In the case of (1), $AA^* = \lambda^2 I_W$.
(3) If $n = m$, then A is a conformal isomorphism.
Proof:
Suppose A is horizontally conformal. Then there exists an orthonormal basis $\underbrace{v_1, \ldots, v_n}_{\ker(A)^\perp}, \underbrace{v_{n+1}, \ldots, v_m}_{\ker A}$ for V, and an orthonormal basis $w_1, \ldots, w_n$ for W, such that $Av_i = \lambda w_i$ for $i = 1, \ldots, n$, $Av_j = 0$ for $j = n+1, \ldots, m$. Then $A = \begin{bmatrix}\lambda I & 0\end{bmatrix}$, $A^* = \begin{bmatrix}\lambda I \\ 0\end{bmatrix}$, so that $AA^* = \lambda^2 I$. ■

Back to Riemannian geometry.

Lemma
Suppose u is weakly horizontally conformal, so that $\langle du \cdot X, du \cdot Y\rangle = \lambda^2\langle X, Y\rangle$ for all $X, Y \in H_p$. Then $\lambda^2 = \frac{2e(u)}{n} = \frac{|du|^2}{n}$. (Contrast this with weakly conformal, where it's $\frac{|du|^2}{m}$.)
Proof:
Picking an orthonormal frame such that $e_1, \ldots, e_n$ span H_p and $e_{n+1}, \ldots, e_m$ span V_p, we have

$$|du|^2 = \sum_{j=1}^m \langle du \cdot e_j, du \cdot e_j\rangle = \sum_{j=1}^n \langle du \cdot e_j, du \cdot e_j\rangle = \sum_{j=1}^n \lambda^2\langle e_j, e_j\rangle = n\lambda^2$$

giving the result. ■

Theorem
A smooth map $u : (M^m, g) \to (N^n, h)$ is a harmonic morphism if and only if it is a harmonic map and weakly horizontally conformal. In particular, harmonic morphisms don't exist unless $m \geq n$.

To prove this, we need an analytic lemma:

Lemma
Let (N, h) be Riemannian. Let $q \in N$. Let $(y^1, \dots, y^n)$ be geodesic normal coordinates centered at q defined on an open neighbourhood V of q. Suppose we have $c_{\alpha\beta} \in \mathbb{R}$ for $1 \leq \alpha, \beta \leq n$ such that $c_{\alpha\beta} = c_{\beta\alpha}$ and $\sum_{\alpha=1}^n c_{\alpha\alpha} = 0$. Then, there exists a harmonic function f on a (possibly smaller) open neighbourhood $\widetilde{V} \subseteq V$ of q such that

$$\frac{\partial f}{\partial y^\alpha}(q) = c_\alpha, \frac{\partial^2 f}{\partial y^\alpha \partial y^\beta}(q) = c_{\alpha\beta}$$

Idea of Proof:
If h is the Euclidean metric, then we can just take the polynomial $f = \sum_\alpha c_\alpha y^\alpha + \frac{1}{2}\sum_{\alpha,\beta} c_{\alpha\beta} y^\alpha y^\beta$, where then $\Delta f = \sum_\alpha c_{\alpha\alpha} = 0$. If h is not Euclidean, near q (because of normal coordinates) it's close to Euclidean, so one can argue by a contraction mapping/fixed point argument that there exists such a f. ■

Proof of Theorem
Suppose u is a harmonic morphism. Recall the composition formula

$$\tau(v \circ u) = dv \circ \tau(u) + \mathrm{tr}_g(\nabla dv(du\cdot, du\cdot))$$

Fixing $p \in M$, let $q = u(p) \in N$. Now, in the composition formula, if we have $v = f : N \to \mathbb{R}$ harmonic on some open neighbourhood V of q, since u is a harmonic morphism, $u^* f = f \circ u$ is a harmonic function, so that in the composition formula, we have

$$0 = df \cdot \tau(u) + \mathrm{tr}_g((\nabla df)(du\cdot, du\cdot))$$

where applying the lemma with f such that $\frac{\partial f}{\partial y^\alpha}(q) = \delta_{\alpha\beta}$, $\frac{\partial^2 f}{\partial y^\alpha \partial y^\beta} = 0$ for $\beta = 1, \dots, n$, inserting this into the composition formula, we get

$$0 = \frac{\partial f}{\partial y^\alpha}(q) \mathrm{d}y^\alpha|_q (\tau(u) \frac{\partial}{\partial y^\beta}|_q) + \frac{\partial^2 f}{\partial y^\alpha \partial y^\beta}(q) \frac{\partial u^\alpha}{\partial x^i}(q) \frac{\partial u^\beta}{\partial x^j}(q) g^{ij}$$

we get $\tau(u)^\beta = 0$ at p for $\beta = 1, \dots, n$, so that $\tau(u) = 0$.
To show u is weakly horizontally conformal, we apply the lemma again with $c_\alpha = 0$, and any choice of $c_{\alpha\beta}$'s with $c_{\alpha\beta} = c_{\beta\alpha}, \sum_{\alpha=1}^n c_{\alpha\alpha} = 0$. Then the composition formula becomes

$$0 = \underbrace{df \cdot \tau(u)}_{=0,\ u \text{ harmonic}} + \mathrm{tr}_g((\nabla df)(du\cdot, du\cdot))$$

wherever $u^* f$is defined, so that at p,

$$0 = c_{\alpha\beta} \frac{\partial u^\alpha}{\partial x^i}(p) \frac{\partial u^\beta}{\partial x^j}(p) g_p^{ij}$$

$$= \sum_{\alpha \neq \beta} c_{\alpha\beta} \frac{\partial u^\alpha}{\partial x^i}(p) \frac{\partial u^\beta}{\partial x^j}(p) g_p^{ij} + \sum_{\alpha \neq 1} c_{\alpha\alpha} \frac{\partial u^\alpha}{\partial x^i}(p) \frac{\partial u^\alpha}{\partial x^j}(p) g_p^{ij} + c_{11} \frac{\partial u^1}{\partial x^i}(p) \frac{\partial u^1}{\partial x^j}(p) g_p^{ij}$$

But $\sum_{\alpha\neq 1} c_{\alpha\alpha} = -c_{11}$, so that

$$0 = \sum_{\alpha\neq\beta} c_{\alpha\beta}\frac{\partial u^\alpha}{\partial x^i}(p)\frac{\partial u^\beta}{\partial x^j}(p)g_p^{ij} + \sum_{\alpha=1}^{n} c_{\alpha\alpha}\left(\frac{\partial u^\alpha}{\partial x^i}(p)\frac{\partial u^\alpha}{\partial x^j}(p) - \frac{\partial u^1}{\partial x^i}(p)\frac{\partial u^1}{\partial x^j}(p)\right)g^{ij}$$

Now, there are several cases we can consider, since the $c_{\alpha\beta}$'s were arbitrary. In the first case, consider $c_{\alpha\alpha} = 0$ for all α, $c_{\alpha\beta} = c_{\beta\alpha} = 1$ for a fixed choice of α, β, and the rest of the $c_{\alpha\beta}$'s 0. Then we get

$$\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}g^{ij} = 0$$

for $\alpha \neq \beta$. In the second case, consider $c_{\alpha\beta} = 0$ if $\alpha \neq \beta$, $c_{11} = 1$, $c_{\alpha\alpha} = -1$ for some $\alpha \neq 1$, and the rest of the $c_{\alpha\alpha}$'s 0. Then we get

$$\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\alpha}{\partial x^j}g^{ij} = \frac{\partial u^1}{\partial x^i}\frac{\partial u^1}{\partial x^j}g^{ij}$$

Together, these imply

$$\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}g^{ij} = \lambda^2(p)\underbrace{h_q^{\alpha\beta}}_{=\delta^{\alpha\beta},\text{ in normal coords}}$$

Multiplying both sides by $\delta_{\alpha\beta}(q) = h_{\alpha\beta}(q)$ and summing, we get

$$|du|^2 = \frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}g^{ij}h_{\alpha\beta} = n\lambda^2$$

so $|(du)_p|^2 = n\lambda(p)^2$ for all $p \in M$. If $\lambda(p) = 0$, then $(du)_p = 0$. Otherwise, if $\lambda(p) \neq 0$, then letting $A = (du)_p : T_pM \to T_qN$, and writing $A_i^\alpha = \frac{\partial u^\alpha}{\partial x^i}(p)$, and $A^*\frac{\partial}{\partial y^\beta} = B_\beta^i\frac{\partial}{\partial x^i}$, then

$$\langle A\frac{\partial}{\partial x^i}, \frac{\partial}{\partial y^\beta}\rangle = \sum_\alpha \frac{\partial u^\alpha}{\partial x^i}h_{\alpha\beta} = \sum_\alpha \frac{\partial u^\alpha}{\partial x^i}\delta_{\alpha\beta} = \frac{\partial u^\beta}{\partial x^i} = \langle \frac{\partial}{\partial x^i}, A^*\frac{\partial}{\partial y^\beta}\rangle$$

$$= \langle \frac{\partial}{\partial x^i}, B_\beta^j\frac{\partial}{\partial x^j}\rangle = B_\beta^j g_{ij}$$

implying $\frac{\partial u^\beta}{\partial x^i}g^{ik} = B_\beta^k$. Moreover,

$$\langle A^*\frac{\partial}{\partial y^\alpha}, A^*\frac{\partial}{\partial y^\beta}\rangle = B_\alpha^k B_\beta^\ell g_{k\ell}$$

$$= \frac{\partial u^\alpha}{\partial x^i}g^{ik}\frac{\partial u^\beta}{\partial x^j}g_{k\ell} = \frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}g^{ij} = \lambda^2\delta_{\alpha\beta} = \lambda^2\langle\frac{\partial}{\partial y^\alpha}, \frac{\partial}{\partial y^\beta}\rangle$$

Thus, when $\lambda(p) \neq 0$, the dual map A^* is a conformal injection, so by the linear algebra lemma above, A is horizontally conformal, so u is weakly horizontal conformal.
Conversely, suppose u is a harmonic map and weakly horizontally conformal. Let $V \subseteq N$ be open and $f \in \mathcal{C}^\infty(V)$ be harmonic. Fix $p \in u^{-1}(V)$, so $q = u(p) \in V$. Then, letting $x^1, \ldots, x^m$ be normal coordinates centered at p and $y^1, \ldots, y^n$ be normal coordinates centered at q, then

$$\tau(u^*f) = df \circ \cancelto{0}{\tau(u)} + \mathrm{tr}_g((\nabla df)(du\cdot, du\cdot)) = \frac{\partial^2 f}{\partial y^\alpha \partial y^\beta}(q) \quad \underbrace{\frac{\partial u^\alpha}{\partial x^i}(p)\frac{\partial u^\beta}{\partial x^j}(p)g^{ij}(p)}_{=\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial u^j}g^{ij}=\lambda^2\delta^{\alpha\beta}\text{ since } u \text{ WHC}}$$

$$= \frac{\partial^2 f}{\partial y^\alpha \partial y^\beta}(q)\lambda^2(p)\delta^{\alpha\beta}(p) \underbrace{=}_{\text{norm coords}} \lambda^2(\Delta f)(q) \underbrace{=}_{f\text{ harmonic}} 0$$

so that u^*f is harmonic. ■

Remark

From the proof, we see u is a harmonic morphism if and only if for all $f \in \mathcal{C}^\infty(V)$,

$$\Delta(u^*f) = \frac{2e(u)}{n}((\Delta f) \circ u)$$

using that $\lambda^2 = \frac{2e(u)}{n}$.

Corollary

Let u be a Riemannian submersion. Then u is a harmonic map if and only if it is harmonic morphism.
Proof:
All Riemannain submersions are weakly horizontally conformal with $\lambda = 1$. ■

Example

Consider the case $n = 1$, so $N = \mathbb{R}$ or $N = S^1$ if N is connected. Then the harmonic morphisms $M \to N$ are exactly the harmonic maps, since every smooth map is weakly horizontally conformal. Indeed, suppose $(du)_p \neq 0$, where then, since $\dim T_{u(p)}N = 1$ and $\dim V_p = m - 1$, we get $\dim H_p = 1$ by rank-nullity. So if X_p is a basis of H_p, and if $Y_p, Z_p \in H_p$, $Y_p = aX_p, Z_p = bX_p$ for $a, b \in \mathbb{R}$. Then $(du)_pX_p = e_q \neq 0$ is a basis for T_qN, so

$$\langle (du)_pY_p, (du)_pZ_p\rangle = ab\langle e_q, e_q\rangle = ab|e_q|^2$$

whereas $\langle Y_p, Z_p\rangle = ab|X_p|^2$, so that

$$\langle (du)_pY_p, (du)_pZ_p\rangle = \underbrace{\frac{|e_q|^2}{|X_p|^2}}_{=\lambda(p)^2}\langle Y_p, Z_p\rangle$$

so u is weakly horizontally conformal.

Example

Suppose $m = n$, and suppose $u : M \to N$ is a harmonic map.
(1) u is a harmonic morphism if and only if u is weakly conformal (the horizontal space is the whole tangent space).
(2) If $m = n \geq 3$, then u is a harmonic morphism if and only if it is a homothety. We already proved that harmonic maps that are weakly conformal are homotheties. Conversely, if $u^*h = cg$, then

$$e(u) = \frac{1}{2}\langle u^*h, g\rangle = \frac{1}{2}\operatorname{tr}_g(u^*h) = \frac{mc}{2}$$

so $\operatorname{div}(S_u) = \operatorname{div}(e(u)g - u^*h) = 0$ and u is a submersion everywhere (since it is a homothety), so u is a harmonic map, and it is already weakly conformal since it's a homothety.

Proposition

Let $u : (M^m, g) \to (N^n, h)$ be a harmonic morphism, and $v : (N^n, h) \to (P, k)$ be smooth. Then
(1) $e(v \circ u) = \frac{2}{n}e(v)e(u)$.
(2) If v is a harmonic map, then $v \circ u$ is harmonic.
(3) If $u(M)$ is dense in N and $v \circ u$ is harmonic, then v is harmonic.
Proof:
Since u is a harmonic morphism, u is weakly horizontally conformal. Then, for all $p \in M$, there exists an orthonormal basis $(e_1, \ldots, e_n, \underbrace{e_{n+1}, \ldots, e_m}_{\ker((du)_p)})$ for T_pM and an orthonormal basis $(f_1, \ldots, f_n)$ of $T_{u(p)}N$

such that $(du)_p e_i = \lambda(p) f_i$ for $1 \le i \le n$, $\lambda(p)^2 = \frac{2e(u)}{n}$, so that at p,

$$2e(v \circ u) = \sum_{i=1}^{n} |d(v \circ u)e_i|^2 = \sum_{i=1}^{n} |(dv \circ du)e_i|^2 = \lambda^2 \sum_{i=1}^{n} |(dv)^2 f_i|^2 = 2\lambda^2 e(u)$$

so that $e(v \circ u) = \frac{2}{n} e(u)e(v)$.
(2) From the composition formula, since u is harmonic,

$$\tau(v \circ u) = \mathrm{tr}_g((\nabla dv)(du\cdot, du\cdot)) = \sum_{i=1}^{m} (\nabla dv)(du \cdot e_i, du \cdot e_i)$$

$$\underbrace{=}_{e_{n+1},\ldots,e_m \in \ker(du_p)} \sum_{i=1}^{n} (\nabla dv)(du \cdot e_i, du \cdot e_i) = \lambda^2 \sum_{i=1}^{n} (\nabla dv)(f_i, f_i) = \lambda^2 \tau(v)$$

so $\tau(v \circ u)_p = \frac{2e(u)}{n} \tau(v)_{u(p)}$, so if $\tau(v) = 0$, then $\tau(v \circ u) = 0$.
(3) Suppose $v \circ u$ is harmonic. Then by the above formula, $0 = e(u)(\tau(v) \circ u)$. Now, as a general analytic fact about harmonic morphisms (analogous to holomorphic maps), non-constant harmonic morphisms are submersions on an open dense subset of M (open is easy by continuity, density is the hard part). Thus if $u(M)$ is dense, u is non-constant, so $e(u)$ is positive on an open dense set by the analytic fact. Thus $\tau(v) \circ u = 0$ on an open dense set, so by continuity, $\tau(v) \circ u = 0$. Then, since $u(M)$ is dense, $\tau(v) = 0$ by continuity. ■

Holomorphic and Harmonic Maps between Kähler Manifolds

Let (M^{2m}, g, J, ω) be a $U(m)$-structure. Recall that it's called almost Kähler if $d\omega = 0$, and Hermitian if $N_J = 0$, and called Kähler if both hold.

Let $u : (M^{2m}, g, J, \omega_M) \to (N^{2n}, h, J, \omega_N)$ be a smooth map. We know we have a decomposition of $TM \otimes \mathbb{C} = (TM)^{1,0} \oplus (TM)^{0,1}$, so we can extend $du : TM \to TN$ complex-linearly to $TM \otimes \mathbb{C}$. Now, we saw $\pi^{1,0} = \frac{1}{2}(\mathrm{Id} - iJ)$ and $\pi^{0,1} = \frac{1}{2}(\mathrm{Id} + iJ)$. We see these add to the identity, so we may write

$$du = \mathrm{Id}_{TN \otimes \mathbb{C}} \circ du \circ \mathrm{Id}_{TM \otimes \mathbb{C}} = (\pi^{1,0} + \pi^{0,1}) \circ du \circ (\pi^{1,0} + \pi^{0,1})$$
$$= \underbrace{\pi^{1,0} du\, \pi^{1,0}}_{:=\partial u} + \underbrace{\pi^{1,0} du\, \pi^{0,1}}_{:=\overline{\partial} u} + \underbrace{\pi^{0,1} du\, \pi^{1,0}}_{:=\partial \overline{u}} + \underbrace{\pi^{0,1} du\, \pi^{0,1}}_{:=\overline{\partial}\overline{u}}$$

(The complex conjugate for these components of du is just notation!) where we can use the definition of $\pi^{0,1}$ and $\pi^{1,0}$ to get

$$\partial u = \frac{1}{4}(du - iJdu - iduJ - JduJ)$$

$$\overline{\partial} u = \frac{1}{4}(du - iJdu + iduJ + JduJ)$$

$$\partial \overline{u} = \frac{1}{4}(du + iJdu - iduJ + JduJ)$$

$$\overline{\partial}\overline{u} = \frac{1}{4}(du + iJdu + iduJ - JduJ)$$

Note that $\overline{\partial} u = 0$ if and only if $du + JduJ = 0$ and $-Jdu + duJ = 0$, but these two conditions are equivalent (multiply by J, since $J^2 = -1$) if and only if $duJ = Jdu$, i.e. du commutes with J. If this happens, we say u is **holomorphic** (note that M, N need not be holomorphic manifolds, but if they are, then $\overline{\partial} u = 0$ is the Cauchy-Riemann equations, and exactly characterizes when u is holomorphic).
Similarly, $\partial u = 0$ if and only if $duJ = -Jdu$, which is called "anti-holomorphic". For shorthand, we'll

call u $\pm$-holomorphic if it is either holomorphic or anti-holomorphic.

Now, in a local frame, let $\{e_1, \dots, e_m, \overline{e}_1, \dots, \overline{e}_m\}$ be a local frame for $TM \otimes \mathbb{C}$, where $e_1, \dots, e_m$ are $(1,0)$, and $\overline{e}_1, \dots, \overline{e}_m$ are $(0,1)$. Similarly, let $\{f_1, \dots, f_m, \overline{f}_1, \dots, \overline{f}_m\}$ be a local frame for $TN \otimes \mathbb{C}$ made of $(1,0)$ and $(0,1)$ vector fields. Then we can write

$$du = \underbrace{u_k^\alpha e^k \otimes f_\alpha}_{=\partial u} + \underbrace{u_{\overline{k}}^\alpha e^{\overline{k}} \otimes f_\alpha}_{=\overline{\partial} u} + \underbrace{u_k^{\overline{\alpha}} e^k \otimes f_{\overline{\alpha}}}_{=\partial \overline{u}} + \underbrace{u_{\overline{k}}^{\overline{\alpha}} e^{\overline{k}} \otimes f_{\overline{\alpha}}}_{=\overline{\partial}\overline{u}}$$

Then we can write the energy density as

$$e(u) = \frac{1}{2}|du|^2 = \frac{1}{2}|\partial u|^2 + \frac{1}{2}|\overline{\partial} u|^2 + \frac{1}{2}|\partial \overline{u}|^2 + \frac{1}{2}|\overline{\partial}\overline{u}|^2 = |\partial u|^2 + |\overline{\partial} u|^2$$

where we use that complex conjugation preserves norm. Then, we have

$$|\partial u|^2 = g^{k\overline{\ell}} h_{\alpha\overline{\beta}} u_k^\alpha u_{\overline{\ell}}^{\overline{\beta}}$$

$$|\overline{\partial} u|^2 = g^{k\overline{\ell}} h_{\alpha\overline{\beta}} u_{\overline{\ell}}^\alpha u_k^{\overline{\beta}}$$

Let $|\partial u|^2 = e'(u)$, and $|\overline{\partial} u|^2 = e''(u)$, so that $e(u) = e'(u) + e''(u)$. We see $e''(u) = 0$ if and only if u is holomorphic, and $e'(u) = 0$ if and only if u is anti-holomorphic.
Now, suppose M is compact (and always oriented, since it has an almost complex structure), where $\mu_g = \frac{\omega_M^m}{m!}$. Define

$$E'(u) = \int_M e'(u)\mu_g \quad E''(u) = \int_M e''(u)\mu_g$$

so that $E(u) = E'(u) + E''(u)$.

Definition
We denote $k(u) := e'(u) - e''(u)$, and let $K(u) := E'(u) - E''(u)$.

Theorem (Lichnerowicz)
If M, N are both almost Kähler with M compact, then $K(u)$ is a smooth homotopy invariant.

Lemma
$k(u) = \langle \omega_M, u^*\omega_N \rangle$. (Recall $e(u) = \frac{1}{2}\langle g, u^*h \rangle$)
Proof:
Both ω_M and ω_N are type (1, 1), so we can write $\omega_M = (\omega_M)_{i\overline{j}} e^i \wedge e^{\overline{j}}$, where $(\omega_M)_{k\overline{\ell}} = \sqrt{-1} g_{k\overline{\ell}}$, and similarly $\omega_N = (\omega_N)_{\alpha\overline{\beta}} f^\alpha \wedge f^{\overline{\beta}}$. Then, we note that $u^* f^\alpha = u_k^\alpha e^k + u_{\overline{k}}^\alpha e^{\overline{k}}$, and similarly $u^* f^{\overline{\beta}} = u_\ell^{\overline{\beta}} e^\ell + u_{\overline{\ell}}^{\overline{\beta}} e^{\overline{\ell}}$, so that

$$u^*\omega_N = (\omega_N)_{\alpha\overline{\beta}} (u^* f^{\overline{\alpha}}) \wedge (u^* f^{\overline{\beta}}) = (\omega_N)_{\alpha\overline{\beta}} (u_k^\alpha u_{\overline{\ell}}^{\overline{\beta}} e^k \wedge e^{\overline{\ell}} - u_{\overline{\ell}}^\alpha u_k^{\overline{\beta}} e^k \wedge e^{\overline{\ell}} + (2,0) + (0,2))$$

(we renamed an index in the last equality).
Taking the inner product with ω_M (which is $(1,1)$), we obtain,

$$\langle \omega_M, u^*\omega_N \rangle = (\omega_M)_{i\overline{j}} (\omega_N)_{\alpha\overline{\beta}} (u_k^\alpha u_{\overline{\ell}}^{\overline{\beta}} - u_{\overline{\ell}}^\alpha u_k^{\overline{\beta}}) \langle e^i \wedge e^{\overline{j}}, e^k \wedge e^{\overline{\ell}} \rangle$$

$$= -g_{i\overline{j}} h_{\alpha\overline{\beta}} (u_k^\alpha u_{\overline{\ell}}^{\overline{\beta}} - u_{\overline{\ell}}^\alpha u_k^{\overline{\beta}})(0 - g^{i\overline{\ell}} g^{\overline{j}k}) = g^{k\overline{\ell}} h_{\alpha\overline{\beta}} (u_k^\alpha u_{\overline{\ell}}^{\overline{\beta}} - u_{\overline{\ell}}^\alpha u_k^{\overline{\beta}})$$

$$|\partial u|^2 - |\overline{\partial} u|^2 = k(u) \quad \blacksquare$$

Lemma
Let $\Phi : [0,1] \times M \to N$ be a smooth map between smooth manifolds. Let σ be any closed 2-form on N. Define $u_t : M \to N$ to be $\Phi(t, \cdot) = u_t$. Then

$$\frac{\partial}{\partial t}(u_t^* \sigma) = d(u_t^*(\frac{\partial u_t}{\partial t} \lrcorner \sigma))$$

Proof:
Since $d\sigma = 0$, $0 = u_t^* d\sigma = d(u_t^*\sigma)$ for all t. Let $\underline{d}$ be the exterior derivative on $[0,1] \times M$, so if $\alpha \in \Omega^k(M)$, then we can think of it as in $\Omega^k([0,1] \times M)$ with $\frac{\partial}{\partial t} \lrcorner \alpha = 0$, so that $\underline{d}\alpha = dt \wedge \frac{\partial \alpha}{\partial t} + d\alpha$. First, we claim

$$\Phi^*\sigma = u_t^*\sigma + dt \wedge \underbrace{u_t^*(\frac{\partial u_t}{\partial t} \lrcorner \sigma)}_{\beta}$$

Since both sides are 2-forms on $[0,1] \times M$, we can evaluate them on vector fields. Let $X, Y \in \Gamma(TM)$, then, since X, Y have no $\frac{\partial}{\partial t}$-part,

$$(\Phi^*\sigma)(X, Y) = \sigma(d\Phi \cdot X, d\Phi \cdot Y) = \sigma(du_t \cdot X, du_t \cdot Y) = (u_t^*\sigma)(X, Y)$$

On the other hand, $dt \wedge \beta(X, Y) = 0$ since X, Y have no $\frac{\partial}{\partial t}$-parts, so both sides of the claim agree on X, Y. Otherwise, let $X = \frac{\partial}{\partial t}$, and $Y \in \Gamma(TM)$. Then, the left hand side is

$$(\Phi^*\sigma)(\frac{\partial}{\partial t}, Y) = \sigma(d\Phi \cdot \frac{\partial}{\partial t}, d\Phi \cdot Y) = \sigma(\frac{\partial u_t}{dt}, du_t \cdot Y) = (\frac{\partial u_t}{\partial \sigma} \lrcorner \sigma)(du_t \cdot Y) = \beta(Y)$$

The right hand side is

$$\underbrace{(u_t^*\sigma)(\frac{\partial}{\partial t}, Y)}_{=0} + (dt \wedge \beta)(\frac{\partial}{\partial t}, Y) = \beta(Y) - 0 = \beta(Y)$$

and the claim follows. Now, we compute, since $d\sigma = 0$,

$$0 = \underline{d}(\Phi^*\sigma) \overset{\text{claim}}{\overbrace{=}} \underline{d}(u_t^*\sigma + dt \wedge \beta) = dt \wedge (\frac{\partial}{\partial t}(u_t^*\sigma) + \cancelto{0}{d(u_t^*\sigma)}) - dt \wedge d(u_t^*(\frac{\partial u_t}{\partial t} \lrcorner \sigma))$$

$$= dt \wedge \left(\frac{\partial}{\partial t}(u_t^*\sigma) - d(u_t^*(\frac{\partial u_t}{\partial t} \lrcorner \sigma)) \right)$$

so taking the interior product with $\frac{\partial}{\partial t}$ we obtain the lemma. ■

Proof of Lichnerowicz's Theorem:
By the first Lemma,

$$k(u)\mu_g = \langle \omega_M, u^*\omega_N \rangle \mu_g = u^*\omega_N \wedge \star\omega_M$$

By the fundamental theorem of calculus and the second Lemma (since $\sigma = \omega_N$ is closed)

$$u_1^*\omega_N - u_0^*\omega_N = \int_0^1 \frac{\partial}{\partial t}(u_t^*\omega_N) dt = \int_0^1 d\left(u_t^* \left(\frac{\partial u_t}{\partial t} \lrcorner \omega_N \right) \right) dt$$

$$= d\left(\int_0^1 u_t^* \left(\frac{\partial u_t}{\partial t} \lrcorner \omega_N \right) dt \right)$$

and so $u_1^*\omega_N - u_0^*\omega_N = d\alpha$ is exact (for α the 1-form on M given by the integral). Thus,

$$k(u_1)\mu_g - k(u_0)\mu_g = (u_1^*\omega_N - u_0^*\omega_N) \wedge \star\omega_M = d\alpha \wedge \star\omega_M = d(\alpha \wedge \star\omega_M)$$

where the last equality is because $\star\omega_M = \frac{1}{m!}\omega_M^{m-1}$ (since we know what μ_g is in terms of ω), so that $d(\star\omega) = d(\frac{1}{m!}\omega_M^{m-1}) = 0$, since $d\omega_M = 0$. Thus $k(u_1)\mu_g - k(u_0)\mu_g$ is exact, so by Stokes' theorem,

$$K(u_1) - K(u_0) = 0 \quad \blacksquare$$

Remark
Note that in the above theorem, we only needed $d(\star\omega) = 0$, and not $d(\omega) = 0$, which is weaker (1-out-of-4 torsion components, the Lee form vanishing, instead of 3 with $d\omega = 0$).

Befoore looking at applications of the Lichnerowicz theorem, let's examine this homotopy invariant $K(u)$ in some special cases.

Proposition
Suppose $u : M \to N$ is a smooth map between almost Kähler manifolds with M compact such that $[u^*\omega_N] = \lambda[\omega_M]$ for some $\lambda \in \mathbb{R}$. Then $K(u) = \lambda m \operatorname{vol}(M)$.
Proof:
We know $u^*\omega_N = \lambda\omega_M + d\alpha$ for some α by our condition on the cohomology classes, so that by the first Lemma,

$$k(u) = \langle u^*\omega_N, \omega_M\rangle = \lambda|\omega_M|^2 + \langle\omega_M, d\alpha\rangle$$

On the other hand, $|\omega_M|^2\mu_g = \omega_M \wedge \star\omega_M = \omega_M \wedge \frac{\omega_M^{m-1}}{(m-1)!}$, but also $|\omega_M|^2\mu_g = |\omega_M|^2\frac{\omega_M^m}{m!}$, so comparing we get that $|\omega_M|^2 = m$, so $k(u) = \lambda m + \langle\omega_M, d\alpha\rangle$. Integrating both sides we obtain

$$K(u) = \lambda m \int_M \mu_g + \int_M \langle\omega_M, d\alpha\rangle\mu_g = \lambda m \operatorname{vol}(M) + \underbrace{\langle\langle\omega_M, d\alpha\rangle\rangle}_{=\langle\langle d^*\omega_M, \alpha\rangle\rangle = 0} = \lambda m \operatorname{vol}(M) \quad \blacksquare$$

Corollary
Suppose $u : M \to N$ is a smooth map between almost Kähler manifolds with M compact, and suppose $[u^*\omega_N] = 0$. Then $K(u) = 0$. Moreover, if $u : M \to N$ is $\pm$-holomorphic, then u is constant.
Proof:
The first part follows since $\lambda = 0$ in the above proposition. For the other part, we have $0 = K(u) = E'(u) - E''(u)$, so $E'(u) = E''(u)$. Thus if u is $\pm$-holomorphic, then $E'(u) = E''(u) = 0$, so $E(u) = 0$, so $du = 0$ so u is constant. $\blacksquare$

If $u_0 \sim u_1$, let u_t be a family of smooth maps $u_t : M \to N$. Then, $E'(u_t) - E''(u_t) = K(u)$ is constant on each connected component of $\operatorname{Map}(M, N)$. Thus, we obtain that $\frac{d}{dt}E'(u_t) = \frac{d}{dt}E''(u_t) = \frac{1}{2}\frac{d}{dt}E(u_t)$.

Corollary
The critical points of E', E'', E all coincide. Moreover, their minima coincide.
Proof:
$\frac{d}{dt}E'(u_t) = \frac{d}{dt}E''(u_t) = \frac{1}{2}\frac{d}{dt}E(u_t)$ from above shows their critical points coincide. To show their minima coincide, suppose $u_0 \sim u_1$, so that $E'(u_1) - E'(u_0) = E''(u_1) - E''(u_0)$, and suppose u_0 is a minimum of E'. Then, $E'(u_0) \le E'(u_1)$ if and only if $E''(u_0) \le E''(u_1)$ by the previous equation, so the minima of E' and E'' coincide. Also, $E = K + 2E''$, where K is constant on homotopy classes, so $(E - 2E'')(u_1) = (E - 2E'')(u_0)$, hence $E(u_1) - E(u_0) = 2(E''(u_1) - E''(u_0))$, which by the same argument as above, implies the minima of E and E'' coincide. $\blacksquare$

Corollary
Let $u : M \to N$ be a smooth map. If u is $\pm$-holomorphic, then it is harmonic, and moreover is an absolute minimum of E in its homotopy class. Hence $\pm$-holomorphic maps are stable (index zero).
Proof:
If u is $\pm$-holomorphic, then either $E'(u) = 0$ or $E''(u) = 0$, which means it's an absolute minimum of E' or E'' (and hence E) respectively. By the first corollary, it follows that u is a minimum of E in its

homotopy class, so u is harmonic. ■

Corollary
In the above setup, assume further that $n = 1$ (so N is a Riemann surface) with compatible metric. Then any $\pm$-holomorphic map is a harmonic morphism.
Proof:
Indeed, by the previous corollary u is harmonic, and since u is $\pm$-holomorphic, at every $p \in M$, $(du)_p$ has rank 0 or 2 (check). If $(du)_p \neq 0$ and $X \in \ker((du)_p)$, then $(du)_p JX = \pm J(du)_p X = 0$, so $JX \in \ker((du)_p)$, so since $\ker(du)_p$ has real dimension $2m - 2$ and is preserved by J, we have $\ker(du)_p \cong \mathbb{C}^{m-1}$, $\ker(du)_p^\perp \cong \mathbb{C}$. On the other hand, restricted to the horizontal space we have an isomorphism $(du)_p : \ker(du)_p^\perp \xrightarrow{\cong} T_{u(p)}N$, which is in fact a conformal isomorphism by complex analysis (since the spaces involved are complex and du (anti)-commutes with J), which shows u is weakly horizontally conformal and thus a harmonic morphism. ■

Corollary
In the above setup, suppose $u_0 : M \to N$ is harmonic and is a minimum of E in its homotopy class. If u_0 is homotopic to a $\pm$-holomorphic map, then u_0 is $\pm$-holomorphic.
Proof:
If $u_0 \sim u_1$, and u_1 is $\pm$-holomorphic, then $E'(u_1)$ or $E''(u_1)$ is zero and a minimum of E in its homotopy class is also a minimum of E' or E''. Thus, since $E(u_0)$ is minimal in its class, but there is u_1 in its class with $E'(u_1) = 0$ or $E''(u_1) = 0$, we must have either $E'(u_0) = 0$ or $E''(u_1) = 0$. ■

Corollary
Let $u_0 : M \to N$ be $\pm$-holomorphic. Suppose $u_t : M \to N$ is a smooth variation of u_0 for $t \in (-\varepsilon, \varepsilon)$ such that u_t is harmonic for all t. Then, for all t, u_t is $\pm$-holomorphic (same $\pm$ as u_0).
Proof:
For all $t \in (-\varepsilon, \varepsilon)$,
$$\frac{d}{dt}E'(u_t) = \frac{d}{dt}E''(u_t) = \frac{1}{2}\frac{d}{dt}E(u_t) = 0$$
since each u_t is harmonic. Thus, E', E'' are constant in t, so that either $E'(u_t) = 0$ or $E''(u_t) = 0$ for all t (because u_0 has this). ■

Remark
This is a kind of stability result: to deform a $\pm$-holomorphic map through harmonic maps, it must be deformed through a $\pm$-holomorphic variation. There are analogous results in other settings. For example, if (M, g) is compact with a special holonomy metric, this is an Einstein metric. In this case, there exists a theorem (Dai-Wang-collaborators) that the only way to deform the metric g through Einstein metrics is through special holonomy metrics.

Corollary
In the above setup, let u_0, u_1 be homotopic maps with u_0 holomorphic and u_1 antiholomorphic. Then u_0, u_1 are both constant. In particular, any $\pm$-holomorphic map homotopic to a constant must be constant.
Proof:
$E''(u_0) = E'(u_1) = 0$, so
$$0 \leq E'(u_0) = K(u_0) = K(u_1) = -E''(u_1) \leq 0$$
so $E'(u_0) = E''(u_1) = 0$, hence $E(u_0) = E(u_1) = 0$, implying u_0, u_1 are constant. ■

Harmonic Maps between Kähler Manifolds

Now, assume (M^{2m}, g, J, ω_M) is compact Kähler and (N^{2n}, h, J, ω_N) are both Kähler. Let $\tau'(u) = 0$ and $\tau''(u) = 0$ be the Euler-Lagrange equations for E' and E''. By the first Corollary, $\tau(u) = 0$ if and only if $\tau'(u) = 0$ if and only if $\tau''(u) = 0$ if and only if u is harmonic.

Proposition
Let $z^1, \dots, z^m$ be holomorphic coordinates on M (indexed by Latin) and $w^1, \dots, w^n$ be holomorphic coordinates on N (indexed by Greek). Then,

$$\tau'(u)^\gamma = 2g^{j\overline{k}}(u^\gamma_{j\overline{k}} + {}^N\Gamma^\gamma_{\alpha\beta} u^\alpha_j u^\beta_{\overline{k}})$$

and

$$\tau''(u)^{\overline{\gamma}} = 2g^{j\overline{k}}(u^{\overline{\gamma}}_{j\overline{k}} + {}^N\Gamma^{\overline{\gamma}}_{\overline{\alpha}\overline{\beta}} u^{\overline{\alpha}}_j u^{\overline{\beta}}_{\overline{k}})$$

where $u^\alpha_i = \frac{\partial u^\alpha}{\partial z^i}$ and $u^\alpha_{k\overline{\ell}} = \frac{\partial^2 u^\alpha}{\partial z^k \partial \overline{z}^\ell}$. Note that this does not involve the Christoffel symbols of M. *Proof:* Recall that the harmonic map equation in the general setting is

$$\tau(u)^P = g^{AB}(u^P_{AB} - {}^M\Gamma^k_{AB} u^P_k + {}^N\Gamma^P_{\alpha\beta} u^\alpha_A u^\beta_B)$$

$$e(u) = \frac{1}{2}|du|^2 = \frac{1}{2} g^{AB} h_{PQ} u^P_A u^Q_B$$

where A, B, C are indices on M and P, Q are indices on N. Since $e'(u) = |\partial u^2| = g^{k\overline{i}} h_{\alpha\overline{\beta}} u^\alpha_k u^{\overline{\beta}}_{\overline{i}}$ and $e''(u) = |\overline{\partial} u|^2 = g^{k\overline{i}} h_{\alpha\overline{\beta}} u^\alpha_{\overline{i}} u^{\overline{\beta}}_k$, it follows that $\tau'(u) = 2\operatorname{div}(\partial u)$ and $\tau''(u) = 2\operatorname{div}(\overline{\partial} u)$, so that we can calculate (with the Kähler symmetries of M),

$$\tau'(u)^\gamma = g^{AB}(u^\gamma_{AB} - {}^M\Gamma^C_{AB} u^\gamma_C + {}^N\Gamma^\gamma_{QR} U^Q_A U^R_B)$$

$$= g^{k\overline{i}}(u^\gamma_{k\overline{i}} - 0 + {}^N\Gamma^\gamma_{\alpha\beta} u^\alpha_k u^\beta_{\overline{i}}) + g^{\overline{k}i}(u^\gamma_{\overline{k}i} - 0 + {}^N\Gamma^\gamma_{\alpha\beta} u^\alpha_{\overline{k}} u^\beta_i)$$

which gives the desired result after renaming an index. The case for $\tau''(u)$ is similar. ■

Now, let $u : (M^{2m}, g, J, \omega_M) \to (N^{2n}, h)$ be a smooth map with domain a compact Kähler manifold.

Because $\mathcal{S}^2(T^*M) \otimes \mathbb{C} \subseteq (T^*M \otimes \mathbb{C}) \otimes (T^*M \otimes \mathbb{C}) = ((T^*M)^{1,0} \oplus (T^*M)^{0,1}) \otimes ((T^*M)^{1,0} \oplus (T^*M)^{0,1})$, we can decompose the stress-energy tensor

$$S_u = S_u^{(2,0)} + S_u^{(1,1)} + S_u^{(0,2)}$$

where $S_u^{2,0}(X, Y) = S_u(\pi^{1,0}X, \pi^{0,1}Y)$ and $S_u^{1,1}(X, Y) = S_u(\pi^{1,0}X, \pi^{0,1}Y) + S_u(\pi^{0,1}X, \pi^{1,0}Y)$ and $S_u^{0,2} = \overline{S_u^{2,0}}$ and $\overline{S_u^{1,1}} = S_u^{1,1}$.
Furthermore, we can decompose

$$\nabla = \nabla' + \nabla''$$

via $\nabla' = \pi^{1,0} \circ \nabla$ (and similarly for ∇'').
Also, recall for any tensor A on M,

$$\nabla^* A = -\operatorname{div}(A) = -\operatorname{tr}_g(\nabla A) = -\operatorname{tr}_g(\nabla' A + \nabla'' A) = (\nabla')^* A + (\nabla'')^* A$$

so that we can write

$$\nabla^* = (\nabla')^* + (\nabla'')^*$$

Then

$$-\operatorname{div}(S_u) = \nabla^* S_u = ((\nabla')^* + (\nabla'')^*)(S_u^{(2,0)} + S_u^{(1,1)} + S_u^{(0,2)})$$

$$= \underbrace{(\nabla')^* S_u^{(2,0)}}_{(1,0)} + \underbrace{(\nabla')^* S_u^{(1,1)}}_{(0,1)} + \underbrace{(\nabla'')^* S_u^{(1,1)}}_{(1,0)} + \underbrace{(\nabla'')^* S_u^{(0,2)}}_{(0,1)}$$

If u is harmonic, then $\mathrm{div}(S_u) = 0$, so that we get

$$(\nabla')^* S_u^{(2,0)} + (\nabla'')^* S_u^{(1,1)} = 0$$

plus the same for the complex conjugate. In the special case $m = 1$, then we have a Riemann surface with a compatible metric, $\mu_g = \omega_M$, and $J = \star$ on Ω^1, so we just have an oriented Riemannian 2-manifold. In the local holomorphic coordinate $z = x + iy$,

$$du = \underbrace{\frac{\partial u}{\partial x}}_{\in \Gamma(u^*TN)} \mathrm{d}x + \frac{\partial u}{\partial y}\mathrm{d}y = \frac{\partial u}{\partial z}dz + \frac{\partial u}{\partial \overline{z}}d\overline{z}$$

We want $\{\frac{\partial}{\partial z}, \frac{\partial}{\partial \overline{z}}\}$ to be the dual basis of $\{dz, d\overline{z}\}$, so that $\frac{\partial}{\partial z} = \frac{1}{2}(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y})$. Now, $S_u = \frac{1}{2}|du|^2 g - u^*h$, and we can calculate

$$\begin{aligned}|du|^2 &= \langle \frac{\partial u}{\partial z}dz + \frac{\partial u}{\partial \overline{z}}d\overline{z}, \frac{\partial u}{\partial z}dz + \frac{\partial u}{\partial \overline{z}}d\overline{z}\rangle_{g^{-1}\otimes u^*h} \\ &= 2\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial \overline{z}}\rangle_{u^*h}\langle dz, d\overline{z}\rangle_{g^{-1}}\end{aligned}$$

Moreover,

$$g = \langle \frac{\partial}{\partial z}, \frac{\partial}{\partial \overline{z}}\rangle dz \otimes d\overline{z} + \langle \frac{\partial}{\partial \overline{z}}, \frac{\partial}{\partial z}\rangle d\overline{z} \otimes dz = 2\,\underbrace{\langle \frac{\partial}{\partial z}, \frac{\partial}{\partial \overline{z}}\rangle}_{f}\, dzd\overline{z}$$

so that

$$g = \begin{bmatrix} 0 & 2f \\ 2f & 0 \end{bmatrix}, g^{-1} = \begin{bmatrix} 0 & \frac{1}{2f} \\ \frac{1}{2f} & 0 \end{bmatrix}$$

Now, suppose (N^{2n}, h, J, ω_N) is also Kähler. Then,

$$h = h_{\alpha\overline{\beta}} dw^\alpha d\overline{w}^\beta$$

so that (omitting the precompose with u's),

$$u^*h = h_{\alpha\overline{\beta}} \left(\frac{\partial u^\alpha}{\partial z}dz + \frac{\partial u^\alpha}{\partial \overline{z}}d\overline{z}\right)\left(\frac{\partial \overline{u}^\beta}{\partial z}dz + \frac{\partial \overline{u}^\beta}{\partial \overline{z}}d\overline{z}\right)$$

$$= \underbrace{h_{\alpha\overline{\beta}}\frac{\partial u^\alpha}{\partial z}\frac{\partial u^{\overline{\beta}}}{\partial z}dzdz}_{2,0} + \underbrace{h_{\alpha\overline{\beta}}\frac{\partial u^\alpha}{\partial \overline{z}}\frac{\partial u^{\overline{\beta}}}{\partial \overline{z}}d\overline{z}d\overline{z}}_{0,2} + \underbrace{h_{\alpha\overline{\beta}}\left(\frac{\partial u^\alpha}{\partial z}\frac{\partial u^{\overline{\beta}}}{\partial \overline{z}} + \frac{\partial u^\alpha}{\partial \overline{z}}\frac{\partial u^{\overline{\beta}}}{\partial z}\right)dzd\overline{z}}_{1,1}$$

so

$$(u^*h)^{2,0} = \langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle dz^2, (u^*h)^{0,2} = \langle \frac{\partial u}{\partial \overline{z}}, \frac{\partial u}{\partial \overline{z}}\rangle d\overline{z}^2, (u^*h)^{1,1} = \langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial \overline{z}}\rangle dzd\overline{z} = \frac{1}{2}|du|^2 g$$

So that

$$S_u^{(2,0)} = -(u^*h)^{2,0} = -\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle dz^2$$

Now, keep in mind we can write

$$\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle = \frac{1}{4}(|u_x|^2 - |u_y|^2 - 2i\langle u_x, u_y\rangle)$$

These calculations give us the setup for:

Proposition
$u : (M^2, g, J, \omega_M) \to (N^{2n}, h, J\omega_N)$ be a smooth map from a compact Kähler manifold to a Kähler manifold.

(i) $(u^*h)^{2,0} = 0$ if and only if u is weakly conformal.
(ii) If u is harmonic, then $(u^*h)^{2,0} = -S_u^{2,0}$ is a holomorphic quadratic differential, meaning it is a section in $\Gamma(\mathcal{S}^2(T^*M)^{1,0})$.
Proof:
(i) $(u^*h)^{2,0} = 0$ if and only if u^*h is $(1, 1)$, i.e.

$$u^*h = (u^*h)^{1,1} = \langle u_z, u_{\overline{z}}\rangle dz d\overline{z} = (|u_x|^2 + |u_y|^2) dz d\overline{z} = 2|u_x|^2 dz d\overline{z} = \lambda^2 g$$

where $\lambda^2 = \frac{|u_x|^2}{f}$, and we use that $(u^*h)^{2,0} = 0$ tells us $|u_x|^2 = |u_y|^2$, $\langle u_x, u_y\rangle = 0$ (so that real and imaginary parts of $\langle u_z, u_z\rangle$ vanish). Thus, u is weakly conformal.
(ii) If u is harmonic, $\mathrm{div}(S_u) = 0$, but S_u has no $(1,1)$ part from the above calculation, so that $\mathrm{div}(S_u) = 0$ if and only if $(\nabla')^* S_u^{2,0} = 0$ if and only if $-\operatorname{tr}_g(\nabla'' S_u^{2,0}) = 0$ if and only if

$$0 = \operatorname{tr}_g(\nabla''(\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle dz^2)) := \operatorname{tr}_g(\frac{\partial}{\partial \overline{z}}\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle d\overline{z} \otimes dz^2)$$

$$= (\text{pos function})(\frac{\partial}{\partial \overline{z}}\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle) dz$$

so u harmonic implies $\mathrm{div}(S_u) = 0$ implies $\frac{\partial}{\partial \overline{z}}\langle \frac{\partial u}{\partial z}, \frac{\partial u}{\partial z}\rangle = 0$ implying $(u^*h)^{2,0}$ is a holomorphic quadratic differential. ■

Example
We claim any harmonic map $(S^2, g_{round}) \to N$ is weakly conformal. Indeed, by the previous proposition, $(u^*h)^{2,0}$ is a holomorphic quadratic differential on S^2. But there are no nonzero such objects. To see this, suppose $\alpha = k(z)dz^2$ is well-defined on S^2 with $k(z)$ holomorphic. Take holomorphic polar coordinates on $S^2 = \mathbb{CP}^1$, z the holomorphic coordinate on $S^2 \setminus \{N\}$, and $w = \frac{1}{z}$ holomorphic coordinate on $S^2 \setminus \{S\}$. In the other chart, we have

$$\alpha = k(1/w)(d(1/w))^2 = k(1/w)\frac{1}{w^4} dw^2$$

so we need $k(1/w)\frac{1}{w^4}$ to be holomorphic on $\mathbb{C} \setminus \{0\}$. Thus we must have a finite limit as $w \to 0$, so k is bounded at infinity, so k is bounded on every ball in $\mathbb{C}$. Thus by Liouville's theorem, k is constant, and $k(\infty) = 0$ implies $k = 0$. Thus, $u^*h = 0$, so u is weakly conformal by the proposition.

For the last thing in this section, suppose we have a smooth map between Kähler manifolds $u : (M^{2m}, g, J, \omega_M) \to (N^{2n}, h, J, \omega_N)$ with M compact. We can simplify the second variation formula in this context.
Then, we saw that the critical points of E, E', E'' coincide, with

$$\frac{1}{2}\frac{\partial E}{\partial t} = \frac{\partial E'}{\partial t} = \frac{\partial E''}{\partial t}$$

Thus, it's enough to consider the second variation of E'', and in a paper of Siu-Yau, they compute explicitly, where s is a complex parameter in some open neighbourhood of 0 in $\mathbb{C}$,

$$\frac{\partial^2 E''}{\partial s \partial \overline{s}} E''(u_s)\Big|_{s=0} \overbrace{=}^{m=1} \int_M (|\nabla_s u_{\overline{z}}|^2 + |\nabla_{\overline{s}} u_{\overline{z}}|^2 + R^N(u_{\overline{s}}, \overline{u_{\overline{s}}}, u_{\overline{z}}, \overline{u_{\overline{z}}}) + R^N(u_s, u_{\overline{s}}, u_{\overline{z}}, \overline{u_{\overline{z}}}) - 2\mathrm{Re}(R^N(u_{\overline{s}}, \overline{u_{\overline{z}}}, u_s, \overline{u_{\overline{z}}})))\mu_g$$

(this works for $m \neq 1$ but is more complicated, and I might have gotten some of the bars wrong).
But because N is Kähler, its curvature tensor has special symmetries involving J_N, and the functional E'' also involves J_N. Using these symmetries and this interaction, they prove that if we choose a particular type of variation, this reduces to a single curvature term, related to the holomorphic bisectional curvature of N. In particular, they prove a stability result of the form:
If u is harmonic and N has positive holomorphic bisectional curvature, plus "topological conditions", then u must be $\pm$-holomorphic. They used this as a first step (together with Sacks-Uhlenbeck's existence

theorem) to prove the Frankel conjecture:

Theorem (Frankel Conjecture, Siu-Yau)
If (N^{2n}, h, J, ω_N) is Kähler with positive holomorphic bisectional curvature, then N is biholomorphic to $\mathbb{CP}^n$ (the metric h need not be Fubini-Study).

The Sacks-Uhlenbeck theorem says: if M is a compact Riemann surface, and N is a compact Riemannian manifold, then a topological assumption on its homotopy class guarantees there is a energy-minimizing harmonic map in that homotopy class.

Smith Maps

We'll define a special and natural class of smooth maps between Riemannian manifolds where one is equipped with a calibration form. There will be two flavours, $u : (L^k, g) \to (M^n, h, \alpha)$ with α a k-calibration, called a Smith immersion, and a Smith submersion $u : (M^n, h, \alpha) \to (L^k, g)$ with α an $(n-k)$-calibration.

We'll need some linear algebra: Let $(V^n, \langle\cdot,\cdot\rangle)$ and $(W^m, \langle\cdot,\cdot\rangle)$ be inner product spaces. Then, for $A : V \to W$ a linear map, we recall $|A|^2 = \operatorname{tr}(A^*A) = \sum_{i=1}^n |Av_i|^2$, where $\{v_i\}$ is an orthonormal basis of V.

Definition
$A : V \to W$ is called a conformal injection if $\langle Av_1, Av_2\rangle = \lambda^2\langle v_1, v_2\rangle$ for some $\lambda > 0$ and all $v_1, v_2 \in V$, if and only if $A^*A = \lambda^2 I$, if and only if $A = \lambda\widehat{A}$, where $\widehat{A} : V \to W$ is some isometric injection. Moreover, necessarily $|A|^2 = n\lambda^2$ by taking the trace of both sides.

Moreover, given $A : V \to W$, we can take $\Lambda^k A : \Lambda^k V \to \Lambda^k W$ by $(\Lambda^k A)(v_1 \wedge \ldots \wedge v_k) = (Av_1) \wedge \ldots \wedge (Av_k) \in \Lambda^k W$ on decomposable elements and extending by linearity.

Lemma (Hadamard's Inequality)
Let $A : V \to W$ be a nonzero linear map. Then

$$|\Lambda^n A| \leq \frac{1}{(\sqrt{n})^n}|A|^n$$

with equality if and only if A is a conformal injection.
Proof:
Let $B = A^*A : V \to V$, which is a nonzero self-adjoint linear operator, so that $\langle Bv, v\rangle = |Av|^2 \geq 0$. Hence, by the spectral theorem, there exists an orthonormal basis $\{v_1, \ldots, v_n\}$ of V consisting of eigenvectors of B with eigenvalues $\mu_i \geq 0$. Then $\{v_1 \wedge \ldots \wedge v_n\}$ is an orthonormal basis of $\Lambda^n(V)$, so that

$$|\Lambda^n A|^2 = \operatorname{tr}((\Lambda^n A)^*(\Lambda^n A)) = \operatorname{tr}((\Lambda^n A^*)(\Lambda^n A)) = \operatorname{tr}(\Lambda^n(A^*A)) = \operatorname{tr}(\Lambda^n B) = \mu_1 \ldots \mu_n$$

Thus by AM-GM,

$$|\Lambda^n A|^2 = \mu_1 \ldots \mu_n \leq \left(\frac{1}{n}\sum_{j=1}^n \mu_j\right) = \frac{1}{n^n}\operatorname{tr}(B)^n = \frac{1}{n^n}\operatorname{tr}(A^*A)^n = \frac{1}{n^n}|A|^{2n}$$

Taking square roots, we obtain the desired inequality. When do we have equality? In AM-GM, if and only if $\mu_1 = \cdots = \mu_n =: \mu$, so that

$$B = A^*A = \mu I$$

for some $\mu > 0$ (because $A \neq 0$), so that $A^*A = \mu I$ is a conformal injection if and only if $|A|^2 = n\mu$. ∎

Now, let (M^n, g) be a Riemannian manifold. Recall a calibration α on M is a closed k-form on M such that for all $p \in M$, $\alpha_p(v_1, \ldots, v_k) \leq 1$ if $v_1, \ldots, v_k \in T_pM$ are orthonormal (called the comass one condition).
Then, we saw that if L is an oriented k-submanifold of M, $\iota : L \to M$ is an immersion, and L is α-calibrated if $\alpha|_L = \iota^*\alpha$ equals μ_L. We note $\iota^*\alpha = \mu_L$ is a fully nonlinear (if $k > 1$) PDE for ι.
Thus, we have a first-order PDE that implies harmonicity (a second order condition), analogous to $\pm$-holomorphic implying harmonicity.

Proposition
If $\alpha \in \Omega^k(M)$ has comass one, and M is oriented, then $\star\alpha \in \Omega^{n-k}(M)$ has comass one.
Proof:
Use the metric to identify $\Lambda^k(TM) \cong \Lambda^k(T^*M)$, and let $L_p \subseteq T_pM$ be an oriented $(n-k)$-plane, so that $L_p = e_1 \wedge \ldots \wedge e_{n-k}$, and $L_p^\perp = e_{n-k+1} \wedge \ldots \wedge e_n$ is an oriented k-plane, such that $e_1, \ldots, e_n$ is an oriented orthonormal basis of T_pM. Then,

$$\begin{aligned}(\star\alpha)(e_1 \wedge \ldots \wedge e_{n-k}) &= \langle \star\alpha, e_1 \wedge \ldots \wedge e_{n-k}\rangle \overbrace{=}^{\star \text{ isometry}} \langle \star\star\alpha, \star(e_1 \wedge \ldots \wedge e_{n-k})\rangle \\ &= \langle \pm\alpha, e_{n-k+1} \wedge \ldots \wedge e_n\rangle = \pm\alpha(e_{n-k+1} \wedge \ldots \wedge e_n) \in [-1, 1]\end{aligned}$$

since α has comass one. ■

Lemma (First Cousin Principle)
Let $\alpha \in \Omega^k(M)$ have comass one. Suppose $L_p \subseteq T_pM$ is an oriented k-plane which is α_p calibrated ($\alpha_p(e_1, \ldots, e_k) = 1$ if $e_1, \ldots, e_k$ is an oriented orthonormal basis of L_p). Let $w \in L_p^\perp$, then $\alpha_p(e_1, \ldots, e_{k-1}, w) = 0$.
Proof:
Define $w_t = e_k \cos t + w \sin t$. Then $\{e_1, \ldots, e_{k-1}, w_t\}$ are orthonormal. Hence $f(t) := \alpha(e_1, \ldots, e_{k-1}, w_t) \leq 1$ for all t, and $f(0) = 1$ since L_p is α_p-calibrated. Moreover, f is certainly a smooth function of t. Thus 0 is a maximum of f, so that $f'(0) = 0$, but $f'(0) = \alpha(e_1, \ldots, e_{k-1}, w)$. ■

It's called the first cousin principle, since if $\iota : L \to M$ is inclusion, then $\iota^*T^*M = T^*M|_L = T^*L \oplus N^*L$, so that

$$\begin{aligned}\iota^*(\Lambda^k(T^*M)) = \Lambda^k(\iota^*(T^*M)) = \Lambda^k(T^*L \oplus N^L) &= \bigoplus_{p+q=k} (\Lambda^p(T^*L) \otimes \Lambda^q(N^*L)) \\ &= \Lambda^k(T^*M) \oplus (\Lambda^{k-1}(T^*M) \otimes N^*L) \oplus \ldots\end{aligned}$$

Thus, if $\alpha \in \Gamma(\Lambda^kT^*M)$, its pullback as a section (NOT AS A FORM) can be decomposed $\iota^*\alpha = (\iota^*\alpha)_{k,0} + (\iota^*\alpha)_{k-1,1} + \ldots$, and the first cousin principle says that if L is α-calibrated, then the "first cousin" $(\iota^*\alpha)_{k-1,1}$ is zero.

From now on, all manifolds are oriented, and if we integrate, that manifold is compact.

Recall from the homework that if $u : (M, g) \to (N, h)$ is smooth, its p-energy is

$$E_p(u) = \frac{1}{(\sqrt{p})^p} \int_M |du|^p \mu_g$$

whose critical points are called p-harmonic points, and they satisfy a first variation formula

$$\tau_p(u) = \operatorname{div}(|du|^{p-2} du) = 0$$

where $\tau_p(u)$ is the p-tension. When $p > 2$, this is degenerate elliptic, and much of the elliptic regularity theory doesn't apply. Recall from Assignment 3, that if $p = \dim M$, then E_p depends only on the conformal class of metrics on M.

Smith Immersions

Theorem
Let $u : (L^k, g) \to (M^n, h, \alpha)$ be smooth, with L compact oriented, and $\alpha \in \Omega^k(M)$ a comass one form. Then $u^*\alpha \le \lambda^k \mu_g$, where $\lambda^2 = \frac{1}{k}|du|^2$, with equality if and only if $u^*h = \lambda^2 g$ (u is weakly conformal) and the image $u(L^0)$ is α-calibrated, where $L^0 \subseteq L$ is the open set where $du \neq 0$.
Proof:
The desired inequality is immediate at points where $du = 0$. Let $x \in L^0$, and let $e_1, \dots, e_k$ be an orthonormal basis of (T_xL, g_x). Then

$$(u^*\alpha)_x(e_1 \wedge \dots \wedge e_k) = \alpha(\Lambda^k(du_x)(e_1 \wedge \dots \wedge e_k)) \overbrace{\le}^{\text{comass one}} |\Lambda^k(du_x)(e_1 \wedge \dots \wedge e_k)| = |\Lambda^k du_x| \le \lambda^k$$

where the last inequality is Hadamard's inequality, and we use that comass one implies that $\alpha(v_1, \dots, v_k) \le |v_1 \wedge \dots \wedge v_k|$. This proves the inequality.
We have equality if and only if both inequalities have equality. If the second inequality is an equality, we know from Hadamard's inequality that this occurs if and only if du_x is a conformal injection, so that $u^*h = \lambda^2 g$ (note that this holds on $L \setminus L^0$ because du is zero there) and u is weakly conformal. For the first inequality, let $e_1, \dots, e_k$ be an orthonormal basis of (T_xL, g_x). Then $\frac{1}{\lambda}(du)_x e_1, \dots, \frac{1}{\lambda}(du)_x e_k$ is an orthonormal basis of $((du)_x(T_xL), h_{u(x)})$, since du_x is a conformal injection. Thus, if the first inequality is also an equality, we have

$$\alpha((du)_x e_1, \dots, (du)_x e_k) = \lambda^k$$

if and only if

$$\alpha(\frac{1}{\lambda}(du)_x e_1, \dots, \frac{1}{\lambda}(du)_x e_k) = 1$$

so that $(du)_x(T_xL)$ is $\alpha_{u(x)}$-calibrated, showing that $u|_{L^0} : L^0 \to M$ is an α-calibrated immersion with respect to the metric $u^*h = \lambda^2 g$ on L^0. ■

Remark
In the proof, we saw that equality holds in the inequality if and only if $u^*\alpha = \frac{1}{(\sqrt{k})^k}|du|^k \mu_g$ and $u^*h = \frac{1}{k}|du|^2 g$. But the first equation automatically implies the second: the first equation is $u^*\alpha = \lambda^k \mu_g$ with $\lambda = \frac{1}{\sqrt{k}}|du|$, which implies $u^*h = \lambda^2 g$.

Definition
A smooth map $u : (L^k, g) \to (M^n, h, \alpha)$ is called a Smith immersion if and only if $u^*\alpha = \lambda^k \mu_g$, where $\lambda^2 = \frac{1}{k}|du|^2$, i.e. where equality occurs in the above theorem. When $k > 1$, this is a first order fully nonlinear PDE for u.

Remark
Smith immersions are not in general immersions, only at points where $\lambda > 0$ (which is open in L).

Theorem (Energy Inequality)
Take (L^k, g) and (M^n, h, α) as before, assume L is compact, and $d\alpha = 0$. If u is a Smith immersion, then u is a critical point of the k-energy, hence a k-harmonic map. Moreover, u is a minimizer of E_k in its homotopy class.
Proof:

$$E_k(u) = \frac{1}{(\sqrt{k})^k}\int_L |du|^k \mu_g = \int_L \lambda^k \mu_g \underbrace{\ge}_{\text{theorem}} \int_L u^*\alpha \underbrace{=}_{d\alpha=0} [\alpha] \cdot u_*[L]$$

where $[a] \in H^k(M)$ is its cohomology class, $u_*[L] \in H_k(M)$ is the pushforward of the fundamental class to a homology class in M, and $\cdot$ is the topological pairing. But we know equality for the inequality holds if and only if u is a Smith immersion, giving that it minimizes in its homotopy class (and any such minimum is a Smith immersion). ■

Remark
Recall that the k-harmonic map equation $\operatorname{div}(|du|^{k-2}du) = 0$ is conformally invariant if the dimension of the domain is k. Moreover, if we take the Smith immersion equation, differentiate it, and use that α is closed, we obtain the k-harmonic map equation directly. Thus, one would expect the Smith immersion equation to be conformally invariant, which is the case: let $\widetilde{g} = f^2 g$, $\widetilde{\lambda}^2 = \frac{1}{k}|du|^2_{\widetilde{g},f} = f^{-2}\lambda^2$, so that $\lambda = f\widetilde{\lambda}$. Thus, since $\mu_{\widetilde{g}} = f^k \mu_g$, $\lambda^k \mu_g = \widetilde{\lambda}^k \mu_{\widetilde{g}}$, and everything transforms correctly.

Smith Submersions

Now let $u : (M^n, h, \alpha) \to (L^k, g)$ with $\alpha \in \Omega^{n-k}(M)$ having comass one.
Let $M^0 \subseteq M$ be the set of points where $(du)_x \neq 0$, so that if u is surjective, $u|_{M^0} : M^0 \to L$ is a submersion.
In this case, given $x \in M^0$, $T_x M = \ker((du)_x) \oplus \ker((du)_x)^\perp = V_x \oplus H_x$, so that given a covariant m-tensor σ on M^0, we can write

$$\sigma = \sigma^{(m,0)} + \sigma^{(m-1,1)} + \cdots + \sigma^{(0,m)}$$

where $\sigma^{(p,q)} \in \Gamma(\bigotimes^p V^*|_{M_0} \otimes \bigotimes^q H^*|_{M_0})$. For example, μ_h is type $(n-k, k)$ and the metric is $h = h^{(2,0)} + h^{(1,1)}$ (no $1,1$ part since vertical and horizontal are orthogonal complements).
In this setting, u is weakly horizontally conformal if and only if $u^* g = \lambda^2 h^{(0,2)}$, where taking traces with respect to h, we obtain

$$|du|^2 = \operatorname{tr}_h(u^* g) = \lambda^2 \operatorname{tr}_h(h^{(0,2)}) = \lambda^2 k$$

so that $\lambda^2 = \frac{1}{k}|du|^2$ (not n).

Remark
If $M^0 = M$, then a weakly horizontally conformal map is a horizontally conformal submersion ($\lambda > 0$ everywhere on M). If $\lambda \equiv 1$, then u is a Riemannian submersion.

Remark
Let $u : (M^n, h) \to (L^k, g)$ be weakly horizontally conformal, so that $u|_{M^0} : M^0 \to L$ is a submersion, so for any $x \in M^0$, the fibre of u containing x, namely $u^{-1}(u(x))$, is a smooth $(n-k)$-submanifold by the implicit function theorem.

Theorem
Let $u : (M^n, h, \alpha) \to (L^k, g)$ with $\alpha \in \Omega^{n-k}(M)$ comass one, and u a smooth surjection. Then $\alpha \wedge u^* \mu_g \leq \lambda^k \mu_h$ ($\lambda^2 = \frac{1}{k}|du|^2$) where equality occurs if and only if (i) $u^* g = \lambda^2 h^{(0,2)}$ (so u is weakly horizontally conformal) and (ii) the fibres of $u|_{M^0}$ are α-calibrated.
Proof:
If $(du)_x = 0$, both sides of the inequality are zero. Now, let $x \in M^0$ so $(du)_x \neq 0$. If $\operatorname{rank}((du)_x) < k$, then $u^* \mu_g = 0$ but $\lambda > 0$, so the inequality is strict. So now suppose $\operatorname{rank}((du)_x) = k$. Let $e_1, \ldots, e_k$ be an oriented orthonormal basis of $H_x = \ker((du)_x)^\perp$, and let $\widetilde{e}_1, \ldots, \widetilde{e}_{n-k}$ be an oriented orthonormal basis of $V_x = \ker((du)_x)$, where the orientations are determined by the isomorphism $(du)_x : H_x \xrightarrow{\cong} T_{u(x)} L$ and the orientation on M given by the ordered splitting $T_x M = V_x \oplus H_x$, so that $\mu_h = \widetilde{e}_1 \wedge \ldots \wedge \widetilde{e}_{n-k} \wedge e_1 \wedge \ldots \wedge e_k$. Since $u^* \mu_g$ is type $(0, k)$, we have

$$(\alpha \wedge u^* \mu_g)(\widetilde{e}_1 \wedge \ldots \wedge \widetilde{e}_{n-k} \wedge e_1 \wedge \ldots \wedge e_k) = \alpha(\widetilde{e}_1, \ldots, \widetilde{e}_{n-k}) \mu_g((du)_x e_1, \ldots, (du)_x e_k)$$

$$\overbrace{\leq}^{\text{comass one}} \quad \mu_g(\Lambda^k (du)_x (e_1, \ldots, e_k))$$

However, $\mu_g(\xi) = \langle \mu_g^\sharp, \xi \rangle = |\xi|$ since μ_g forms an orthonormal basis of the top exterior power, so that we get the above is equal to

$$= |\Lambda^k((du)_x)(e_1 \wedge \ldots \wedge e_k)| = |\Lambda^k((du)_x)| \overbrace{\leq}^{\text{Hadamard}} \lambda^k = \lambda^k \mu_h(\widetilde{e}_1 \wedge \ldots \wedge \widetilde{e}_{n-k} \wedge e_1 \wedge \ldots \wedge e_k)$$

so $\alpha \wedge u^*\mu_g \leq \lambda^k \mu_h$. Now, suppose equality holds, so that equality holds in the two inequalities. When equality holds in the second inequality, we have that $(du)_x : H_x \to T_{u(x)}L$ is a cofnormal injection, so that $u^*g = \lambda^2 h^{(0,2)}$, so u is weakly horizontally conformal. Let $x \in M^0$, and let $\widetilde{e}_1, \ldots, \widetilde{e}_{n-k}$ be an oriented orthonormal basis of $V_x = \ker((du)_x) = T_x(u^{-1}(u(x)))$. If the first inequality is an equality, then $\alpha(\widetilde{e}_1, \ldots, \widetilde{e}_{n-k}) = 1$, so $u^{-1}(u(x))$ is α-calibrated. ■

Definition
$u : (M^n, h, \alpha) \to (L^k, g)$ is a Smith submersion if $\alpha \wedge u^*\mu_g = \frac{1}{(\sqrt{k})^k}|du|^k \mu_h$, which implies $\mu^*g = \frac{1}{k}|du|^2 h^{(0,2)}$. When $k > 1$, this is a first order fully nonlinear PDE for u.

Remark
A Smith submersion is only a submersion on M^0.

Remark
One can show $\alpha \wedge u^*\mu_g = \lambda^k \mu_h$ if and only if $u^*\mu_g = \lambda^k (\star\alpha)^{(0,k)}$ and $u^*g = \lambda^2 h^{(0,2)}$.

Theorem (Energy Inequality)
Let $u : (M^n, h, \alpha) \to (L^k, g)$ be as before with M compact, and suppose $d\alpha = 0$. Then if u is a Smith submersion, then u is k-harmonic. Moreover, it is minimizing for the k-energy in its homotopy class.
Proof:
We have

$$E_k(u) = \frac{1}{(\sqrt{k})^k} \int_M |du|^k \mu_h = \int_M \lambda^k \mu_h \overbrace{\geq}^{\text{theorem}} \int_M \alpha \wedge u^*\mu_g = ([\alpha] \cup u^*[\mu_g]) \cdot [M]$$

where $\cup$ is the cup product, $[\alpha] \in H^{n-k}(M), u^*[\mu_g] \in H^k(M)$, and $[M] \in H_n(M)$. But the right hand side is constant on the homotopy class of u (since homotopic maps induce the same pullback on cohomology), so we get a universal lower bound for E_k on each homotopy class, and it is attained precisely at the Smith submersions. ■

Remark
Conformality in this case is slightly different: for maps $u : M^n \to L^k$, the n-energy is conformally invariant, not the k-energy. So a Smith submersion is not a solution of a PDE which only depends on the conformal class of the metric on M. It is invariant under a "horizontal" conformal change: write $h = h^{(2,0)} + h^{(0,2)}$, and define a new metric $\widetilde{h} = h^{(2,0)} + f^2 h^{(0,2)}$ for $f > 0$ a positive function. Then, since $du = 0$ on the vertical part,

$$\widetilde{\lambda}^2 := \frac{1}{k}|du|^2_{\widetilde{h},g} = f^{-2}\lambda^2$$

so that $\lambda = f\widetilde{\lambda}$ and $\mu_{\widetilde{h}} = f^k \mu_h$, so that

$$\alpha \wedge u^*\mu_g = \lambda^k \mu_h = \widetilde{\lambda}^k \mu_{\widetilde{h}}$$

and similarly $u^*g = \lambda^2 h^{(0,2)} = \widetilde{\lambda}^2 \widetilde{h}^{(0,2)}$.

Why should we care?
If $(M^n, h, \alpha) = (M^{2m}, h, \omega)$ is an almost Kähler manifold, so that $k = 2$ and (L^2, g) is a 2-dimensional

Kähler manifold, then the Smith immersion equation reduces to the J-holomorphic map equation (maybe up to a $\pm$), and so the Smith immersion equation generalizes it. What is the J-holomorphic map equation good for? In the 90s, something called Gromov-Witten invariants were developed, where one "counts" the J-holomorphic maps $(\Sigma^2, g) \to (M^{2m}, \omega, J, g)$ of a compact Riemann surface into a symplectic (i.e. almost Kähler) manifold. This "count" depends only on the symplectic structure. Similarly, we can try to "count" Smith immersions for other calibrations. Cheng-Karigiannis-Madnick did a lot of the analysis necessary for this "counting" to be well-defined, but still missing understanding the critical set $L \setminus L^0 = \{x \in L : (du)_x = 0\}$ for a given Smith immersion u. In the J-holomorphic case, this is finite (proved with unique continuation, which we don't know holds if $k > 2$).
For Smith submersions, away from $M \setminus M^0$ we get what's called a smooth calibrated fibration.
Why do we care about these?
Mirror symmetry is a physical prediction about how two Calabi-Yau manifolds are related: the symplectic geometry of one and its Gromov-Witten invariants has something to do with its mirror pair's complex geometry. The SYZ (Strominger-Yau-Zaslow) "conjecture" is a differential geometric explanation for this. Given a Calabi-Yau threefold M^6 (its calibrated submanifolds are special Lagrangians), there should exist a calibrated fibration $M^6 \to L^3$ with some singular fibres, where then we can construct the mirror pair by (1) removing singular fibres, (2) taking duals of the smooth fibres, (3) "compactify" (add back in the singular ones) somehow to get the mirror. The Smith submersion equation is the first example of a natural geometric PDE whose solutions give calibrated fibrations.
The GYZ conjecture (Gukov-Yau-Zaslow) is the analogous thing for M^7 a G_2-manifold and the fibres coassociative submanifolds.

Eells-Sampson Theorem

Let (M^m, g) and (N^n, h) be Riemannian manifolds. Given the energy map $E : \mathrm{Map}(M, N) \to \mathbb{R}$, we saw for u_t a smooth variation of some $u : M \to N$, if $\frac{d}{dt}\Big|_{t=0} u_t = v \in \Gamma(T^*M \otimes u^*TN)$, then

$$\frac{d}{dt}\Big|_{t=0} E(u_t) = -\langle\langle \tau(u), v \rangle\rangle$$

We can interpret this as saying $(\operatorname{grad} E)_u = -\tau(u)$. But the gradient is the direction of steepest ascent of the function, so what if we try to move in the opposite direction? This motivates:

Definition
Fix some map $f \in \mathrm{Map}(M, N)$. The harmonic map heat flow, starting at f, is the PDE $\frac{\partial}{\partial t} u_t = \tau(u_t)$ (evolution equation) with initial condition $u_0 = f$. This flow (if it exists) would stop at a harmonic map.
In local coordinates $x^1, \ldots, x^m$ on M and $y^1, \ldots, y^n$ on u, this says

$$\frac{\partial u^\gamma}{\partial t} = g^{ij} \left(\frac{\partial^2 u^\gamma}{\partial x^i \partial x^j} - {}^M\Gamma_{ij}^k \frac{\partial u^\gamma}{\partial x^k} + ({}^N\Gamma_{\alpha\beta}^\gamma \circ u) \frac{\partial u^\alpha}{\partial x^i} \frac{\partial u^\beta}{\partial x^j} \right)$$

where $u^1, \ldots, u^n$ are functions of $x^1, \ldots, x^m, t$. This is a nonlinear "heat equation", quadratic in first derivatives (it is of the form $\frac{\partial u^\gamma}{\partial t} = \Delta_M u^\gamma +$ quadratic terms).

Theorem (Eells-Sampson)
Suppose M, N are both compact and that the sectional curvature of (N, h) is nonpositive (so $R^N(X, Y, Y, X) \leq 0$). Then, for any initial condition f, the harmonic map heat flow has a unique smooth solution for all $t \in [0, \infty)$ and converges to a harmonic map u_∞ as $t \to \infty$.

Corollary
If M, N are compact and the sectional curvature of (N, h) is nonpositive, then every homotopy class of maps $M \to N$ has a harmonic representative.

Compare this with a theorem from Riemannian geometry which says that every homotopy class of maps $S^1 \to (N, h)$ with N compact (no curvature assumption), has a harmonic representative (i.e. is represented by a geodesic).

Remark
The condition that the sectional curvature of (N, h) is nonpositive is sufficient, but not necessary.

Remark
The harmonic map heat flow was the first geometric flow to be studied (1960s), kicking off a whole industry like:
Ricci flow, $\frac{\partial g_t}{\partial t} = -2\operatorname{Ric}(g_t)$, introduced by Hamilton in the 80s (proving that if the initial metric has positive scalar curvature, it converges to a constant sectional curvature method) and used by Perelman to resolve the Poincaré conjecture and geometrization of 3-manifolds.
Mean curvature flow, for an immersion $i : L^k \to (\mathbb{R}^n, g)$, $\frac{\partial i}{\partial t} = -H(i)$.
Yang-Mills flow, Yamabe flow, G_2-Laplacian flow, etc.

The proof (and many geometric flow proofs) has three key steps:
(1) Short-time existence and uniqueness. That is, do there exist unique smooth solutions to the flow, at least for $t \in [0, T)$ for some $T > 0$. This is usually only (parabolic) PDE theory.
(2) Long-time existence. Can we take $T = \infty$?
(3) If we have (2), does $t \to \infty$ converge to anything? Both (2) and (3) are both analytic and geometric. We'll sketch (1) and work through (2) and (3) in detail.

Sketch of Step (1):
In local coordinates, take
$$\frac{\partial u^\gamma}{\partial t} = \Delta_M u^\gamma + ({}^N\Gamma^\gamma_{\alpha\beta} \circ u)\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}$$
Since $u_0 = f$, for small time, we can approximate the non-linear part by replacing u with f, giving the linear parabolic PDE
$$\frac{\partial \widehat{u}^\gamma}{\partial t} = \Delta_M \widehat{u}^\gamma + ({}^N\Gamma^\circ_{\alpha\beta} \circ f)\frac{\partial f^\alpha}{\partial x^i}\frac{\partial f^\beta}{\partial x^j} \quad (\star)$$
There is a good theory of linear parabolic PDEs, giving a unique solution $\widehat{u}$ to $\star$ with $\widehat{u}_0 = f$. We expect $\widehat{u} \approx u$ for small time. Consider the differential operator
$$P : u^\gamma \mapsto \Delta_M u^\gamma - ({}^N\Gamma^\gamma_{\alpha\beta} \circ u)\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}$$
so that $P(u) = 0$ at each t if and only if u is a solution to the harmonic map heat flow. Let
$$Q(z) = P(\overbrace{\widehat{u} + z}^{u}) - P(\widehat{u})$$
where $Q(0) = 0$, where Q is a map $X \to Y$, where X and Y are appropriate Hölder spaces $C^{k,\alpha}$. Then one shows $Q'(0)$ is an isomorphism (of Banach spaces) and applies the inverse function theorem for Banach spaces to conclude Q is a homeomorphism between sufficiently small neighbourhoods of 0 in X, Y. Thus, given $k \in Y$ with $|k|$ small, then there exists a unique $z \in X$ with $|z|$ small such that $Q(z) = k$, $z_0 = 0$, $\left.\frac{\partial z}{\partial t}\right|_{t=0} = 0$. Let $u = \widehat{u} + z$, and $w = P(\widehat{u})$. Then,
$$Q(z) = P(\widehat{u} + z) - P(\widehat{u}) = k$$
implying $P(u) = w + k$, and we know $u_0 = f$. Now, let $\zeta : \mathbb{R} \to \mathbb{R}$ be a smooth cutoff function that is 1 on $[0, \varepsilon)$ and 1 on $[2\varepsilon, \infty)$. Choose $k = -\zeta w$, so that this k has sufficiently small norm for ε small enough (since w is the linear solution), so that $P(u) = w + (-\zeta)w = 0$ for $0 \leq t < \varepsilon$. By elliptic regularity,

the solution u is smooth since f is. Thus, there exists $\varepsilon > 0$ such that there exists a unique solution to harmonic map heat flow with initial condition f (this sketch does not discuss uniqueness). ε depends only on $(M,g),(N,h),f$ and the Hölder exponent.
Thus, given an initial condition f, there is a maximal time interval $[0,T)$ on which there exists a unique smooth solution to $\frac{\partial u}{\partial t} = \tau(u)$ with $u_0 = f$ for some $T \in (0,\infty]$.

For steps 2 and 3, we'll need Weitzenböck-type formulas for harmonic map heat flow.

Given $u \in \mathrm{Map}(M,N)$, we work on the vector bundle $E = T^*M \otimes u^*TN$ over M. E has a fibre metric and a connection $\nabla : \Gamma(E) \to (T^*M \otimes E)$, where given $\sigma \in \Gamma(E)$, recall we showed

$$(\nabla^2\sigma)(X,Y) = \nabla_X(\nabla_Y\sigma) - \nabla_{\nabla_X Y}\sigma \quad (\dagger)$$

If $X,Y,Z \in \Gamma(TM)$, then $\sigma(Z) = \Gamma(u^*TN)$, so that the Leibniz rule takes the form

$$(R^\nabla(X,Y)\sigma)(Z) = (R^{u^*\nabla}(X,Y))(\sigma(Z)) - \sigma(R^M(X,Y)Z) \quad (\dagger\dagger)$$

Proposition (Ricci Identity)
Let $\sigma = \sigma^\alpha_k \mathrm{d}x^k \otimes u^*\frac{\partial}{\partial y^\alpha} \in \Gamma(E)$, and write $\nabla^2\sigma = \nabla_i\nabla_j\sigma^\alpha_k \mathrm{d}x^i \otimes \mathrm{d}x^j \otimes \mathrm{d}x^k \otimes u^*\frac{\partial}{\partial y^\alpha}$. Then,

$$\nabla_i\nabla_j\sigma^\alpha_k - \nabla_j\nabla_i\sigma^\alpha_k = -\,{}^M R^\ell_{ijk}\sigma^\alpha_\ell + {}^N R^\alpha_{\beta\gamma\delta}\frac{\partial u^\beta}{\partial x^i}\frac{\partial u^\gamma}{\partial x^j}\sigma^\delta_k$$

Proof:
Write $e_i = \frac{\partial}{\partial x^i}$. By $(\dagger\dagger)$,

$$\begin{aligned}(R^\nabla(e_i,e_j)\sigma)(e_k) &= (R^{u^*\nabla}(e_i,e_j))(\sigma(e_k)) - \sigma(R^M(e_i,e_j)e_k) \\ &= \left({}^M R^\alpha_{\beta\gamma\delta}\frac{\partial u^\beta}{\partial x^i}\frac{\partial u^\gamma}{\partial x^j}\sigma^\delta_k - {}^M R^\ell_{ijk}\sigma^\alpha_\ell\right) u^*\frac{\partial}{\partial y^\alpha}\end{aligned}$$

On the other hand, by $(\dagger)$,

$$(\nabla^2\sigma)(e_i,e_j,e_k) = \nabla_i\nabla_j\sigma^\alpha_k u^*\frac{\partial}{\partial y^\alpha} = (\nabla_i(\nabla_j\sigma))(e_k) - (\nabla_{\nabla_i e_j}\sigma)(e_k)$$

so

$$(\nabla^2\sigma)(e_i,e_j,e_k) - (\nabla^2\sigma)(e_j,e_i,e_\ell) = \left(\nabla_i(\nabla_j\sigma) - \nabla_j(\nabla_i\sigma) - \nabla_{\underbrace{\nabla_{e_i}e_j - \nabla_{e_j}e_i}_{=[e_i,e_j]=0}}\sigma\right)(e_k) = (R^\nabla(e_i,e_j)\sigma)(e_k) \quad \blacksquare$$

Now let $\sigma = du \in \Gamma(E)$, so that $\sigma^\alpha_k = \frac{\partial u^\alpha}{\partial x^k}$, so that the Ricci identity becomes

$$\nabla_i\nabla_j\frac{\partial u^\alpha}{\partial x^k} - \nabla_j\nabla_i\frac{\partial u^\alpha}{\partial x^k} = -\,{}^M R^\ell_{ijk}\frac{\partial u^\alpha}{\partial x^\ell} + {}^N R^\alpha_{\beta\gamma\delta}\frac{\partial u^\beta}{\partial x^i}\frac{\partial u^\gamma}{\partial x^j}\frac{\partial u^\delta}{\partial x^k}$$

Definition
Let $u = u_t$ (depends on t). Let $\kappa(u) = \frac{1}{2}\left|\frac{\partial u}{\partial t}\right|^2$ be the kinetic energy density, and $K(u) = \int_M \kappa(u)\mu_g$ be the kinetic energy. These definitions are valid on $[0,T)$.

Proposition (Weitzenböck formulas for harmonic map heat flow)
Suppose $\frac{\partial u}{\partial t} = \tau(u)$ on $[0,T)$. Then,

$$\frac{\partial}{\partial t}e(u) = \Delta(e(u)) - |\nabla du|^2 - g^{ij}\langle (du)(\mathrm{Ric}^M(e_i)), (du)e_j\rangle + g^{ij}g^{k\ell}R^N((du)e_i,(du)e_k,(du)e_\ell,(du)e_j)$$

and
$$\frac{\partial}{\partial t}\kappa(u) = \Delta(\kappa(u)) - \left|\nabla_t \frac{\partial u}{\partial t}\right|^2 + g^{ij}R^N((du)e_i, \frac{\partial u}{\partial t}, \frac{\partial u}{\partial t}, (du)e_j)$$

Proof:
Since all the connections involved are metric compatible, $\nabla_i(g^{jk}u^*h_{\alpha\beta}) = 0$ and $\nabla_t(g^{jk}u^*h_{\alpha\beta}) = 0$. With this, we compute

$$\frac{\partial}{\partial t}e(u) = \nabla_t\left(\frac{1}{2}g^{ij}u^*h_{\alpha\beta}\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j}\right) = g^{ij}u^*h_{\alpha\beta}\nabla_t\left(\frac{\partial u^\alpha}{\partial x^i}\right)\frac{\partial u^\beta}{\partial x^j} \quad (\star)$$

By assignment 1,
$$\nabla_t\frac{\partial u}{\partial x^i} = \nabla_{\frac{\partial}{\partial t}}(du)(\frac{\partial}{\partial x^i}) = \nabla_{\frac{\partial}{\partial x^i}}(du)(\frac{\partial}{\partial t}) = \nabla_i\frac{\partial u}{\partial t}$$

so that $(\star)$ becomes

$$\frac{\partial}{\partial t}e(u) = g^{ij}u^*h_{\alpha\beta}\left(\nabla_i\frac{\partial u^\alpha}{\partial t}\right)\frac{\partial u^\beta}{\partial x^j} = g^{ij}\langle\nabla_i\frac{\partial u}{\partial t}, (du)(e_j)\rangle$$

For the Laplacian, by the same calculation as above and then the product rule,

$$\Delta(e(u)) = g^{k\ell}\nabla_k\nabla_\ell(e(u)) = g^{k\ell}\nabla_k(g^{ij}u^*h_{\alpha\beta}\nabla_\ell\frac{\partial u^\alpha}{\partial x^i}\frac{\partial u^\beta}{\partial x^j})$$
$$= |\nabla du|^2 + g^{k\ell}g^{ij}u^*h_{\alpha\beta}\left(\nabla_k\nabla_\ell\frac{\partial u^\alpha}{\partial x^i}\right)\frac{\partial u^\beta}{\partial x^j}$$
$$\overbrace{=}^{\nabla du \text{ symmetric}} |\nabla du|^2 + g^{k\ell}g^{ij}u^*h_{\alpha\beta}\left(\nabla_k\nabla_i\frac{\partial u^\alpha}{\partial x^\ell}\right)\frac{\partial u^\beta}{\partial x^j}$$
$$\overbrace{=}^{\text{Ricci identity}} |\nabla du|^2 + g^{ij}\underbrace{\langle\nabla_i\tau(u), du(e_j)\rangle}_{g^{k\ell}\nabla_i\nabla_k\frac{\partial u^\alpha}{\partial x^\ell}\text{ from Ricci}} + \underbrace{(\mathrm{Ric}\,M)_i^r u^*h_{\alpha\beta}g^{ij}\frac{\partial u^\alpha}{\partial x^r}\frac{\partial u^\beta}{\partial x^j}}_{g^{k\ell}(-{}^MR^r_{ki\ell}\frac{\partial u^\alpha}{\partial x^r})\text{from Ricci}} + g^{k\ell}g^{ij}R^N((du)e_k, (du)e_i, (du)e_\ell, (du)e_j)$$
$$= |\nabla du|^2 + g^{ij}\langle\nabla_i\tau(u), (du)e_j\rangle + g^{ij}\langle(du)(\mathrm{Ric}^M(e_i)), (du)(e_j)\rangle - g^{ij}g^{k\ell}R^M((du)e_k, (du)e_i, (du)e_j, (du)e_\ell)$$

Using that $\frac{\partial u}{\partial t} = \tau(u)$, we see that

$$\frac{\partial}{\partial t}e(u) - \Delta(e(u)) = -|\nabla du|^2 - g^{ij}\langle(du)(\mathrm{Ric}^M(e_i)), (du)e_j\rangle + g^{ij}g^{k\ell}R^N((du)e_i, (du)e_k, (du)e_\ell, (du)e_j)$$

The second identity is the same arguments only with $\kappa(u)$ in place of $e(u)$ and using that $\mathrm{Ric}^{(0,T)\times M}(\frac{\partial}{\partial t}, \frac{\partial}{\partial x^i}) = 0$. ■

Corollary
Let $u : [0, T) \times M \to N$ be a harmonic map heat flow.
(1) If $K^N \leq 0$ and if there exists $c > 0$ such that $\mathrm{Ric}^M \geq -cg$ (always true if M is compact), then $\frac{\partial}{\partial t}e(u) \leq \Delta(e(u)) + 2ce(u)$.
(2) If $K^N \leq 0$, then $\frac{\partial}{\partial t}(\kappa(u)) \leq \Delta(\kappa(u))$.
Proof:
(2) is immediate, as $-|\nabla_t\frac{\partial u}{\partial t}|^2$ is nonpositive, and $g^{ij}R^N((du)e_i, \frac{\partial u}{\partial t}, \frac{\partial u}{\partial t}, (du)e_j)$ is nonpositive by our constraint on the sectional curvature, so that the Weitzenböck formula above gives the result.
For (1), since $\mathrm{Ric}^M \geq -cg$, we have

$$\langle du(\mathrm{Ric}^M(e_i)), (du)(e_j)\rangle \geq -c\langle(du)e_i, (du)e_j\rangle$$

so that

$$-g^{ij}\langle (du)(\mathrm{Ric}^M(e_i)), (du)e_j\rangle \leq cg^{ij}\langle (du)e_i, (du)e_j\rangle = 2ce(u)$$

and the result follows by the Weitzenböck formula the same way as (2). ■

Corollary
Let $u : [0,T) \times M \to N$ be a harmonic map heat flow, with M compact. Then
(1) $\frac{d}{dt}E(u) = -2K(u) \leq 0$.
(2) If $K^N \leq 0$, then $\frac{d^2}{dt^2}E(u) = -2\frac{d}{dt}K(u) \geq 0$.
Proof:

$$(1) \quad \frac{d}{dt}E(u) = -\langle\langle \frac{\partial u}{\partial t}, \tau(u)\rangle\rangle = -\langle\langle \frac{\partial u}{\partial t}, \frac{\partial u}{\partial t}\rangle\rangle = -K(u) \leq 0$$

For (2), using (1),

$$\frac{d}{dt}K(u) = \frac{d}{dt}\int_M \kappa(u)\mu_g = \int_M \left(\frac{\partial}{\partial t}\kappa(u)\right)\mu_g \overbrace{\leq}^{\text{above}} \int_M (\Delta(\kappa(u))\mu_g = 0$$

by the divergence theorem. ■

Theorem (Step 3 of Eells-Sampson Theorem)
If harmonic map heat flow has long-time existence ($T = \infty$), then $\lim_{t\to\infty} u_t = u_\infty$ is a harmonic map.
Proof:
Let $F = E(u) : [0,\infty) \to \mathbb{R}$. Then $F \geq 0$, $\frac{d}{dt}F \leq 0$, and $\frac{d^2}{dt^2}F \geq 0$ by the previous corollaries. Thus, F is a nonnegative nonincreasing convex function, implying by single variable calculus that $\lim_{t\to\infty} F'(t) = 0$. Indeed, let $G = F'$. The fact G is nonpositive but always increasing means that it has to have a horizontal asymptote as $t \to \infty$. If this asymptote is negative, then since $F(x) = \int_0^x G(t)dt + C$, it would imply F is negative, a contradiction. Thus, since $F' = -2K(u)$, we have $K(u_\infty) = 0$. Thus, since

$$\frac{\partial}{\partial t}(u) = \tau(u)$$

for all t and $K = \int_M |\frac{\partial u}{\partial t}|^2\mu_g$, as $t \to \infty$, the left hand side goes to zero, so that $\tau(u_\infty) = 0$ by continuity, so u_∞ is harmonic. ■

It remains to prove step 2 (long-time existence).

Lemma (Maximum Principle)
Suppose $u : [0,T) \times M \to \mathbb{R}$ is smooth with M compact such that $(\Delta^M - \frac{\partial}{\partial t})u \geq 0$ on $(0,T) \times M$. Then

$$\max_{[0,T)\times M} u = \max_{\{0\}\times M} u$$

In particular, the maximum on the left hand side exists. $\Delta - \frac{\partial}{\partial t}$ is called the heat operator.
Proof:
Let $\varepsilon_1, \varepsilon_2 > 0$. Let $\widehat{u}(t,x) = u(t,x) - \varepsilon_1 t$. Let $Q = [0, T-\varepsilon_2] \times M$, which is a compact manifold with boundary. Since $\widehat{u}$ is continuous on Q, it attains a maximum at some $(t_0, x_0) \in Q$. We claim $t_0 = 0$, which completes the proof since $\varepsilon_1, \varepsilon_2$ are arbitrary so that we can let them go to zero to get the result. Thus, suppose $t_0 > 0$. Then $(\Delta - \frac{\partial}{\partial t})u \geq 0$, implying

$$\Delta\widehat{u} - (\frac{\partial \widehat{u}}{\partial t} + \varepsilon_1) \geq 0$$

so

$$\frac{\partial \widehat{u}}{\partial t} \leq \Delta\widehat{u} - \varepsilon_1 = g^{ij}\left(\frac{\partial^2 \widehat{u}}{\partial x^i \partial x^j} - \Gamma^k_{ij}\frac{\partial \widehat{u}}{\partial x^k}\right) - \varepsilon_1$$

At (t_0, x_0), $\widehat{u}$ has a maximum, so $\frac{\partial \widehat{u}}{\partial x^i}(t_0, x_0) = \frac{\partial \widehat{u}}{\partial t}(t_0, x_0) = 0$, and $\frac{\partial^2 \widehat{u}}{\partial x^i \partial x^j}(t_0, x_0)$ is negative semi-definite, so that the above inequality tells us at (t_0, x_0), $0 \leq -\varepsilon_1$, a contradiction. ■

Proposition
Let $u : [0, T) \times M \to N$ be a harmonic map heat flow, $u_0 = f$, with $K^N \leq 0$, and $\mathrm{Ric}^M \geq -cg$ for some $c > 0$ (automatic if M is compact). Then, for all $(t, x) \in (0, T) \times M$,
(1) $e(u(t, x)) \leq \exp(2ct) \sup_{y \in M}(e(f)(y))$.
(2) $|\frac{\partial u}{\partial t}(t, x)| \leq \sup_{y \in M} |\frac{\partial u}{\partial t}(0, y)|$.
This says we have bounds on growth of energy density and kinetic energy density in terms of our initial data.
Proof:
To prove (1), by the Corollary above, we have

$$(\Delta - \frac{\partial}{\partial t})e(u) \geq -2ce(u)$$

Let $v = \exp(-2ct)e(u)$, so that

$$\frac{\partial v}{\partial t} = -2c \exp(-2ct)e(u) + \exp(-2ct)\frac{\partial}{\partial t}e(u)$$

and

$$\Delta v = e^{-2ct}\Delta(e(u))$$

so that

$$\Delta v - \frac{\partial v}{\partial t} = \exp(-2ct)(\Delta(e(u)) - \frac{\partial}{\partial t}e(u) + 2ce(u)) \geq 0$$

by our bound above and the exponential being positive. Thus, by the maximum principle applied to v,

$$v(t, x) = \exp(-2ct)(e(u))(t, x) \leq \max_{y \in M} v(0, y) = \max_{y \in M}(e(f))(y)$$

implying

$$(e(u))(t, x) \leq \exp(2ct) \max_{y \in M}(e(f))(y)$$

For (2), we had $(\Delta - \frac{\partial}{\partial t})(\kappa(u)) \geq 0$, so by the maximum principle applied to $\kappa(u)$, we have

$$\frac{1}{2}\left|\frac{\partial u}{\partial t}(t, x)\right|^2 \leq \max_{y \in M} \frac{1}{2}\left|\frac{\partial u}{\partial t}(0, y)\right|^2 \quad ■$$

Theorem (Long-Time Existence of Harmonic Map Heat Flow)
In the above setup, $T = \infty$ and the limit $t \to \infty$ of u_t exists and is smooth.
Proof:
Suppose $T < \infty$. By (1) above, $e(u(t))$ is uniformly bounded on $[0, T) \times M$, so that $|du|$ is uniformly bounded on $[0, T) \times M$. Also, $|\frac{\partial u}{\partial t}|$ is uniformly bounded on $[0, T) \times M$ by (2).
These give uniform bounds on u in the appropriate Hölder spaces. Let t_i be a sequence converging to T. Consider the family of maps $\{u(t_i, \cdot) : M \to N\}_i$, so that because we have uniform bounds, this is a uniformly equicontinuous family, hence by Arzela-Ascoli, we can pass to a convergent subsequence which converges (in our Hölder space) to some $u(T, \cdot)$, with $\frac{\partial u}{\partial t}(t_i, \cdot) \to \frac{\partial u}{\partial t}(T, \cdot)$. Since $\frac{\partial u}{\partial t} = \tau(u)$ for all $t \in [0, T)$, we have $\frac{\partial u}{\partial t}(T, \cdot) = \tau(u(T, \cdot))$. But by short-time existence and uniqueness, we can then do the harmonic map heat flow starting at $u(T, \cdot)$, we can continue the flow past time T, contradicting maximality. A similar argument shows the limit as $t \to \infty$ of u_t exists and is smooth. ■